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Adaptive Damper Control Through Real-Time Driver Behaviour Identification

Enhancing Vehicle Dynamics and Passenger Comfort Through Simulation in CarMaker

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Adaptive Damper Control Through Real-Time Driver Behaviour Identification

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Abstract

It is not uncommon for drivers to make mistakes which jeopardize the comfort or the handling of the vehicle. Mistakes such as braking too aggressively at a red light can lead to discomfort for the driver and passengers. Using a set of driver inputs, such as pedal positions, these mistakes can be identified and categorised based on the type of the mistake. This was done through retrieving telemetry from recorded drives from a Tesla Model 3 in Assetto Corsa. These logged runs were exported to MATLAB and then co-simulated using Simulink and IPG CarMaker.

These identifications were then used to improve comfort and handling through active damper tuning based on the active identification. The damper changes activate once the identification is completed and remains active until the event is over.

The objective of the project was to develop an algorithm that could accurately identify these driver behaviours and modify the dampers individually. The goal being, to either increase comfort or improve the handling, depending on which mode of the algorithm is selected.

The evaluation for the comfort scenarios was accomplished using a factorial test design where each damper setting was modified individually, after which the run was simulated and the results were exported. This was repeated three times for each driver behaviour, between each test damper changes were made to increase the comfort. The results showed that the final setup could significantly improve the comfort, for a majority of the behaviours. The largest advancements were made in the pitch velocity, while smaller improvements were made in vertical acceleration and longitudinal jerk.

For the dynamic scenarios' evaluation a number of participants performed a blind test in the driving simulator at Brembo N.V. After each run the drivers filled out a questionnaire about how the car felt to drive. The study of the dynamic scenarios resulted in inconclusive results as the simulator's calibration was subpar.

Keywords
Adaptive damper control, driver behaviour, driver in the loop

Sammanfattning

Det är inte ovanligt för förare att göra misstag som förvärrar bilens komfort eller kördynamik. Misstag som till exempel att bromsa för aggresivt vid ett rödlyse kan skapa obekvämlighet för föraren och passagerare. Genom att använda en uppsättning av förarparmeterar, såsom pedalernas position, kan sådana felbeteenden bli identifierade och kategoriserade baserat på typen av fel. Detta gjordes genom att hämta telemetri från inspelade körningar med en Tesla Model 3 i Assetto Corsa. Data från dessa körningar exporterades sedan till MATLAB och cosimulerades med Simulink och IPG CarMaker.

Dessa identifieringar har sedan använts i ett försök att förbättra bekvämlighet eller kördynamik genom att göra aktiva dämpningsändringar baserat på den aktiva identifieringen. Dessa dämparändringar aktiveras när en identifiering har genomförts och kvarstår tills händelsen har passerat.

Syftet med detta projekt var att utveckla en algoritm som kan identifiera dessa förarfelbeteenden och ändra dämparna därefter. Med målet att antingen förbättra komfort eller kördynamik beroende på det valda läget i algoritmen.

För att värdera komfortscenariorna användes ett faktoriellt test, där varje dämparinställning ändrades individuellt. Därefter simulerades varje körning och resultated exporterades. Detta upprepades tre gånger för varje beteende, mellan varje test gjordes dämparändringar för att förbättra komforten. Resultaten visade att den rekommenderade förändringen av dämparegenskaperna kunde öka komforten signifikant i flertalet identifieringar. De största minskningarna fanns i nigvinkelhastighet medans mindre förbättringar uppnådes i vertikalaccelerationer och longitudinellt ryck.

För det dynamiska körfallet gjordes ett blindtest på ett antal deltagare i en körsimultor hos Brembo N.V. Efter varje körning fyllde förarna i en enkät om hur körupplevelsen var. Studien för det dynamiska körfallet resulterade dock i inkonklusiva resultat då simulatorrattens kalibrering ej uppfyllde tillfredställande nivå.

Nyckelord
Adaptiv dämparkontroll, förarbeteende, förare i loopen

Preface

Use of AI
During the development of this project AI has been deployed in multiple aspects. AI has most prominently been used as a tool when developing the algorithm, it has been used to find coding solutions as well as generation of repetitive code when the solution had been found, i.e. copying a algorithm that been created for one state onto another state. There has been no use of AI when writing the report or as a source for information.

Dividing of Responsibilities
During the majority of the project, throughout development of the algorithm and conceptualization of states has the work been conducted equally and in unison between the two authors. Towards the end of the project some of the work was partitioned, during which David Sipos put a majority focus on post-processing results while Ted Degernäs focused on report writing.

Acknowledgments

We would like to express our appreciation to Öhlins Racing AB for the support and organization of this project, and also in involving us in group meetings and activities, which was our first taste of working as engineers.

A special gratitude to Emil Sällberg, our supervisor at Öhlins, guiding us throughout this study, giving very helpful insights on how the dampers act in the different manoeuvres. Also, we would like to thank Mats Widmark, a simulations engineer at Öhlins, who not only helped with the implementation of the car model in CarMaker but also reached out to Brembo N.V.'s simulator team, which gave us the opportunity to test our model in their environment. While performing the experiment at Brembo, Allessandro Manzoni, Riccardo Fornasari and Pietro Caresia were a big help with running the simulator and extracting the data.

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Acronyms

AB Aggressive Braking
ABS Anti-lock Braking System
AC Assetto Corsa
AIT Accelerating Into Turn
AT Aggressive Throttle
BAT Brake After Throttle
BT Brake Tapping
CM CarMaker
DiL Driver in the Loop
ESC Electronic Stability Control
EV Electric Vehicle
FL Front Left
FR Front Right
GUI Graphical User Interface
JS Jerky Steering
KPI Key Performance Indicators
NEO Not Easing Off
RL Rear Left
RMS Root Mean Square
RR Rear Right
SB Steering and Braking
SM Speed Mismatch With Road Surface
TAB Throttle After Braking
TC Traction Control

UB Uneven Braking

UI User Interface

UT Uneven Throttle

VA Varying Steering Angle in Constant Turn Radius

Symbols

a	Acceleration	$[\mathrm{m/s^2}]$
b_f	Front Track Width	$[\mathrm{m}]$
b_r	Rear Track Width	$[\mathrm{m}]$
C_D	Coefficient of Drag	$[-]$
C_L	Coefficient of Lift	$[-]$
C_S	Coefficient of Side-force	$[-]$
deg	Degree	$[^\circ]$
g	Gravitational Acceleration	$[\mathrm{m/s^2}]$
h	High Value During Factorial Tests	$[-]$
j	Jerk	$[\mathrm{m/s^3}]$
L	Wheelbase	$[\mathrm{m}]$
l	Low Value During Factorial Tests	$[-]$
x	Along Vehicle's Longitudinal Direction	$[-]$
y	Along Vehicle's Lateral Direction	$[-]$
z	Along Vehicle's Vertical Direction	$[-]$
θ	Pitch Angle	$[\mathrm{rad}]$
$\dot{\theta}$	Pitch Velocity	$[\mathrm{rad/s}]$
$\ddot{\theta}$	Pitch Acceleration	$[\mathrm{rad/s^2}]$
φ	Roll Angle	$[\mathrm{rad}]$
$\dot{\varphi}$	Roll Velocity	$[\mathrm{rad/s}]$
$\ddot{\varphi}$	Roll Acceleration	$[\mathrm{rad/s^2}]$
ψ	Yaw Angle	$[\mathrm{rad}]$
$\dot{\psi}$	Yaw Rate	$[\mathrm{rad/s}]$
$\ddot{\psi}$	Yaw Acceleration	$[\mathrm{rad/s^2}]$

1 Introduction

1.1 Background

Öhlins Racing AB is a renown and successful damper manufacturer that since 1974 has produced high quality dampers for road and track vehicles [1]. In 1984 they patented their semi-active continuously controlled electronic suspension (CES), which enables continuous electronic control of the dampers in order to create a smooth ride on the highway while still having dynamic handling characteristics in the curves. Although impressive, Öhlins are currently missing one key variable in their semi-active damping, the driver. The current technology changes damper characteristics based on vehicle rur. This means that the system makes its adjustments based on parameters such as vehicle velocity, vehicle acceleration, damper velocity and pitch angle, but not based on driver parameters such as throttle, brake and steering wheel rate. Adding the driver parameters to the already employed vehicle parameters could result in even better driving characteristics but also an improved understanding about how different drivers want to drive their vehicles. As a result, safety, performance, and comfort might be improved if the vehicle is more predictable, enjoyable, and easier to drive.

1.2 Objective and Limitations

The goal of this project is to investigate if it is possible to improve both objective and subjective driving characteristics of a vehicle in comfort and dynamic driving separately. Firstly, in a standard road driving condition also denoted as comfort, wherein urban, suburban and highway daily driving such as commuting and leisure is included. Secondly, a performance condition also denoted as dynamic, used on a racetrack. In standard, the objective will be to find which parameters has the largest contribution on comfort and what changes are possible by adjusting the dampers. While in performance mode the subjective driving parameters including vehicle feel and ease of driving will be under scrutiny. To achieve both of these metrics, the goal is to develop an algorithm that identifies driver behaviours and corrects for them by adjusting the vehicle's damper settings. These behaviours will be investigated further at a later stage in this report, however, some examples include a driver braking too aggressively, or steering hastily.

Some limitations had to be set in order to be able to achieve the finished model, these were the following:

- The simulations should be based on a Tesla Model 3, though the damper tuning should be able to scale to the current vehicle in the simulation.
- No effects of gear shifts will be analysed.
- All assists such as traction control (TC), electronic stability control (ESC), and anti-lock braking system (ABS) are turned off.
- Regenerative braking is turned off.
- Stochastic unevenness such as potholes will not be taken into account

1.3 Research Questions

The following research questions are the basis of this thesis work:

- For the chosen driving conditions, which metrics should be used?
- For the chosen vehicle and driving conditions, is it possible to create an algorithm that identifies driver characteristics?
- Is it possible to change the perceived comfort and handling by changing damper settings based on the algorithm identified situation?
- How should the dampers be tuned in each scenario?
- How much can the subjective/objective performance of the car be improved by utilising the algorithm and changing the damper characteristics?

1.4 Sustainability

The sustainability of this project will be evaluated from an economic, societal, and environmental viewpoint. From a societal perspective, it aims to develop the sustainability of transport through identifying flaws in the driver's behaviour and adjusting the dampers in order to result in a more controlled, comfortable, and safe riding experience. Economically and environmentally, this adaptive damper solution is developed through simulation software and simulators instead of testing on physical vehicles, which would lead to unnecessary fatigue on parts, consumption of fuel, and other financial costs of testing. The sustainable development of this solution could lead to a future with safer public transport, which makes the project align with the United Nations Sustainable Development Goal 11.2 [2].

2 Theory and Literature Review

2.1 Dampers

This section will describe the role of the damper in the vehicle system, its purpose and how different subsystems of the damper can be altered in order to change vehicle behaviour.

2.1.1 How Dampers Work

The dampers in combination with the springs are the main connection between the tyres and the carbody, with the suspended parts often called the sprung mass [3] and the rest described as the unsprung mass. The suspension system has to withstand and translate all forces from body to wheel and vice versa, whilst it also provides sufficient grip with the road in a multitude of driving conditions. For this to be achieved, adequate vertical load has to be applied on all four wheels in order to provide ample grip in lateral and longitudinal cases. In addition to the driving performance, comfort also has to be taken into account, especially in everyday driving situations. Here, the transmitted road roughness and noise have to be minimized in the sprung mass, especially in high-frequency scenarios, as those are the most disturbing for the passenger [4]. The combination of the vehicle's handling performance and comfort is important for the assessment of vehicle behaviour, however, optimizing this contradicting nature could lead to compromises.

In layman's terms, the suspension system primarily consists of one spring and one damper at each wheel, creating a so-called spring-damper system. For smaller displacements and velocities, a reasonable approximation is that the force of the spring is proportional to spring displacement, while the force of the damper is proportional to damper velocity, i.e., the difference in velocity between the wheel and the carbody. A suspension system of only a spring is technically possible, but it is impractical as it is the damper that dissipates energy. Excitation of a spring-only subsystem would result in perpetual movement, only slowing down due to internal friction losses or external forces. The damper will dissipate energy due to friction inflicted in small orifices within its chamber, converting the kinetic energy into thermal energy, which is dispersed into its surroundings.

2.1.2 Damper Setup

In order to understand the characteristics of different dampers, a damper curve is often utilized [5]. This curve displays force as a function of velocity. In Figure 1, a generic damper curve is presented. This curve can be broken down into two directions and two velocity thresholds. Firstly, all positive values, i.e. those in the first quadrant, represent compression, when the wheel is forced upwards towards the vehicle, while the negative values in the third quadrant represent rebound, when the wheel is moving down or away from the vehicle. Secondly, the nonlinearity, in this example at $200\frac{mm}{s}$ and $-200\frac{mm}{s}$ is often referred to as ”the knee”, and exists to have different damper characteristics for slow and fast movements. For slower movements, such as pitch from acceleration and braking or roll from lateral acceleration, sufficient force is needed to stop the movement, and therefore, the slope is higher in this area. The larger derivative in this area is also due to the force relation to velocity, meaning that lower speeds inherently result in lower forces. However, for high velocity transient movements such as hitting a pothole, the continuation of the high slope would result in very large forces and thus high accelerations in the cabin, in turn leading to decreased comfort. The areas before and after ”the knee” will henceforth be referred to as slow and fast compression for positive area, as well as slow and fast rebound for the negative values. The placement of ”the knee” and the slope of slow and fast compression/rebound will make a big difference in vehicle characteristics and must be evaluated carefully.

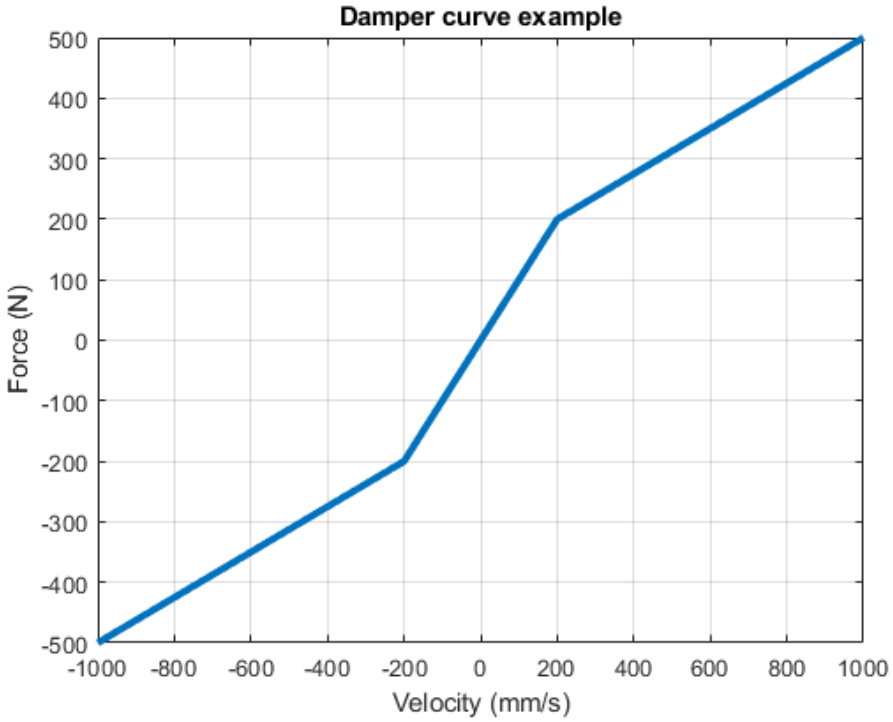


Figure 1: Damper curve example

As discussed above, the damper curve affects how the vehicle responds to

motion in different events. Semi-active dampers allow for the internals to be adjusted and can simply be thought of as expanding or shrinking the orifices that the fluids inside have to flow through. This in turn increases or decreases the friction and thereby the heat dissipated, which results in a modified damper curve.

When a damper's curve is designed, it is generally a compromise between the requirements on the handling and comfort of the vehicle. Typically, when comfort is prioritized, the damper curve is set to give less force at equivalent velocities, which leads to more wheel travel and lower propagating force into the cabin. On the other hand, when handling and response are prioritized, the damper curve is commonly defined to give higher forces at the equivalent velocities. This, in turn, means less wheel travel until the force of the load is met, thereby giving quicker feedback to the driver.

2.2 Literature Review

In this chapter, a few of the most important sources for this thesis are presented and discussed.

2.2.1 Understanding the Data

Throughout this project, time-based graphs of various driver inputs and vehicle parameters will be analysed in order to identify different driver behaviours. Driver inputs are used as they are signals that directly represent the driver's behaviour. Besides this, the vehicle parameters are also analysed as they are a result of the driver's inputs. For example, when a person understeers in a curve it is a consequence of over saturating the front tires [6], which could be due to the driver inputting too much brakes while also steering but it is also dependant on the speed the driver chose to enter the curve with. This is why all available driver input parameters such as throttle position, brake position, steering angle are all considered, along with all the available vehicle parameters including velocity, acceleration in the x, y, and z directions, and slip values. Interpreting these different data channels is most effectively done by plotting time or displacement on the x-axis and the corresponding parameter on the y-axis. These graphs can then be aligned along a shared x-axis, making it easier to analyse multiple parameters simultaneously and understand their correlations. An example of this from the software MoTeC i2 Pro, which is used to convert the telemetry, being driver inputs, from Assetto Corsa's (AC) format to Matlab's format, is shown in Figure 2.

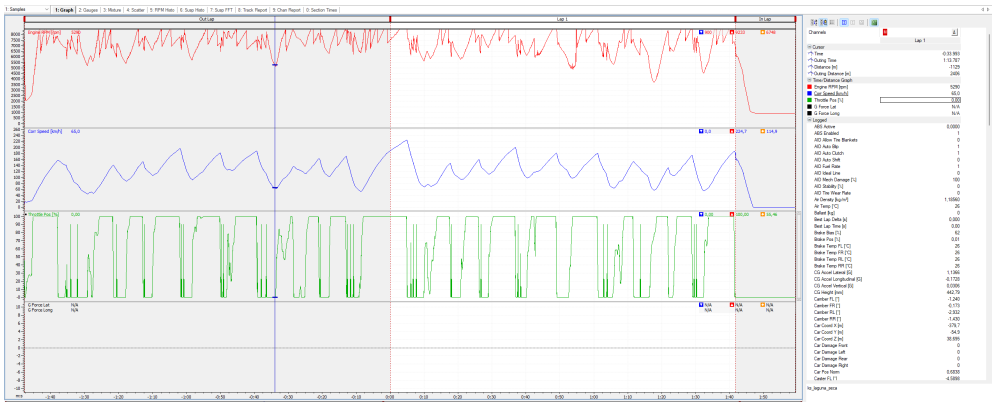


Figure 2: Telemetry example

This type of telemetry analysis is crucial in motorsport, since it is very difficult for a driver to accurately feel the exact grip limit while on track [7]. Data can be used to compare a driver's own lap times and inputs either with previous laps to optimize performance by approaching a personal best in every corner, or with faster drivers to identify where time can be gained [7]. In "A Delta Guide: Understanding Telemetry Data in Cornering Types", Davin Sturdivant expands on this concept and highlights which data channels are most important for different types of cornering manoeuvres [8]. For example, in a high-speed curve, the brake trace may be less important than throttle application, while in a low-speed curve, the greatest gains are often made through effective braking.

2.2.2 Studies on Parameter Influence in Different Driving Scenarios

2.2.2.1 Sensitivity analysis of human driving characteristics on road and driving conditions for active vehicle control systems

In "Sensitivity analysis of human driving characteristics on road and driving conditions for active vehicle control systems" Schnelle and Wang evaluated the sensitivity of humans for a variety of manoeuvres and driving parameters [9]. A human driver was modelled and verified against CarSim's baseline driver. This model controlled the steering and was integrated into CarSim's DiL co-simulation in combination with Simulink. In CarSim's DiL, a human is able to control the inputs of the simulation, in their study, the human was instead replaced with the modeled driver. The driver model's goal was to follow a path specific to each of the six studied manoeuvres. For each of them, a number of vehicle and driver parameters were changed independently, and the driver's deviation from the path was measured. This deviation was defined by Schnelle and Wang as the error, and a function of this was used to determine the sensitivity of the human driver model to each parameter in their study.

When they evaluated the acceleration sensitivity, the starting velocity was decreased to 10% and 50%, for high and low acceleration respectively, of the baseline's initial velocity, v_0, and the driver accelerated up to v_1 during the manoeuvre. Similarly, testing the sensitivity to deceleration was carried out by starting at v_0 and braking to 10% and 50% for the high and low deceleration respectively, during the manoeuvre. Both, changing the acceleration and de-

celeration, impacts the driver model's ability to track the path, as with varying velocities the distance the model looks ahead also changes. The article concluded that the human driver model is most sensitive to changes in acceleration, deceleration, and steering gain. This specific insight was the only element carried forward from their study into the design of the current work.

2.2.3 Studies on Evaluation of Comfort

Evaluating comfort is a complex task as it is subjective from person to person. In order to examine if the algorithm has tuned the dampers for more or less comfort, it is important to find a relation between measurable parameters and the human feeling of comfort.

2.2.3.1 Standards for passenger comfort in automated vehicles: Acceleration and jerk

In the study "Standards for passenger comfort in automated vehicles: Acceleration and jerk" conducted by de Winkel et al., a deeper investigation is made in the relation between passenger comfort and different levels of acceleration and jerk in the vehicle [10]. The study placed 23 blindfolded participants, 7 female and 16 male, in a 3D actuated driving simulator, see Figure 3. The so-called DAVSi simulator (Delft Advanced Vehicle Simulator) has a range of motion of $\pm 0.51m$ and velocity $\pm 0.81 \frac{m}{s}$ in the longitudinal and lateral axes. Thus, it can produce a range of acceleration and jerk profiles with a maximum acceleration of $a_{max} = 2 \frac{m}{s^2}$ and a maximum jerk of $j_{max} = 15 \frac{m}{s^3}$. During the experiment, participants were exposed to 136 different stimuli, consisting of a range of acceleration between $0.2 \frac{m}{s^2}$ to $2 \frac{m}{s^2}$ and a range of jerk between $0.5 \frac{m}{s^3}$ and $15 \frac{m}{s^3}$, as well as two different types of movement, sinusoidal acceleration or triangular acceleration. The participants then rated their comfort based on seven different verbal qualifiers (Excellent, Very Good, Good, OK, Bad, Very Bad, and Terrible). Results were then studied by Ksander N. de Winkel et al., and through postprocessing, threshold values for each verbal qualifier were determined. These can be seen in Table 1.



Figure 3: DAVSi Simulator

Table 1: Findings of study: "Standards for passenger comfort in automated vehicles: Acceleration and jerk" [10]. Values are presented in $\frac{m}{s^2}$

$(\frac{m}{s^2})$	Excellent	Very Good	Good	OK-Threshold	Bad	Very Bad	Terrible
Longitudinal Acceleration	0,28	0,56	0,89	1,23	1,89	2,12	2,12
Longitudinal Deceleration	0,18	0,46	0,79	1,13	1,79	2,02	2,02
Lateral Acceleration	0,03	0,31	0,64	0,98	1,64	1,87	1,87

This study was particularly valuable because it presented clearly quantifiable thresholds for different levels of comfort. That is especially useful in this study, since obtaining qualitative input from a passenger during simulations is very difficult. However, the values presented seem quite low, as the terrible levels of acceleration could easily be achieved on multiple occasions on a normal drive. An example of this can be seen in "Quantifying the effect of roadway, driver, vehicle, and location characteristics on the frequency of longitudinal and lateral accelerations" by Ali et al. where accelerations in everyday driving are analysed [11]. Here, lateral accelerations and longitudinal deceleration of up to $0.5g$ $(4.9\frac{m}{s^2})$ as well as longitudinal accelerations of up to $0.4g$ $(3.92\frac{m}{s^2})$ occur on somewhat regular basis of once per 10 miles $(16km)$. Whilst accelerations of $0.3g$ $(2.94\frac{m}{s^2})$ or above occur commonly in all directions, multiple times every mile $(1.6km)$. This is probably due to the nature of the experiment. Although blindfolding the participants increases the objectivity of which accelerations are uncomfortable, it might also change the behaviour of participants from real-world scenarios. For example, could a passenger be more comfortable with higher accelerations in situations such as roundabouts because they will know the direction of the acceleration beforehand and can thusly prepare

by either holding onto something or tense up.

2.2.3.2 Ride comfort of Passenger Cars on Two-Lane Mountain Highways Based on Tri-axial Acceleration from Field Driving Tests

In "Ride Comfort of Passenger Cars on Two-Lane Mountain Highways Based on Tri-axial Acceleration from Field Driving Tests" by Xu et al., another approach to quantifying ride comfort is conducted [12]. Instead of using a simulator with three-dimensional motion, participants are tested in a real-world scenario on a Chinese mountain road. 16 subjects with driving experience ranging from 2 to 31 years were given a route through a mountainous area but were not told how to drive, in order to achieve as similar driving characteristics as possible to their personal style.

Based on earlier studies, such as "Insight into the lateral ride discomfort thresholds of young-adult bus passengers at multiple postures: Case of Singapore" by Nguyen et al. and ISO 2631-1, the authors establish a comfort threshold, which is then refined by the results of the experimental study [13, 4]. Adjustments made to the thresholds were based on a combination of data from a triaxial accelerometer, a gyroscope, the driver's heart rate, and the driver's subjective experience. The updated comfort values can be observed in Table 2.

Table 2: Findings of study "Ride comfort of Passenger Cars on Two-Lane Mountain Highways Based on Tri-axial Acceleration from Field Driving Tests"[12]

$(\frac{m}{s^2})$	Comfortable	Relatively Comfortable	Uncomfortable	Unbearable
Longitudinal Acceleration	<1,0	<1,3	>1,3	
Longitudinal Deceleration	<1,5	<2,7	>2,7	
Lateral Acceleration	<1,65	<2,85	<4,05	>4,05
Vertical Acceleration	<0,5	<1,0	<1,6	> 1,6

The comfort values presented in this study appear more reasonable, as they are somewhat higher and are based on real-world driving experiences. Compared with the thresholds reported by de Winkel et al., a substantial difference in comfort levels is observed, particularly for lateral acceleration. Although the comparison is not fully direct because different rating scales are used, with "uncomfortable" in one study compared against "very bad" in the other, the accepted lateral acceleration levels are still more than twice as high, probably due to the removal of the blindfold. In a real-world driving scenario, high lateral acceleration will not be as uncomfortable as most people have some experience with "reasonable" levels of acceleration in the curves. In the longitudinal direction, acceleration matches quite well between the two studies, however, deceleration is once again much larger in the study by Xu et al. At first reflection, this might feel odd, as during acceleration, passengers are "pushed" backwards into their seats, being offered a lot of support, while at deceleration, passengers are "hanging" in the seat belt without the padding of the seat.

However, these results are based on real-world test runs in which larger accelerations most often occur in deceleration, as indicated by Xu et al., mentioning that $max(a_x) = 2.85\frac{m}{s^2}$ while $min(a_x) = -7.13\frac{m}{s^2}$. Meaning that the reason for the higher acceptable values for deceleration is probably due to the number of occurrences, that deceleration is necessary from a safety perspective, as well as the fact that the study takes personal experience into account.

2.2.3.3 Fundamental Study of Jerk: Evaluation of Shift Quality and Ride Comfort

This study by Huang and Wang involved 16 passengers who investigated transient and durative jerks while driving in a straight line [14]. The transient tests were used to correlate the peak jerk J_{Peak} with comfort, while the durative jerk was performed to find the relation between the root mean squared (RMS) jerk J_{RMS} and the perceived comfort.

For the transient tests, drivers were asked to start in either 1st or 2nd gear, shift up to the top gear, and then shift back down to 1st and halt. This was carried out with different throttle inputs and shifting patterns to vary the magnitude of the jerks.

For the durative jerks test, the drivers were asked to shift up one gear and then back down 10 times consecutively. This was done with varying starting gears, resulting in the 4 shifting patterns of: (1,2,1), (2,3,2), (3,4,3) and (4,5,4).

After which, the authors created a formula that combines RMS and the peak value of jerk during 2-3 minutes of measurement in order to calculate the average jerk value (AJV) according to the Equation 2.1 below.

$$AJV = 0.004{j_{Peak}}^2 + j_{RMS} \tag{2.1}$$

The value of the calculated AJV can be evaluated according to the following:
≤ 2: Satisfied
$2 \leq 5$: Can do
$5 \leq 8$: Somewhat uncomfortable
> 8: Very uncomfortable

The results in this report also show that only when j_{Peak} is greater than 10 $\frac{m}{s^3}$ is it uncomfortable for certain, below this value, it was indecisive. Also, for the durative jerk assessment it was concluded that under 1.67 $\frac{m}{s^3}$ was comfortable and over 2.23 $\frac{m}{s^3}$ was uncomfortable for all participants. Since there are only a few studies examining jerk in relation to passenger vehicle comfort, this source was particularly intriguing. As jerk is the time derivative of acceleration, jerky behaviour always occurs when driving, with it most commonly being felt during stop and start, as that is when the largest changes in acceleration occur. When evaluating vehicle comfort, a study of jerk in addition to studying acceleration will be essential for a holistic view.

2.2.4 Studies on Vehicle Performance

To achieve the goal of improving performance and identifying mistakes made in a dynamic context, literature for this context will also have to be taken into

account. Here, focus will be on how to adjust dampers in order to boost performance and driver satisfaction.

2.2.4.1 Understanding your Dampers

”Understanding Your Dampers” by Jim Kasprzak is a comprehensive guide on how to set up dampers in a dynamic context [3]. It encompasses everything from how dampers work and what types of dampers exist to how vehicle characteristics change when various damper adjustments are made. This last point has been the most useful, helping with a deeper understanding of how damper adjustments affect vehicle performance and feel, as well as what part of the race track each change presents itself. Kazprzak introduces a particularly useful sectioning of each corner into 6 parts, braking, turn-in, corner entry, mid corner, corner exit and acceleration. This makes it possible to make targeted adjustments based on vehicle behaviour. An example of this is Kazprazak’s explanation of how to maximize traction during the acceleration phase, during which weight transfer to the rear is essential (considering a rear wheel drive vehicle). However, if weight transfer is too quick, then the dynamic spring nature of rear compression might worsen traction performance. If this is the case, an increase in front rebound might instead be the solution to weight transfer control.

2.2.4.2 Performance Braking

When evaluating performance, the largest time gains available are most often under braking. Braking late and aggressively can be a significant advantage over competitors, both lowering lap-time and facilitating overtakes. However, knowing which Key Performance Indicators (KPI) to look for is tricky. In ”Hitting the brakes: is your driver aggressive enough?”, Claude Rouelle is interviewed by Joao Antunes about how braking data can be analyzed in different ways to understand braking performance [15]. Here, Rouelle defines the brake pressure aggression KPI by measuring how often a specific brake pedal velocity is reached during a lap; the more frequently it occurs, the more aggressive the driver is considered. Rouelle adds that this way of measuring aggressiveness is effective but has to be taken in a larger context, as an aggressive driver who keeps locking the wheels will not be as fast and worse on the tyres than a less aggressive driver. In another article by OptimumG, the smoothness of brake release is mentioned as another important factor for optimum braking performance [16]. Here, a way to calculate brake release smoothness by comparing the driver’s actual brake release to a smoothed brake trace is presented. A lower number means a smoother brake release, although this comes with caveats, as a very poor brake release, where the brake pressure fluctuates multiple times, might result in an artificially low brake release number.

This investigation of brake data is useful and will be brought into the performance driving case. In particular, the quantification of driver behaviour will become useful as a measurement of each driver’s quality. However, it also has to be in context with wheel speed, where aggressive braking behaviour results in worsening braking performance as soon as the slip exceeds the optimum slip angle of the tyre. Concerning the brake release number, the risk of unreliable

readings exists, especially if the data comes from inexperienced racing drivers; however, using the brake release smoothness can still be a useful tool to find good examples and well-performing drivers through the algorithm.

2.3 Analysis of Recorded Data

This section covers the theory behind how the driving data in the time domain was analysed, which was a crucial part of constructing the algorithm.

2.3.1 The Brake Curve

The brake curve can be used to estimate the optimal braking zone, thus helping to identify early and late braking. The braking curve is comprised of the brake pedal position or deceleration, which is proportional to the pedal position under non-slip conditions. When analysing the braking curve during the entry into a turn, the change in velocity can be found by integrating the curve. For better understanding, one can study the units, the braking curve is measured in $\frac{m}{s^2}$, which is calculated over the time the braking occurs, measured in s. This results in $\frac{m}{s^2} \cdot s = \frac{m}{s}$, which is the velocity decrease that occurred for that braking.

Given that the entry speed and the speed at the apex of the corner are acceptable, the optimal braking zone can be calculated simply by adjusting the braking point. The area under the braking curve is the same as what the recorded braking has, but in as short a time as possible while maintaining grip with the track. In other words, the speed decrease should be the same for the optimal braking as for the braking where the driver lifts off the brakes, but the optimal will slow down in as short a distance as possible. In Figure 4, this concept is presented with the recorded braking curve as the bold line and the optimal braking as the dotted line, where the area under each curve is the same.

Using this method, recorded manoeuvres can be categorized as braking too early if the braking point is earlier than the optimal, or braking too late if the braking point is after the optimal.

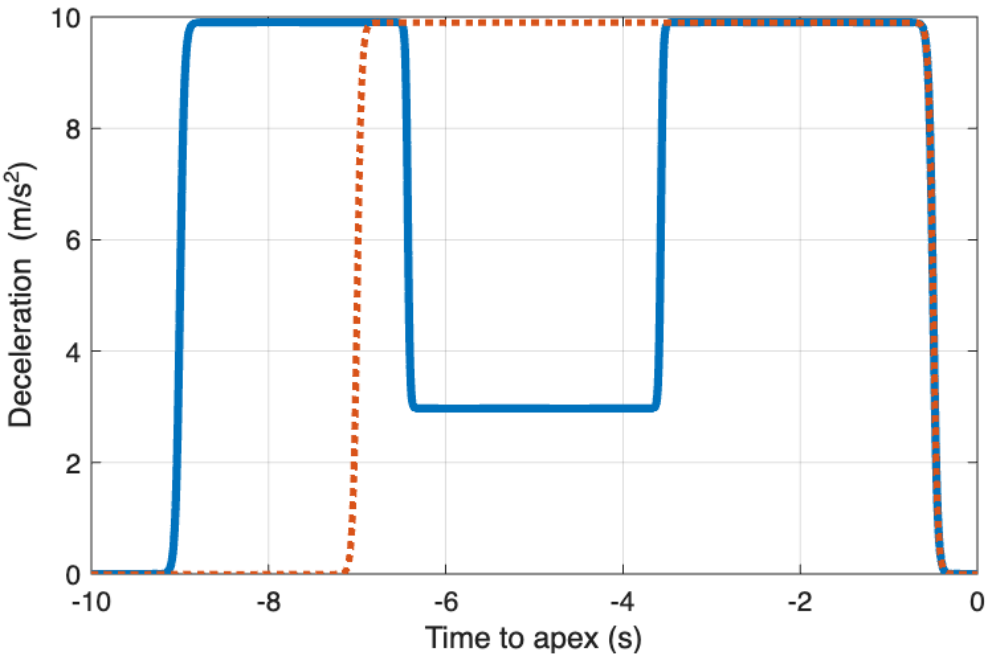


Figure 4: Deceleration vs time plot, braking into a corner. Blue is suboptimal and dotted is optimal braking.

3 Methodology

3.1 Process of Development

For developing the algorithm, MATLAB/Simulink 2023a was used due to the author's earlier experiences and a broad usage of the software at Öhlins, leading to a more efficient future integration at the end of development. As well as its comprehensive capabilities and simple integration with other software.

3.1.1 Hardware

During the study, a multitude of simulations have been run in Assetto Corsa in order to verify the CarMaker (CM) model as well as to record telemetry data from the driver. This has been done on one of the author's personal computers with a Moza R5 racing wheel and Moza SR-P lite pedals as input. A picture of the equipment can be seen in Figure 5.



Figure 5: Moza R5 racing wheel and Moza SR-P lite pedals

3.1.2 IPG CarMaker

CarMaker is a physics engine software that allows simulated changes of car parameters within a range of driving conditions. It has been utilized in several ways, as an introduction to damper tuning, possible driver behaviour, and real-time co-simulation. It has also been very useful as a scenario creator, where specific scenarios have been created in order to study the corresponding driver errors. As an example, during the early parts of the project, a longitudinal test track for the comfort criteria was created. The scenario included all non-lateral cases from Table 3 and was constructed using the built in scenario creator. Some examples include tests for Speed Mismatched With Road Surface, which used different types of uneven terrain, such as speed bumps at varying velocities. While Uneven Braking was investigated in three different ways, first with constant braking, used as a reference for even braking, then manually changing the brake position values, and lastly with CarMakers built-in driver model, configured as an aggressive driver. This made it possible to start creating conditions for each driver error and establish a starting position.

3.1.3 Assetto Corsa, ACTI and MoTeC i2

In order to retrieve the driver inputs from the driven Assetto Corsa laps, a software called ACTI (Assetto Corsa Telemetry Interface) was used. This saved the telemetry from the recorded driving session, which then could be loaded in another software called MoTeC i2, where the ACTI files could be converted into .mat files, which were imported into MATLAB. Once imported, the formatting was changed to timeseries and could then be accessed from the Simulink model that is used to co-simulate between the Simulink model and the CarMaker simulation.

3.2 Simulink Model

The Simulink model is where the algorithm is implemented, and IPG CarMaker was used to simulate different scenarios and manoeuvres in order to evaluate it. Within the model, the main focus was on the Suspension Forces block, where the damper forces are calculated based on the inputs of the driver and vehicle parameters, as can be seen in Figure 6.

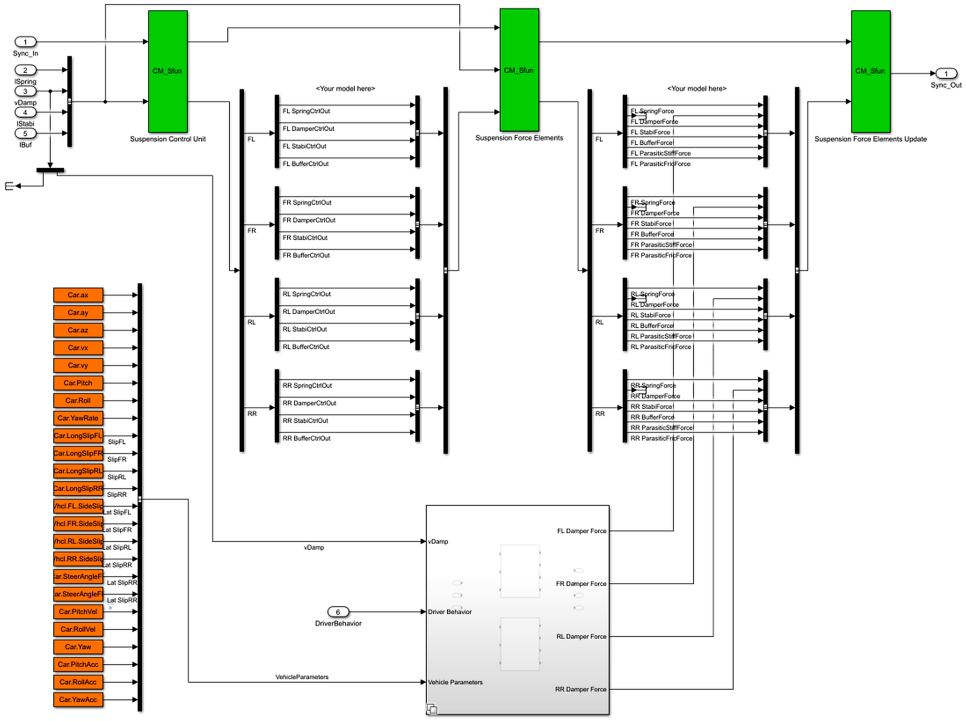


Figure 6: An overview of the Suspension Forces block inside the Simulink model.

Inside the function block, there is a subsystem for the comfort scenario and one for the dynamic scenario. Each of these includes a function block for the algorithm, where the current state is identified based on the array of inputs, then the active state is fed into a subsystem for each quarter car. Within each of these subsystems lies a MATLAB Function block where, depending on the active state, current damper velocity, and lateral acceleration, it decides how the damper curve needs to be scaled in order to get the desired results. The factors that trigger each state and the scaling factors for the damper changes are all coded in a parameter form, this way, if they need to be modified, one can simply set a new value for any parameter and save it to the workspace. Since all of these parameters in Simulink are retrieved from the workspace, their values will change upon the next run.

3.3 Driver Behaviour

The driver should always feel in control of their vehicle, no matter if the goal is to reduce lap time or to have a more comfortable ride. The algorithm will need to be able to analyse the inputs from the driver and identify what they are looking for. Depending on the situation, the algorithm will then adjust the dampers to achieve the goal of the driver.

The inputs that are available to the model are, among others, the pedal inputs, brake and throttle, the steering wheel angle (SWA), and the steering wheel rate (SWR). There are also other relevant inputs from the vehicle used in the identification, such as velocity and acceleration of the car along the x, y, and

z axes and rotations around them in the form of pitch, roll and yaw. Also, the four wheels' speeds and their longitudinal and lateral slip.

3.3.1 Comfort Scenario

In regular driving, comfort is a top priority, and the algorithm is designed to identify whenever the driver makes aggressive or uncomfortable manoeuvres and tunes the dampers so that the passengers experience a smoother ride. Table 3 presents the uncomfortable driver behaviours and the parameters used to identify them.

Table 3: All comfort identifications and utilized parameters.

	Throttle pos	Throttle vel	Brake pos	Brake vel	Steering pos	Steering vel	vx	ax	az	Tyre slip
Aggressive Throttle (AT)	X	X								
Uneven Throttle (UT)	X	X								
Aggressive Braking (AB)			X	X			X	X		
Uneven Braking (UB)			X	X						
Throttle After Braking (TAB)	X	X	X							
Braking After Throttle (BAT)	X		X	X						
Not Easing Off Brakes (NEO)							X	X		
Speed Mismatch to Road Surface (SM)									X	
Accelerating Into Turn (AIT)					X	X	X	X		
Varying Steering Angle (VA)					X					
Jerky Steering (JS)						X				
Too Much Steering During Braking (SB)			X		X		X			X

3.3.2 Dynamic Scenario

In dynamic, lap time and driver feel are top priority, and the algorithm is designed to identify whenever the driver makes manoeuvres that lose time and tunes the dampers so that the driver remains in control and adjusts for their mistakes. Table 4 presents dynamic driver behaviours that lose lap time and the parameters used to identify them.

Table 4: All dynamic identifications and utilized parameters.

	Throttle pos	Throttle vel	Brake pos	Brake vel	Steering pos	Steering vel	vx	ax	ay	az	Tyre slip
Aggressive Throttle (AT)	X				X						X
Aggressive Braking (AB)			X		X						X
Uneven Braking (UB)			X	X							
Gas and Brake (GAB)	X		X								
Sticky Brake Foot (SBF)		X	X	X				X			
Sticky Gas Foot (SGF)	X	X		X				X			
Varying Steering Angle (VA)					X	X				X	
Throttle and Steering (TS)	X	X			X	X					
Braking Too Late (BTL)			X	X	X	X					
Braking Too Early (BTE)	X		X		X						
Throttle Understeer (TUS)					X		X	X	X		X
Brake Understeer (BUS)					X		X	X	X		X
Steering Understeer (SUS)					X		X	X	X		X
Throttle Oversteer (TOS)					X		X	X	X		X
Brake Oversteer (BOS)					X		X	X	X		X
Steering Oversteer (SOS)					X		X	X	X		X
Too Late on Throttle (TLT)	X	X	X		X						
No Trail Brake (TB)	X		X		X			X	X		

3.4 The Manoeuvres

In the development of the algorithm, different driving manoeuvres were tested, and this was done in several ways. The first method was to create the test track and the driving sequence within CarMaker. The second option was to use the recorded driver inputs from a driving session in Assetto Corsa and import the dataset into CarMaker and simulate the run there, as discussed in Chapter 3.1.3. In the latter, a complex hurdle had to be overcome, which was that the car model used in Assetto Corsa and the one CarMaker used were different. This resulted in different vehicle outputs, such as velocities and accelerations, although the driver inputs were the same. The changes made to the CM car model are discussed in detail in Chapter 3.5.

3.5 Vehicle Parameters

The vehicle model that was used for the simulations was based on a Tesla Model 3, this was due to Öhlins already having done tests on that particular model and thus possessing knowledge about its parameters. Initially, the Tesla Model S was chosen from the CarMaker menu as it was the closest alternative available to the Model 3, though it was quickly found that there were still several parameters that needed to be adjusted. Öhlins provided a vehicle data sheet, which was used in order to transform our vehicle model from a Model S to a Model 3. Main changes concern the categories of assembly, body, suspension, tyres, driveline, brakes and steering. These will be discussed in more detail below, and a model can be seen in Figure 7.



Figure 7: The modeled Tesla Model 3 still using a Tesla Model S bodywork.

3.5.1 Assembly

In assembly, the changes required according to the Öhlins data sheet are the wheelbase and track width, as well as a smaller change to wheel and wheel carrier masses. This resulted in a wheelbase as well as front and rear track-widths as shown in Table 5. Carmaker assigns the weight of wheel and wheel carriers separately, and only the total weight was available, so an assumption was made, setting each wheel carrier's weight to $20kg$ and each wheel's weight to $16.5kg$ and $17.5kg$ for the front and rear wheels, respectively.

Table 5: Table of assembly parameters for Tesla Model 3.

Tesla Model 3	**Unit**
Wheelbase (L)	$2875\,\mathrm{mm}$
Trackwidth front (b_f)	$1586\,\mathrm{mm}$
Trackwidth rear (b_r)	$1581\,\mathrm{mm}$

3.5.2 Body

Under the body section in CarMaker, changes were made to the simulated body dimensions as well as aerodynamic properties. The vehicle's weight, center of gravity, and inertias were adjusted according to Tesla's datasheet [17]. Aerodynamic properties were adjusted by doing virtual roll-out tests with the vehicle and adjusting accordingly against the Assetto Corsa model. Drag properties such as C_D, C_S, and C_L can be adjusted at different angles with respect to the car, τ, however, this was deemed too labor-intensive with a marginal model improvement and thus disregarded.

3.5.3 Suspension

As expected, a multitude of changes had to be made to the suspension model. These included the spring stiffness, motion ratio for the springs and dampers, camber, toe, and caster angles, and the bump stop's stiffness. The values of which were taken from Öhlins data sheet and modified to fit CarMakers user interface (UI). Some smaller changes, such as unit conversions and sign changes, had to be made, but no major adjustments were made. However, adjustments

to compensate for motion ratios between wheel and suspension were more troublesome, as it was denoted as "Deflection" in CM and was thus overlooked, but with some help from the supervisor, it was corrected. Damping curves and adjustments are disregarded in the UI, as those are adjusted within the Simulink co-simulation and overwritten, making a change in the UI pointless.

3.5.4 Tyres

For the tyre, model parameters such as the tyre dimensions, rolling resistance, as well as static and dynamic radius had to be set. The tyre used on the Öhlins test vehicle was a $235/35R20$, this tyre is unfortunately not available in CarMaker, leaving the only option of adjusting another existing tyre. Starting with the closest possible option, a $235/55R18$, modifications were made to the aspect ratio and kinematic tyre radius to align with the data sheet. Rolling resistance was also adjusted, it was done in a similar way to aerodynamic resistance, but instead of high-speed roll-outs, lower speed ones were conducted in the area where rolling resistance is the major resistance. Both high and low speeds were then inspected in order to find the right balance between rolling and aerodynamic resistance.

3.5.5 Driveline

The driveline presented some complexity, as the original idea was to include regenerative braking into the model, as the car in AC was driven with it active. It was later decided to deactivate regenerative braking, ESP, and ABS, as those settings seemed to be the most problematic ones when comparing the two simulations together. After this, the velocities matched up significantly better, but the throttle mapping caused issues even though both CM and AC used linear settings (10 % throttle leads to 10 % of the torque, 20 % throttle leads to 20 % of the torque, and so on). The issue was that, when the throttle input was high, the AC modelled car was faster than CM and at low throttle inputs the CM modelled vehicle achieved higher accelerations. The cause of this was found to be that AC was modelling vehicle torque proportional to its maximum available torque at the current motor RPM, while CM was using the torque that was proportional to the maximum the motor could produce. For an example of a motor's power curve, see Figure 8. This problem was resolved through changing the throttle mapping to a 2-dimensional lookup table where the torque was dependent on both the motor RPM and the throttle pedal's position, matching the model from AC.

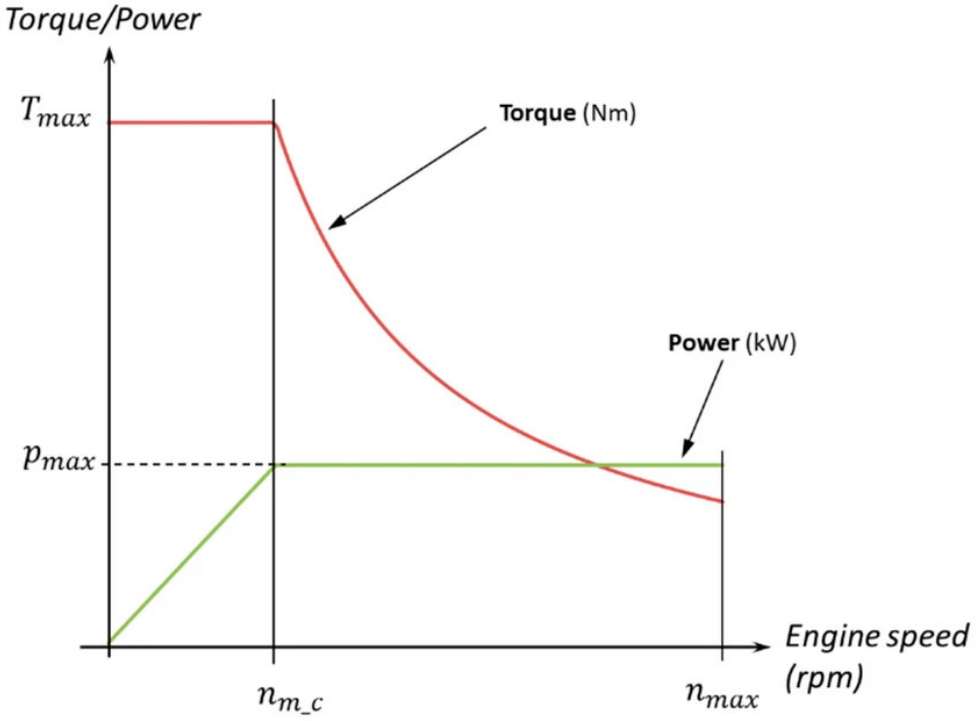


Figure 8: An example of an electric vehicle's power curve where up to engine speed, n_{m_c}, the torque is constant and after it the power is constant [18].

Several other changes were also made to the driveline. The gearbox was modified to use only one gear for forward motion and one for reverse. In addition, both the gearbox and the final drive gear ratios were adjusted to match those used in the AC model, since these values were not included in the data sheet from Öhlins.

3.5.6 Brakes and Steering

The only change made to the brakes was adding a simple amplification factor for the purpose of matching the simulated deceleration to the one recorded in AC. For the steering, a lookup table was added where the steering pinion's angle was defined with varying rack displacement. Although this was not straightforward, as the data sheet from Öhlins contained the steering wheel's angle as a function of the wheel's angle. In order to make this conversion, the Ackermann steering was omitted, and instead, the average angle between the inner and outer wheel was used. Knowing that the steering pinion is directly coupled to the steering wheel, these could be interchanged for the use of the following transformation.

Firstly, the steering wheel angle and the resulting average wheel angle were measured through Simulink for the current setup. Secondly, the corresponding rack travel and the steering pinion to rack displacement ratio were found by the lookup table that was in use in CarMaker. Now, for a specific steering wheel angle, the wheel angle and the rack displacement are known with the current steering settings. Finally, the rack travel to wheel angle ratio was found and used together with the data sheet received from Öhlins, the steering pinion to

rack displacement could thus be computed, and the CarMaker's steering setup was updated accordingly.

4 Algorithm Development

The algorithms are where the most time has been spent during this project. It consists of two MATLAB setup files as well as a Simulink block, which is co-simulated with CarMaker.

Two MATLAB files that are needed run to initialize the environment, address variables, and load damper curves. In order to perform the co-simulation between Simulink and CarMaker the cmenv.m file has to first be run in order to establish a connection. Afterwards, multiple setup and identification parameters, as well as the driver inputs and damper curves, are saved to the workspace by running the initialization.m file.

In the last file, run_experiment.m, the simulation is executed. Here, a couple of different modifications to the type of test can be made. Firstly is the option to run a single test run or a factorial experiment where compression and rebound are changed in front and rear, which will be further explained in the results section. High and low values for those tests are also decided according to the test being performed, these scale the already mentioned damper curve in order to change the resulting damper forces. Finally, the tested driver error can be chosen in order to check how changes during that particular error affect the vehicle.

4.1 Comfort Identification

In the comfort scenario, the goal is to identify driver behaviours that create an uncomfortable experience for the driver and passengers. These thirteen errors, see Table 3 and Chapters 4.1.1 to 4.1.13, have been identified as everyday mistakes made by drivers of all skill levels. The selection of driver errors has been made by the authors based on their personal experiences with the assistance of the project supervisor and personnel at Öhlins. In this section, a short explanation of each identified scenario will take place, describing how they were identified and why they were included.

4.1.1 Aggressive Throttle

Aggressive Throttle is triggered when the driver quickly presses the throttle pedal. This creates a quick increase in acceleration, especially in a high-performance electric vehicle, like the Tesla Model 3 used for the experiment. A quick change in acceleration not only affects vehicle occupants through the increased acceleration but also with a high jerk, pushing them backwards into the seat.

4.1.2 Aggressive Braking

In the same way as Aggressive Throttle but inverted, Aggressive Braking is also considered an uncomfortable experience, especially as higher absolute acceleration is often achieved in braking compared to throttle. The fact that passengers are not pushed into their chair, instead the forward-facing force has to be counteracted by either holding onto something or hanging in the seatbelt. Identification is done in a similar way as the above-mentioned case, where brake position and velocity are considered.

4.1.3 Uneven Throttle

Uneven Throttle is a case where the driver tries to keep a constant speed but is either not paying attention to the undulation of the road or the position of their right foot. Another reason for Uneven Throttle might be slow driving in queues where sudden velocity changes might occur. This creates small jerks in the longitudinal direction, leading to discomfort. Identification is performed by investigating the throttle pedal position as well as a moving average of the vehicle velocity in order to not trigger this case during similar behaviours.

4.1.4 Uneven Braking

Just like Uneven Throttle, Uneven Braking looks at the change in pedal position; however, Uneven Braking's focus is more on how the driver slows down. Here, the optimal is to first come on the brake slowly and then off slowly with a consistent movement. Uneven Braking is triggered when this is not the case, especially when the driver brakes, comes off the brake, and then brakes again, resulting in a jerky longitudinal movement. An example of such a brake curve can be seen in Figure 9. Identification is done by looking at pedal position as well as whether other peaks in braking have occurred during the same braking event.

4.1.5 Brake Tapping

Another longitudinal case that has been identified is Brake Tapping. This means shorter brake instances, which in many cases could have been prevented with well-planned driving. An example of this is if sufficient distance to the car ahead is not given, then a tap of the brakes might be needed when the vehicle ahead brakes, however, if more distance is given, then the driver might be able to roll off the speed instead of having to brake and thus creating a jerky longitudinal movement. Brake Tapping is identified by being a short brake event with a high brake velocity.

4.1.6 Braking After Throttle

The Braking After Throttle case identifies whether braking occurs shortly after an acceleration event. It is achieved by checking the time since the last brake occurrence, when an acceleration event begins. This means that the driver has not correctly predicted the traffic flow. Not only is this disruptive for vehicle occupants who are thrown back and forth in their seats, but it is also detrimental

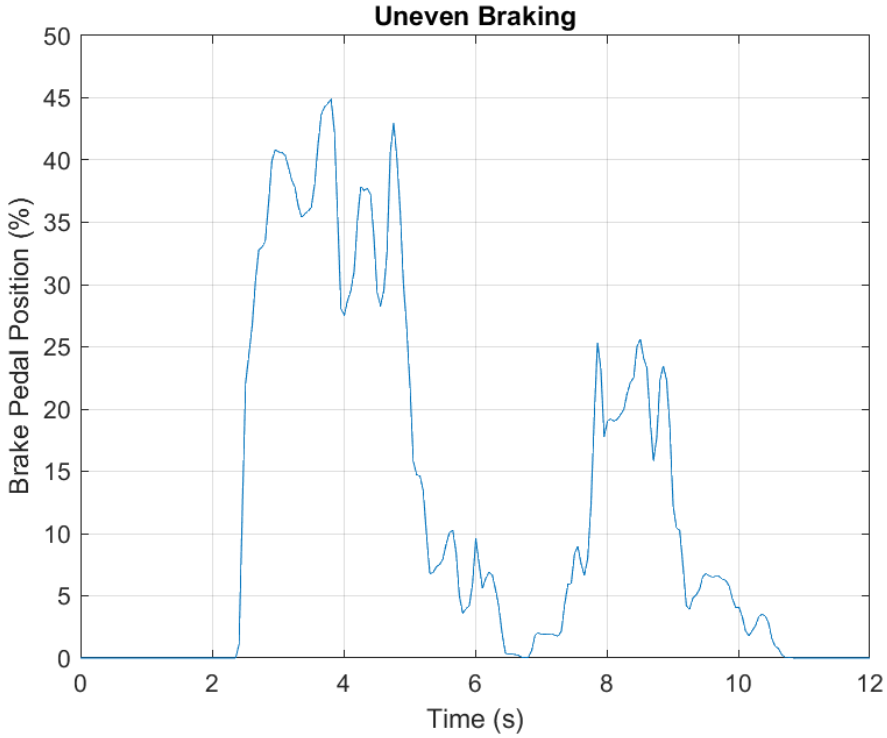


Figure 9: An example of where Uneven Braking is triggered.

for fuel efficiency, as the energy from the acceleration event is converted to heat in the brake discs right away.

4.1.7 Throttle After Braking

Similar to the above-mentioned Braking After Throttle, Throttle After Braking is an uncomfortable behaviour that creates a jerk for the driver and passenger. It is identified in the same way but inverse, meaning that when an acceleration event happens, a check is done in order to see if a braking event has recently occurred.

4.1.8 Not Easing Off

Not Easing Off aims to reduce the uncomfortable jerk that occurs when coming to a complete stop without progressively releasing the brakes. The origins of the jerk are due to the front suspension compression occurring when slowing down a vehicle. If the brake pedal is not released progressively and instead is kept depressed, then the springs in the suspension will return to their equilibrium quickly. If the driver instead slowly releases the brake pedal, the weight transfer towards the rear will occur progressively, reducing the jerky feeling when coming to a complete stop. Identification is done by checking the current velocity against the earlier velocity as well as the acceleration in the x-direction of the vehicle.

4.1.9 Speed Mismatch to Road Surface

This case pertains to when a driver drives over a speed bump or a bumpy road surface at too high a speed. This causes vibrations and pitch oscillations inside the vehicle that are uncomfortable for the occupants of the car. The trigger for this behaviour is when the vertical acceleration exceeds a certain limit in both the positive and negative directions.

4.1.10 Accelerating Into Turns

This lateral and longitudinal case focuses on catching when a driver accelerates right before entering a turn. This leads to increased lateral acceleration compared to if the same turn were taken without accelerating. The resulting larger lateral acceleration leads to an uncomfortable experience for the passengers in the car. The trigger for this is accomplished by studying a combination of the steering angle and velocity, the longitudinal velocity, and acceleration. The state's activation was designed in a way to exclude the situations where the car is starting from a standstill, like it would after a red light in a turning lane.

4.1.11 Varying Angle in Constant Radius Turn

This case identifies when one is driving in a long, constant radius turn, such as ones commonly seen on highways, and the driver weaves left and right within their lane. This can cause motion sickness for the passengers or just simple discomfort due to the repeated switching of direction of the lateral acceleration. To identify this behaviour, the steering angle is analysed.

4.1.12 Jerky Steering

Jerky Steering is a driver behaviour that refers to whenever a quick steering event is performed, this could be an aggressive lane change, avoiding an unexpected object, and so on. These are to be detected as they can be uncomfortable and also to give the driver better control of the vehicle, which is especially important as jerky steering input often is a result of an evasive manoeuvre. To trigger this case, the steering wheel's acceleration is evaluated.

4.1.13 Combined Steering and Braking

Combined Steering and Braking is a case where the driver brakes and steers above a set threshold, this can lead to overloading of the front tyres. When these tyres are fully saturated, they can no longer transfer the desired forces, leading to slip and the wheels locking up. Damper changes are made in order to keep control of the vehicle and keep passengers safe. To identify this dangerous behaviour, the brake pedal's position, steering angle, velocity in the longitudinal direction, and slip percentage were used to trigger and stay active during the entire steering or braking event.

4.2 Dynamic Identification

The dynamic identification aims to detect the driver behaviours that lead to the driver being slower around the track and to help with unstable dynamic conditions. A list of 18 cases, see Table 4 and Chapter 4.2.1 to 4.2.14 was compiled by the authors, together with the help of Öhlins.

4.2.1 Aggressive Throttle

Aggressive Throttle is a case that identifies when a driver is too quick on the throttle and causes the tyres to slip. This loss of traction results in a time loss as the driver needs to correct by decreasing the amount of throttle in order to regain the grip. Here, scenarios where a combination of throttle and steering is not taken into account, only straight-line driving is looked at in this case. To identify this case, the steering angle, throttle pedal position, and longitudinal slip on all wheels were used.

4.2.2 Aggressive Braking

This case identifies when a driver is braking aggressively, meaning too quickly on the brake pedal, and this leads to the wheels locking up. Lock-ups usually happen on either the front axle or rear axle, which led to this case being split into two separate identifications, where one checks the front wheels and the other the rear wheels. Due to the hysteresis of tyres, the force transferred through the wheels must go below the point where the lock-up occurred in order to recover the grip. In practice, this means that the driver has to lift off the pedal and go back on more controlled, which leads to a time loss and no longer being on the intended racing line. Similarly to aggressive throttle, only straight-line driving is looked at in this case. Here, the parameters analysed were steering angle, brake pedal position, and longitudinal slip on all the wheels.

4.2.3 Uneven Braking

Uneven Braking is a behaviour which is designed to detect when a driver realizes that they braked too early and so they lift off, coast into the corner, or even go on throttle, after which they brake once again. An example of this braking can be seen in Figure 9. This causes a loss in lap time as it would be beneficial to delay the braking point, which would carry more speed on the corner entry, while the speed in the corner is the same. This comparison is visualized in Figure 4. For the identification of this case, only the brake pedal position was studied.

4.2.4 Gas and Brake

The driver error of pressing the gas and the brake simultaneously above a certain threshold was decided to be identified. This behaviour is typically unwanted as the result is that the vehicle is neither accelerating nor decelerating optimally. On the other hand, most racing drivers use their left foot for braking and their right for pressing the throttle, this can often lead to a small and

quick overlap of the two, but this is taken into account and will not be flagged as a mistake. To detect this behaviour, the positions of the throttle and brake pedals were used.

4.2.5 Sticky Brake Foot

As an extension of the gas and brake, Sticky Brake Foot detects if the driver accidentally leans on the brake while pressing the throttle. It is not uncommon that, for example, bumps on a straight can lead to the driver unintentionally touching the brake pedal. This is identified by small brake readings during high- to full-throttle periods.

4.2.6 Sticky Gas Foot

In the same vein as above, accidental throttle presses can appear during braking, these can appear from bumps or the weight shifting forward during braking. Sticky Gas Foot is identified by inverting the Sticky Brake Foot identification.

4.2.7 Varying Angle in Constant Turn Radius

Varying Angle in Constant Turn Radius is a behaviour in which the driver, in an attempt to find the sweet-spot in grip, starts sawing at the wheel. Such a behaviour is most likely due to a setup in which the driver struggles to find the optimal slip angle of the front tyres or the vehicle feels understeered, and adjustments to counteract this have to be made. Making the driver more comfortable can improve lap time but also tyre wear by minimizing degradation and heat build-up. In order to identify this behaviour, a direction counter was implemented. It checks if the current steering angle crosses the mean steering angle over the last $0.5\ s$, if this happens three times within the same corner, then the error is triggered.

4.2.8 Increased Steering Angle During Increased Throttle

An increase in steering angle in conjunction with an increase in throttle will, in almost all cases, point to a non-optimal lap time. This can occur in two different situations throughout a corner. The first case occurs when the driver brakes too early and has to accelerate while entering the corner, meaning they could have braked later. The second case is one in which the driver is too optimistic with the rotation on corner exit, in which extra steering angle has to be applied late in the corner in order not to brake track limits. Here, identification relies on checking the throttle and steering traces over a shorter time span, and if both are continuously positive, the state triggers.

4.2.9 Braking Too Late

In the event that a driver has braked too late, time will be lost due to the longer distance that the vehicle travels. In an extreme case, could late braking in extension lead to going off the track or hitting a wall. Identification of late braking was a tricky task as the algorithm has no context of the vehicle position on the

track, which means certain areas cannot be used as a marker for late braking. Instead, the driver's behaviour can be studied at such an event, as it is common for the driver to try to keep to the original line and later try to turn in; however, because they brake too late, velocity is still high, and they will miss the apex. Which means this behaviour can be identified by looking at continuous braking as well as an increasing steering velocity.

4.2.10 Braking Too Early

In many ways, Braking Too Early can be associated with Uneven Braking, as they are both, in essence, too early on the brakes. However, these are defined in different ways, and Braking too Early is a more serious mistake, as in Uneven Braking, the driver realises the mistake early on during the braking procedure, quickly lifts off, and then gets back on again. Whilst in Braking Too Early, the driver slows down to the corner speed too early and then rolls into the corner, this is a graver mistake as more time is spent at a lower speed, in turn leading to slower lap times. To identify such behaviour, a check is made to see if the driver has braked in the last second and the last time in the past five seconds, while they are now starting to turn.

4.2.11 Too Much Demand, Front

This case groups several driver behaviours, each of which results in too much demand on the front wheels. This either leads to excess longitudinal slip, as in the case of using too much brakes or throttle, or it could also be due to too high lateral slip, which is caused by too much steering being requested. In all these states, the front wheels lose grip with the track and the car will understeer, resulting in less control and losing lap time.

When identifying for too much demand on the tyres, the longitudinal-, and side slip were analysed and the slip with the peak transferable force was determined for each. This was later used to determine the percentage of tyre utilization, which was then applied in the identification of this case.

4.2.11.1 Throttle

For this state to activate, the demand on the front wheels must mainly come from longitudinal acceleration. This is done by confirming the car is currently accelerating, comparing the utilization of the front and rear axles and comparing the longitudinal slip and the lateral slip to determine that the majority of the demand is coming from the throttle input.

4.2.11.2 Brake

This case becomes activated when the majority of the demand on the front wheels comes from longitudinal deceleration. The identification is done similarly to the previous state, but instead of accelerating, the vehicle needs to be decelerating so that the majority of the demand is coming from the brake input.

4.2.11.3 Steering

This identification is activated when the demand on the front wheels mainly comes from lateral forces. This is achieved by comparing the car's utilization of the front and rear axles and comparing the longitudinal slip and the lateral slip to determine that the majority of the demand is coming from the steering input.

4.2.12 Too Much Demand, Rear

In the same way as the previous case, this state groups several driver behaviours together, wherein each of them results in too much demand on the rear wheels. This either leads to excess longitudinal slip, as in the case of using too much brakes or throttle, or it could also be due to too high lateral slip, which is caused by too high steering input while the rear tyres have bad traction. In all of these states, the rear wheels lose grip with the track and the car will oversteer, resulting in less control and loss of lap time.

Here, the identification used the same procedure as for Section 4.2.11.

4.2.12.1 Throttle

Here, the throttle needs to be the dominating factor for having excessive demand on the rear tyres for this case to go active. For the identification, the same approach was used as for the front.

4.2.12.2 Brake

In this case, the activation requires the brake to be the main contributor to having excessive demand on the rear tyres. For the identification, the same approach was used as for the front.

4.2.12.3 Steering

For this state, the activation requires the steering's contribution to the lateral load to be the majority of the demand on the rear tyres. For the identification, the same approach was used as for the front.

4.2.13 Too Late on Throttle

In this case, the algorithm is designed to detect when the driver is too late on the throttle or is on very low throttle for a long time on corner exit. Ideally, the throttle application should begin just after the steering wheel starts its motion of returning to the straight position. In order to identify this behaviour, parameters such as steering angle and velocity, throttle and brake pedal, and the time derivative of the gas pedal were analysed.

4.2.14 No Trail Braking

Trail braking is a concept within racing in which braking is progressively released all the way into the tightest part of the corner, whilst steer angle is proportionally added. If perfected, the driver can spend the whole corner at the

edge of the traction circle, thus maximizing available grip. This is an advanced technique, which means it has a low driver error priority; however, if a few other mistakes present themselves, a lot of lap time can be found by optimizing trail braking. Identification of trail braking is done by checking braking, steering, and the friction circle, not allowing the driver to get too close to the centre of the friction circle.

4.3 Damper Adjustments

In order to simulate active adjustments of the dampers within the CarMaker co-simulation, the conventional damper forces, which are based on the damper curves presented in the CarMaker Graphical User Interface (GUI), are terminated within Simulink and replaced with the output of the algorithm. Here, damper forces are computed by retrieving the damper velocities each timestep from Simulink and parsing them through a lookup table with an actively modified damper curve. The modification of the damper curve occurs based on the active states at that time, where the unmodified damper curve is either scaled up or down using predetermined values for each case.

4.3.1 Comfort Scenario

In the comfort scenario, the focus is unsurprisingly on the solace of the passengers. After an identification of an uncomfortable driver behaviour is made, the given state is transmitted to each wheel's damper tuning function, in which a modification of the damper curve is made. These changes are done by scaling the force of the damper with a predetermined scalar, either in compression or rebound, at the front or rear wheels. The modified damper curve is then forwarded to the lookup table to calculate the actual damper force at the wheel.

4.3.2 Dynamic Scenario

The dynamic scenario uses the same type of damper changes as comfort, however, with another step introduced. As some of the identifications made in the dynamic scenario are difficult or impossible to identify and correct for within the time that they take place, these are instead counteracted by making passive damper changes, which are based on the number of times a mistake has been made. In that way, a driver who, for example, consistently fails to trail brake will be given a damper change at an earlier stage in the corner when the mistake has presented itself multiple times. These might not be as accurate, but as the driver has already made the mistake numerous times, it can be predicted to happen again. In addition, a few driver mistakes have been identified within the racing scenario that are viewed as not fixable with damper changes, such as braking on the straight. These are added to the algorithm as indicators to help the driver after a session, but will not affect the damper settings. Presented in Table 6 are all identifications and their respective categorizations.

Table 6: Type of activation for all 18 dynamic driving errors.

Driver error	**Type of activation**
Aggressive braking	Direct
Uneven braking	Direct
Aggressive throttle	Direct
Gas and brake	None
Sticky brake foot	None
Sticky gas foot	None
Braking too late	Direct
Braking too early	Indirect
Varying angle in constant turn radius	Direct

Driver error	**Type of activation**
Throttle understeer	Direct
Brake understeer	Direct
Steering understeer	Direct
Throttle oversteer	Direct
Brake oversteer	Direct
Steering oversteer	Direct
Too late on throttle	Direct
No trail braking	Indirect
Increased throttle and steering	Direct

5 Evaluation of the Algorithm for Damper Adjustments

The experiment is an essential part of evaluating the effectiveness of a newly developed product, and the same goes for algorithms. In order to achieve trustworthy results, the choice of experiment is paramount. As this project evaluates whether driver behaviours can be identified and corrected for by changing the dampers, both comfort and dynamic settings will require two different types of evaluations. As a real-life experiment, where the algorithm is implemented on a vehicle, is not economically viable at this early of a stage in development, these experiments will instead have to be done with simulations. These simulations will differ between the two driving scenarios, in comfort a quantitative approach will be taken, where data such as acceleration, pitch, and roll taken from the mass centre will be investigated in order to reduce uncomfortable vehicle behaviours. On the other hand, the dynamic experiment will be more subjective, as there are no quantifiable measures that show how enjoyable a vehicle is to drive.

5.1 Evaluation of the Comfort Algorithm

When evaluating the quality of comfort, as mentioned earlier, quantitative values of motion will be evaluated in order to make conclusions about ride quality. These will be dependent on the type of identification, as they can hint at what needs to be prioritized. As an example, pitch and longitudinal jerk might be interesting parameters when testing the Not Easing Off state, as what is intended to be reduced here is the sudden jolt, which is common when coming to a complete stop. However, a parameter such as roll is irrelevant here, and even though a 50% reduction of roll might seem positive, due to the values already being small, making a change like this is considered irrelevant. Meanwhile, in a state such as jerky steering, roll velocity and lateral acceleration are major factors of comfort, whilst longitudinal acceleration and jerk might be unnecessary. These choices of relevant parameters will be done on a case-by-case basis and discussed more in the results.

5.1.1 Factorial Design

In order to achieve reliable results within the comfort experiment, an independent variable test will be conducted. In this study, a type of factorial experimental design presented in Vehicle Dynamics by Erik Wennerström et.al. [19]

will be applied. This is an effective way to quantify the effects of each parameter change with the aim of clear and understandable results. First, a matrix is produced, the shape of the matrix depends on the number of levels and the number of factors that are tested. Levels, which are the number of different values used for each parameter, and factors, which are the number of parameters. In this study, two scalings of damping force will be applied to four different types of damping, A, B, C, and D. The matrix size is given by the number of factors on the x-axis and the number of levels to the power of the number of factors, in this case $(2^4 = 16)$, on the y-axis. Then each element in the matrix is filled with the levels of each factor, making each row unique. The factorial design table can be seen in Table 7.

Table 7: Factorial design table.

Trial	A	B	C	D
1	-	-	-	-
2	-	-	-	+
3	-	-	+	-
4	-	-	+	+
5	-	+	-	-
6	-	+	-	+
7	-	+	+	-
8	-	+	+	+
9	+	-	-	-
10	+	-	-	+
11	+	-	+	-
12	+	-	+	+
13	+	+	-	-
14	+	+	-	+
15	+	+	+	-
16	+	+	+	+

When the experiments are concluded, columns will be added to the right-hand side of the table with the results of the tested variables. The effect (c_A) of each factor is then calculated as the difference between the average result for $A(+)$ also described as A set to h and the corresponding average result of $A(-)$ also mentioned as A set to l. An example of this is shown in Equation 5.1, where T_n is the resulting value from each evaluation. This is then repeated for all other factors. Interactions between the effects, such as $c_A \cdot c_B$, can also be studied, but these will be overseen during this thesis due to the amount of data that is analysed.

$$c_A = -\frac{(T_1, T_2, .., T_8)}{8} + \frac{(T_9, T_{10}, .., T_{16})}{8} \tag{5.1}$$

5.1.2 Evaluation Method

The comfort experiments are conducted through CarMaker and Simulink by first doing runs in Assetto Corsa as mentioned in Chapter 3.1.3. These outings were conducted in a way that purposefully triggered all the states at least once, except for Not Easing Off (NEO) and Speed Mismatched With Road surface

(SM). These runs were driven at three different velocities, $50\frac{km}{h}$, $80\frac{km}{h}$ and $120\frac{km}{h}$ and were recorded as separate outings. As NEO and SM were not triggered in the previously mentioned recorded run, a new run was recorded for NEO where the focus was solely on sudden and smooth stops, whereas for SM a different method had to be used. Here, instead of driving on a track in AC the decision was taken to instead simulate an uneven road in CM and run the experiments there. Every outing is then run through the algorithm 17 times for each case, the first 16 using high and low values as seen in Table 7 and then one time with standard damping values as given by Öhlins. The 17th run is used as a reference point when measuring the change between the starting point and the final result.

Due to the varying velocities, each state will be run three times. After each run is conducted, maximum values of evaluating factors such as a_x, a_y, roll velocity, pitch velocity, etc., are extracted to Microsoft Excel and post-processed. All 13 states will be run in this manner, repeating for each of the speeds, starting with a high or $+$ scaling of 1 and a low or $-$ scaling of 0.5. The scalars will then be individually modified for each state by checking the results and trying to enhance positive changes and reduce negative changes.

5.2 Evaluation of the Dynamic Algorithm

As the goal of the dynamic experiment is different from that of the comfort scenarios, another type of evaluation method is required. As the ”feel” of the vehicle is a very subjective and fuzzy way of evaluating performance, a questionnaire will be presented to several participants in order to understand how they feel about the vehicle's characteristics. A blind experiment will be conducted, in which the participants will get three outings, the first one being 8 warm-up laps, where the algorithm will be on or off. After which, two runs of 5 laps will be performed, where one will be with active damping changes, and one will be without, randomly choosing which is which. The experiment will take place on the Idiada Dry Handling course, as this is deemed the most suitable of the available licenced tracks. Due to it being quite short, a laptime of about 70 seconds, thus easier to learn. The track also has a varying combination of fast and slow corners, therefore making it possible to evaluate the algorithm in a few different types of corners. The driver will be asked a few questions about their experience at the beginning of the experiment, and after each of these two outings, the questionnaires are presented in Table 8 and Table 9.

The choice of driver is a combination of experienced and inexperienced drivers. This was done in order to understand if a difference in experience would sway the result in one way or another. This resulted in a driver lineup of 9 people, 8 male and one female, of which three can be considered experienced.

Table 8: Questions asked before the start of the experiment.

Question	**Answer scale**
How many years of driving experience do you have?	Number of years
How much experience do you have driving on a track in real life?	[None, under 10 hours, 10–100 hours, over 100 hours]
How is your experience driving in a simulator?	[None, under 10 hours, 10–100 hours, over 100 hours]

Table 9: Questions asked after each of the five lap outings.

Question	**Answer scale**
How difficult/easy was the vehicle to drive?	[0-10]
How "in control" did you feel of the vehicle?	[0-10]
How stable was the vehicle under braking? (Did you experience any locking wheels or moving under braking)	[0-10]
How responsive was the vehicle on turn-in? (Did the vehicle respond well when you started rotating the wheel)	[0-10]
How stable was the vehicle under acceleration? (Did you experience oversteer or understeer?)	[0-10]
How predictable did you feel the vehicle was? (Did it do what you expected it to?)	[0-10]
In the case you lost control (oversteer/sliding), how difficult was it to regain control?	[Yes, No], [0-10]
Did you feel fast or slow while driving the vehicle?	[0-10]
Did you notice any changes in the vehicle's behaviour during the outing? If so, how noticeable was it?	[0–10]
In your own opinion, how fun was the vehicle to drive?	[0–10]

Initial conditions applied to each state have been discussed in collaboration with the supervisor at Öhlins, with the intent to improve the driving experience when doing dynamic driving on a track. This means that a state which is triggered from understeering will have a change with the intended outcome to make the vehicle more oversteered, and vice versa.

However, in the dynamic driving scenario, a differentiation from comfort has to be made. Which is that the inner and outer wheels are considered separate. This is because, within dynamic driving, lateral weight transfer is key, and in order to control it as effectively as possible, the inner and outer wheels' damping have to operate independently. Thus is a new naming convention for each parameter introduced, what in comfort was A is distinguished between inside and outside, becoming E and F for inner and outer wheel respectively. In Table 10, all initial conditions can be seen. However, these might be modified slightly as some trial time is allocated on the simulator, before the start of the experiment, in order to evaluate the changes.

Once the experiment is complete, the responses of the participants will be compiled and evaluated in order to determine whether



Table 10: Damper scaling changes for different driving behaviours in dynamic driving

Driving behaviour	**Damper scaling change**			
Aggressive Braking Front	E = 1.20	F = 1.20	K = 1.20	L = 1.20
Aggressive Braking Rear	K = 0.80	L = 0.80		
Uneven Braking	G = 1.20	H = 1.20		
Aggressive Throttle	G = 1.20 K = 0.80	H = 1.20 L = 0.80	I = 1.20	J = 1.20
Varying Angle in Constant Turn	F = 1.05	K = 1.15		
Throttle and Steering	G = 1.20	H = 1.20	F = 1.20	
Brake Too Late	E = 1.20 G = 0.80	F = 1.20 H = 0.80	I = 1.20 K = 0.80	J = 1.20 L = 0.80
Brake Too Early	E = 1.20	F = 1.20	K = 0.80	L = 0.80
Throttle Understeer	G = 1.20	H = 1.20		
Brake Understeer	E = 1.20 K = 0.80	F = 1.20 L = 0.80	I = 0.80	J = 0.80
Steering Understeer	F = 1.05	K = 1.15		
Throttle Oversteer	I = 1.20	J = 1.20		
Brake Oversteer	K = 1.20	L = 1.20		
Steering Oversteer	G = 0.80	H = 0.80	K = 0.80	L = 0.80
Too Late on Throttle	I = 1.20	J = 1.20		
No Trail Braking	E = 1.20 I = 0.90	F = 1.20 J = 0.90	K = 0.90	L = 0.90

the implementation of the damper changes made a significant impact on the drivers' impression of the car. Drivers with similar experience levels will be grouped, as inexperienced drivers will presumably make graver mistakes and trigger the cases more often, and perhaps the damper adjustments will have greater significance. The responses of these groups will be averaged, and the results between the outing with active and inactive damper changes will be assessed.

The experiments will be run on, the parent company of Öhlins Racing, Brembo N.V.'s simulator in Bergamo, Italy, in a driver-in-the-loop simulator based on VI Grade simulation, see Figure 10. Porting the CarMaker simulation to VI Grade will be done in collaboration with Mats Widmark, who is a simulation engineer at Öhlins.

6 Results From Evaluation of Comfort Algorithm

In this section, the results of the comfort algorithm evaluation will be presented. These are divided into different identifications and further split into the different test runs. Each case focuses on a few different parameters, which are explained further below. The aim is to identify which damper setting combinations improve comfort, as well as the parameters where the greatest potential for improvement exists. An example of a parameter that makes a big difference in comfort is a_x; however, an issue with a_x is that small to no change is possible with damper changes due to the origin of longitudinal acceleration being almost solely from acceleration and deceleration at the wheels. Instead, a parameter such as pitch acceleration might be more interesting to investigate, as damper changes have a larger impact on this. A large pitch acceleration or velocity of the vehicle can still affect the occupant's comfort negatively by throwing their heads in either direction.

6.1 Longitudinal Cases

When considering results for the following states; Aggressive Throttle, Aggressive Braking, Throttle After Braking, Braking After Throttle, Not Easing Off, Brake Tapping, Uneven Throttle, and Uneven Braking, identification focus has been placed on a few factors. As mentioned above, changes in longitudinal acceleration and deceleration, a_x, will be very small by only varying the damper settings. This leads to the jerk in the longitudinal direction, also remaining mostly unchanged, but this will still be taken into the analyses. However, larger reductions can instead be made in pitch. In pitch, the focus will ideally be on reducing the pitch velocity and pitch acceleration as these are where excitation on the body happens. Although in a majority of the cases, the pitch acceleration varies so little that it is disregarded. The differences in pitch angle will likely not play a role in the passengers' comfort as long as it stays to a low value, and thus the exclusion of pitch angle was decided based on its low value of around $1 - 2^\circ$ and the fact that a static pitch angle will not introduce accelerations in the body.

6.2 Lateral Cases

When considering results for the following states; Jerky Steering, Varying Angle in Long Constant Turn Radius, and Accelerating Into Turns, the factors roll velocity and acceleration, lateral acceleration, and lateral jerk will be considered the most important. Acceleration in this case, a_y, is included here, whilst not in the longitudinal cases, due to damper changes having a greater impact on lateral dynamics. The dampers could, for example, introduce some understeer on turn-in, which makes the Jerky Steering case less uncomfortable.

6.3 Combined Cases

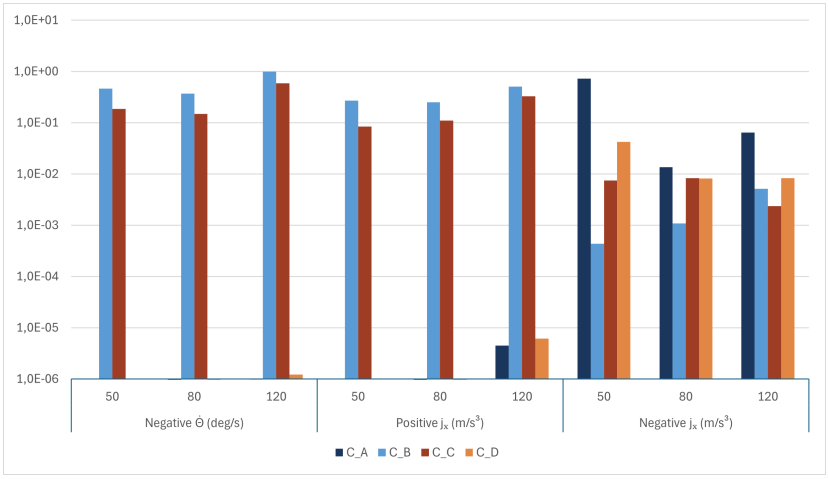
When considering results for Steering and Braking, there are many factors are important at the same time. Here, a combination of longitudinal and lateral acceleration is present, as well as, the pitch and roll velocity and longitudinal and lateral jerk are considered, all in both positive and negative direction due to the steering taking place in both left and right directions. These parameters are investigated because they are important in the ranges where the dampers can make meaningful change.

6.4 Aggressive throttle

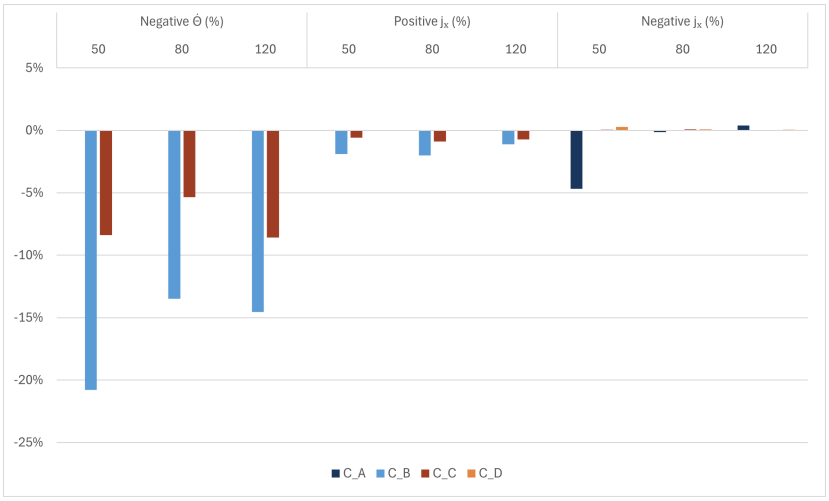
The following tests have been conducted for Aggressive Throttle.

6.4.1 First Test

After completion of the first test run with the base damper values $h = 1$ and $l = 0.5$ as mentioned in Chapter 5.1.2, one parameter stood out, this was pitch velocity. As seen in Figure 11, pitch velocity had a substantial effect, indicating that changes to the damper during the Aggressive Throttle state can result in pitch velocity decrease of up to 20%.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 11: Results from the first test in Aggressive Throttle.

For the second test the damper scaling was therefore increased on B and C as an increase will result in a decrease in pitch velocity for those damper change parameters. A and D are left unchanged as they seems to have minimal affects. Changing the damper settings to $h = 1.3$ and $l = 0.8$ was seen as reasonable, as this keeps the distance between high and low the same, making results easier to compare between test runs.

6.4.2 Second Test

Here, the results clearly show that increased B and D reduce the negative pitch velocity, $\dot{\theta}$, at all measured speeds, albeit with diminishing returns when compared to the first test. The other result that sticks out is A's effect on negative jerk, j_x, although this is only seen at 50 $\frac{km}{h}$, which could simply be an outlier, but it was decided to investigate this by increasing its scaling. The results from the second test can be found within Appendix A in Figure 30 in more detail.

For the third test is was decided it increase B and D even more to $l = 1.1$ and $h = 1.6$ and A's to $l = 0.8$ and $h = 1.3$.

6.4.3 Third Test

The results from the third test can be found within Appendix A in Figure 31. The optimal setup from the third test was found through analysing the relevant parameters for each run of the last test. It was found that run 16 had the best result. Jerk was prioritized and then pitch velocity, although pitch velocity the biggest contributor between tests. The final setup, as well as every step, can be seen in Table 11. The final results taken from the third test, run 16, can be viewed in Table 12.

Table 11: Testing and final setup for Aggressive Throttle

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.8, High = 1.3	1.3
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
D	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	1

Table 12: Normalized final result parameter values and their percentage change in Aggressive Throttle

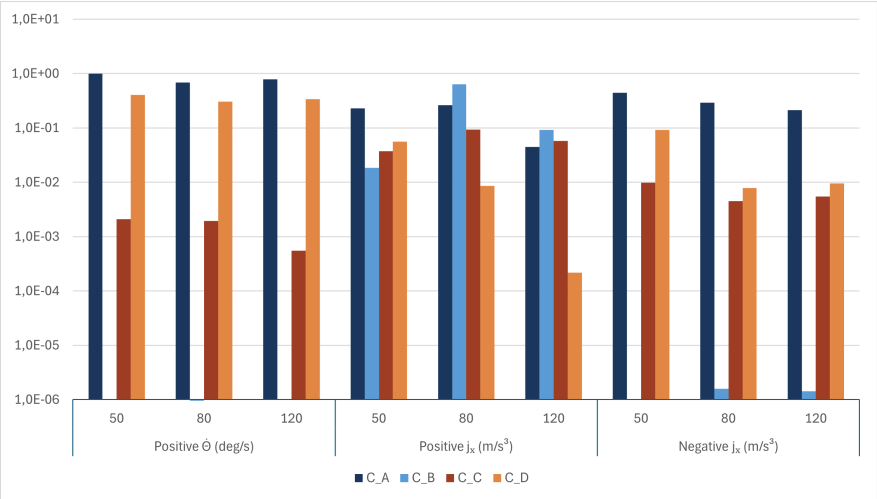
Parameter	Speed	Reference setup	Final setup	Percentage change (%)
Negative $\dot{\theta}$ $(\mathrm{deg/s})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.323	-0.292	9.64
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.401	-0.385	3.98
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.784	21.5
Positive j_x $(\mathrm{m/s^3})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	0.315	0.315	0
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	0.279	0.279	0
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.000	0.032
Negative j_x $(\mathrm{m/s^3})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.926	-0.909	2.00
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.530	-0.530	-0.039
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.002	0.15

6.5 Aggressive Braking

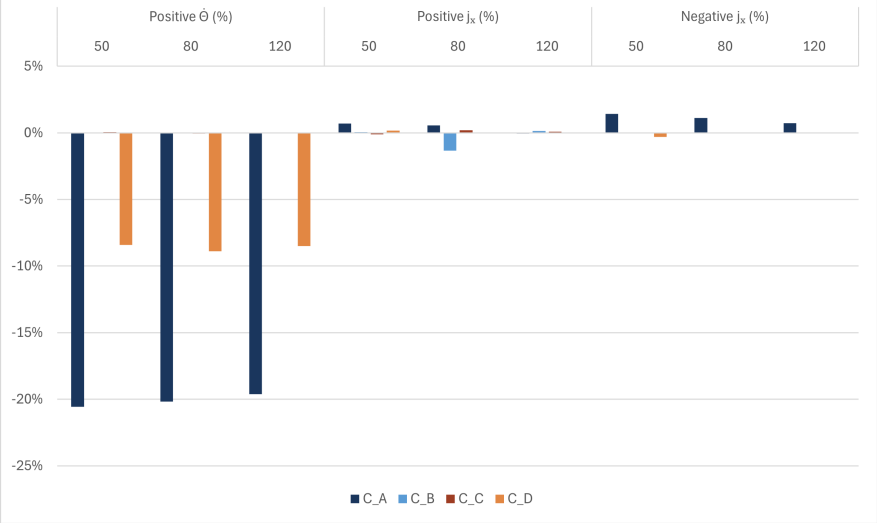
The following tests have been conducted for Aggressive Braking.

6.5.1 First Test

When testing the Aggressive Braking state, a clear correlation between pitch velocity and A and D can be seen in Figure 12. Positive pitch velocity of about $2\frac{deg}{s}$ and a decrease of 20% when applying the high scaling for A, indicating that this should be the focus. As jerk in both directions can be seen as low and unreliable, only showing a small indication that reduced scaling slightly helps j_x. With this in mind the decision was taken to increase A and D to $h = 1.3$ and $l = 0.8$, while keeping B and C in their standard configuration.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 12: Results from the first test in Aggressive Braking.

6.5.2 Second Test

The results are very similar as in the first test, just with lower percentage changes and C-values. Besides that, the results are almost identical, and thus the same approach will be taken as after the first test, which is to increase A and D even more, in this case, it will be to $l = 1.1$ and $h = 1.6$. More details can be found within Appendix A in Figure 32.

6.5.3 Third Test

Effects for the third test can be viewed within Appendix A in Figure 33. Here, the final setup was determined in the same way as Aggressive Throttle, by analysing all runs of the third test and prioritizing jerk. In general jerk was worsened as can be seen in Table 14, however, a large improvement was found in pitch velocity. The final setup can be seen in Table 13.

Table 13: Testing and final setup for Aggressive Braking

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
B	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	1
C	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	0.5
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1

Table 14: Normalized final result parameter values and their percentage change in Aggressive Braking

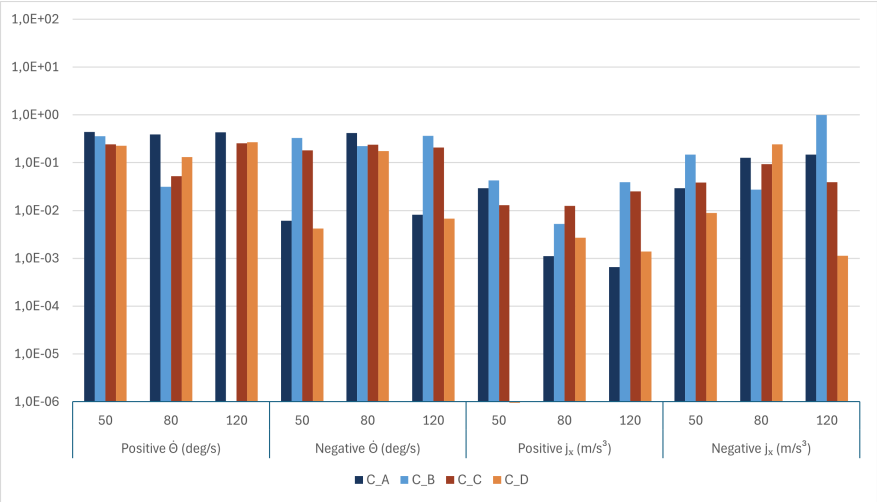
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.808	19.2
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	0.702	0.547	22.2
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	0.822	0.700	14.7
Positive $j_x\,(\mathrm{m/s^3})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	0.471	0.474	-0.485
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	0.697	0.697	-0.37
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.991	0.175
Negative $j_x\,(\mathrm{m/s^3})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.008	-0.77
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.839	-0.845	-0.775
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.930	-0.936	-0.547

6.6 Uneven Throttle

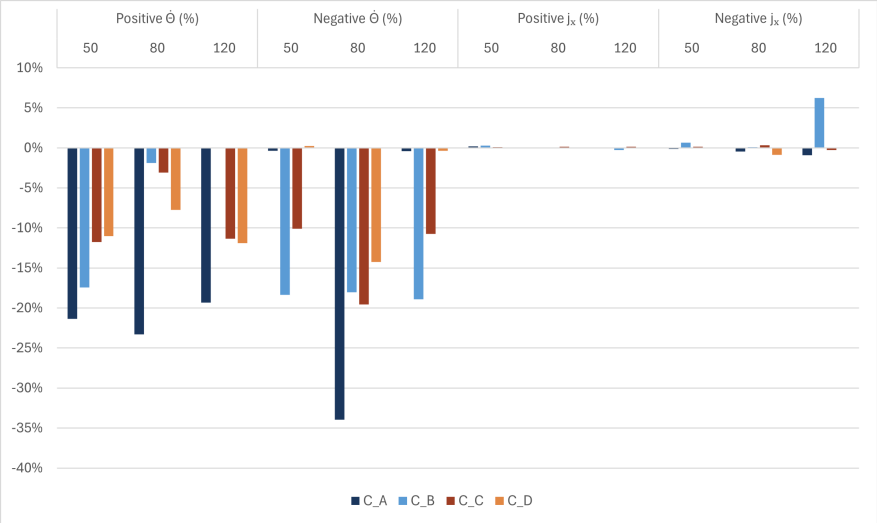
The following tests have been conducted for Uneven Throttle.

6.6.1 First Test

Following the first test in Uneven Throttle a clear correlation was established between pitch velocity and increased scaling, as can be seen in Figure 13. Once again, pitch acceleration was left out as the damper changes had minimal effects. In addition, a couple of outlying data points be seen, A and D appear prominently in negative pitch velocity at $80\frac{km}{h}$ but were missing on $50\frac{km}{h}$ and $120\frac{km}{h}$, probably due to some acceleration being recorded within the state which might not happen in the others. A small outlier can also be seen in j_x where an increased B seems to have a positive effect on jerk. This will be further considered in the second test, however, in the meanwhile a increase of all parameters is made in order to minimize pitch velocity. The next test will be run with all parameters set $h = 1.3$ and $l = 0.8$.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 13: Results from the first test in Uneven Throttle.

6.6.2 Second Test

Similarly to Aggressive Braking, for Uneven Throttle, the results stayed mainly the same, with the difference that the percentage change is diminishing. This is why it was decided to keep the process the same and increase all scalings. For the third test, the damper settings will be set to $l = 1.1$ and $h = 1.6$. Presented within Appendix A in Figure 36 are the details for this second test.

6.6.3 Third Test

Within Appendix A in Figure 35, the results for the third test on Uneven Throttle are shown. After the test, run 9 was found to be the best performing, resulting in the final setup shown in Table 15. The results from run 9 of the third test can be seen in Table 16.

Table 15: Testing and final setup for Uneven Throttle

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1

Table 16: Normalized final result parameter values and their percentage change in Uneven Throttle

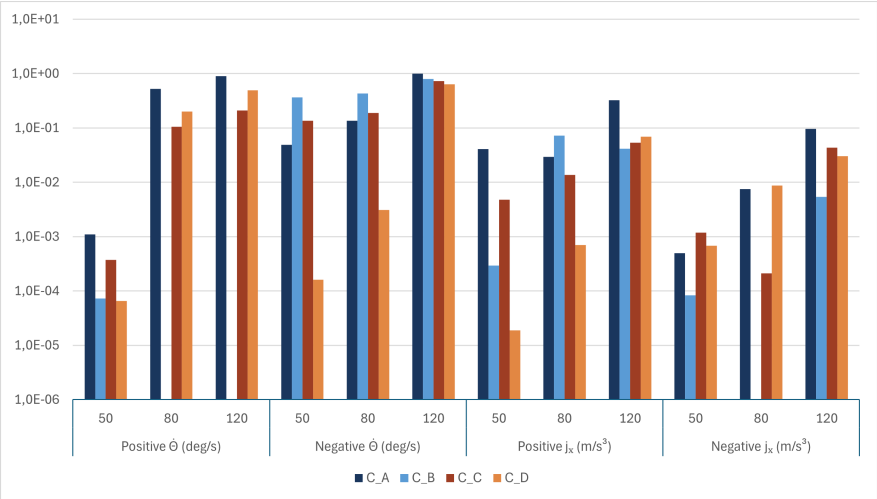
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.910	0.738	19.0
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.749	0.623	16.8
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.816	18.4
Negative $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.930	-0.872	6.13
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.635	-0.569	10.5
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.945	5.60
Positive $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.966	0.973	-0.454
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.601	0.605	-0.418
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.000	0.124
Negative $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.824	-0.822	0.101
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.995	0.504
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.582	-0.575	1.19

6.7 Uneven Braking

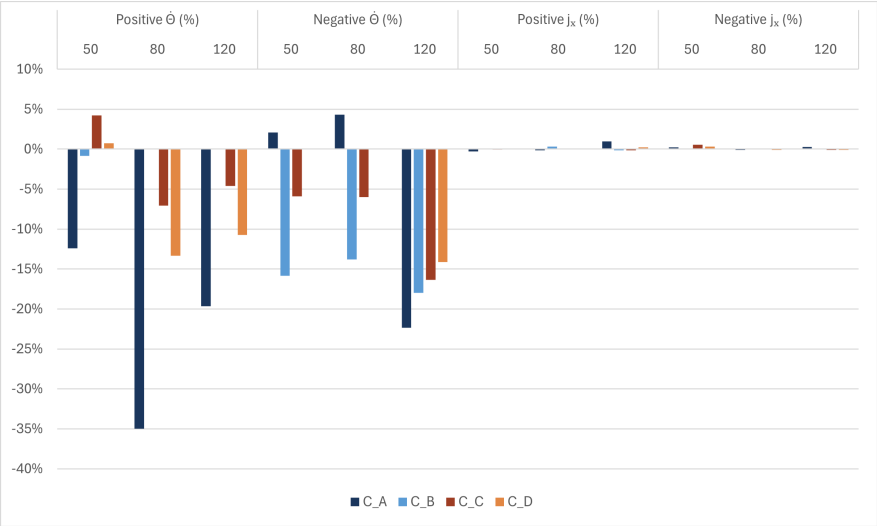
The following tests have been conducted for Uneven Braking.

6.7.1 First Test

The results from the first test of Uneven Braking can be seen in Figure 14. In general, an improvement is made when increasing all scaling, although some outlying results can be seen in both directions of pitch velocity. This could be due to some differences between outings at $50\frac{km}{h}$, $80\frac{km}{h}$, and $120\frac{km}{h}$, however, these are small, and the underlying trend is that increased scaling is positive. Resulting in a damper change where all parameters are set to $h = 1.3$ and $l = 0.8$.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 14: Results from the first test in Uneven Braking.

6.7.2 Second Test

Similarly to the analysis of the first test, within Appendix A in Figure 36 it can clearly be seen that increasing the scaling of all the damper settings would decrease all the parameters at nearly all the speeds, with some minor exceptions. This led to all the settings being increased once again, to $h = 1.6$ and $l = 1.1$.

6.7.3 Third Test

Within Appendix A, Figure 37 shows the effects from the final test of Uneven Braking. Here, run 8 was picked as the best performing, resulting in the final setup seen in Table 17. Results for run 8, including all considered parameters, are shown in Table 18.

Table 17: Testing and final setup for Uneven Braking

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6

Table 18: Normalized final result parameter values and their percentage change in Uneven Braking

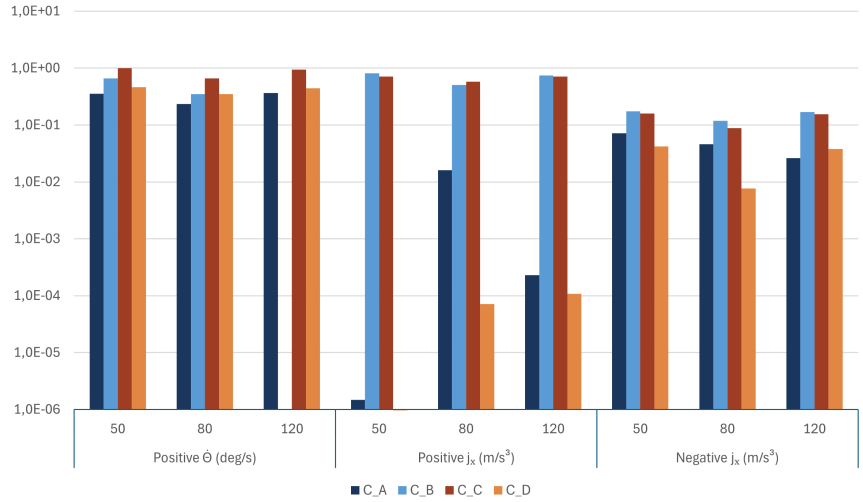
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.002	0.002	8.37
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.325	0.288	12.0
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.863	13.8
Negative $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.519	-0.416	20.0
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.702	-0.571	18.5
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.639	36.1
Positive $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.419	0.419	-0.113
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.641	0.644	-0.306
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.000	0.096
Negative $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.006	-0.006	-0.271
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.265	-0.266	-0.287
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.003	-0.258

6.8 Brake Tapping

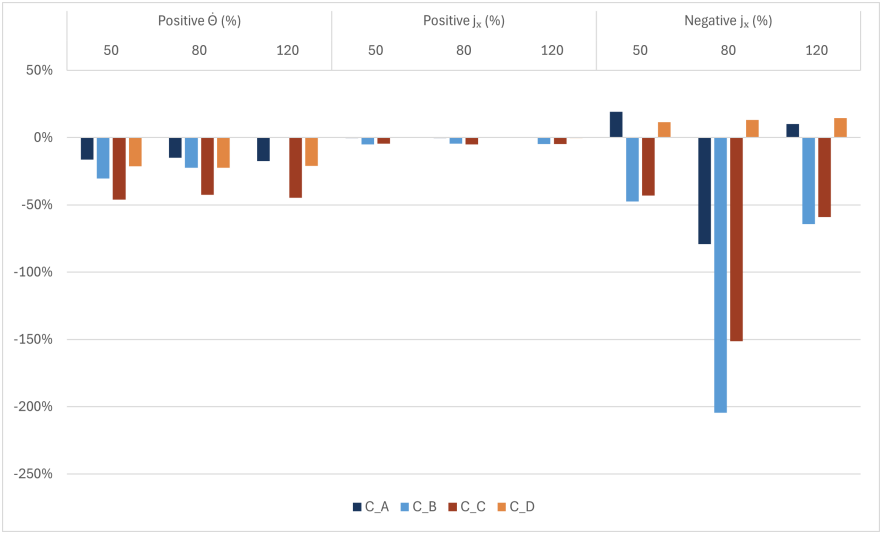
The following tests have been conducted for Brake Tapping.

6.8.1 First Test

In Figure 15, the results can be seen. Important to notice is that although percentage shifts were very large in negative jerk, the absolute value was quite small, making the results misleading. From this information the decision was taken to increase B and C to $l = 0.8$ and $h = 1.3$, and keep A and D the same in order to strike a balance between pitch velocity and jerk.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 15: Results from the first test in Brake Tapping.

6.8.2 Second Test

Although small values in general, an improvement between the tests can be seen within Appendix A in Figure 38. Especially, the positive jerk has decreased by about one order of magnitude. As there is no negative effect for making additional increases to B and C, they will be set to $l = 1.1$ and $h = 1.6$. The setting A will also be increased due to one large effect in negative jerk as well as smaller effects in positive pitch velocity; it will be changed to $l = 0.8$ and $h = 1.3$. While D is kept unchanged due to effects in both directions.

6.8.3 Third Test

Presented within Appendix A in Figure 39 the results of the final test of Brake Tapping. After analysing these results, it was decided that the settings in run 8 were the best, this setup can be seen in Table 19. Also, the final results of the relevant parameters are shown in Table 20.

Table 19: Testing and final setup for Brake Tapping

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.8, High = 1.3	0.8
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
D	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	1

Table 20: Normalized final result parameter values and their percentage change in Brake Tapping

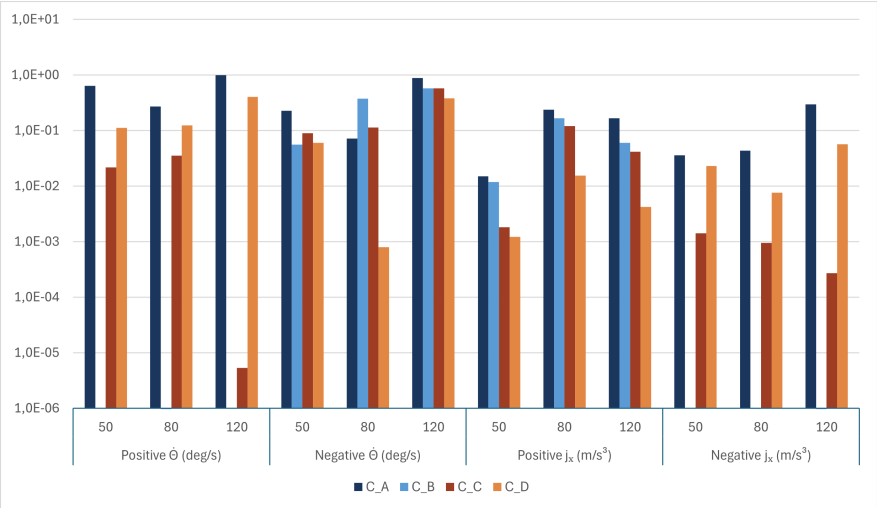
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.414	58.5
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.717	0.368	48.6
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.962	0.514	46.7
Positive $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.000	0.00
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.708	0.708	0.00
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.950	0.950	0.00
Negative $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.273	74.7
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.182	-0.045	71.5
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.727	-0.409	46.1

6.9 Braking After Throttle

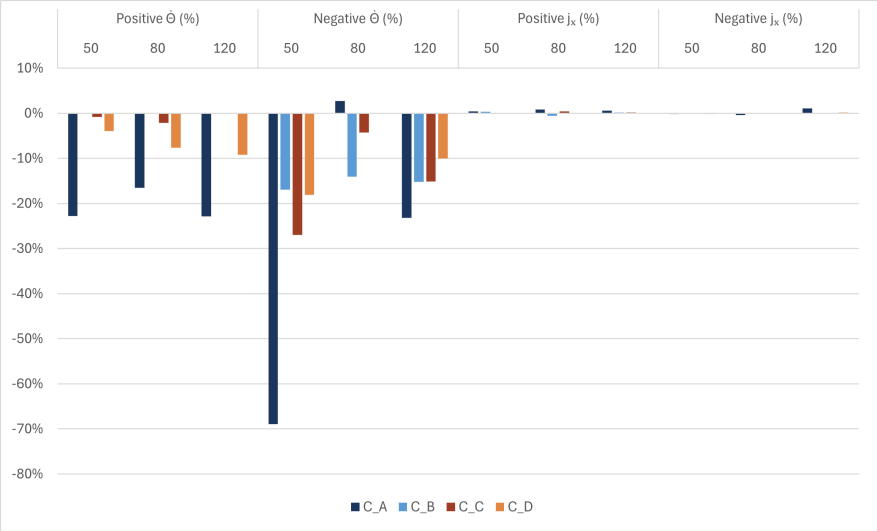
The following tests have been conducted for Braking After Throttle.

6.9.1 First Test

Presented in Figure 16 the results clearly show that for all the damper settings the higher scaling will decrease all relevant parameters and thus, the decision was made to increase all scalings to $l = 0.8$ and $h = 1.3$ for the 2nd test.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 16: Results from the first test in Braking After Throttle.

6.9.2 Second Test

For the second test of Braking After Throttle, similar effects to test one can be seen within Appendix A in Figure 40. With this in mind, the same scaling changes will be made once again. Increasing all parameters to $l = 1.1$ and $h = 1.6$ respectively.

6.9.3 Third Test

Here, run 13 was chosen as the best performing run, the respective damper settings can be found in Table 21, and the data that came from this run can be seen in Table 22. The results from this test can be viewed in more detail within Appendix A in Figure 41.

Table 21: Testing and final setup for Braking After Throttle

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1

,

Table 22: Normalized final result parameter values and their percentage change in Braking After Throttle

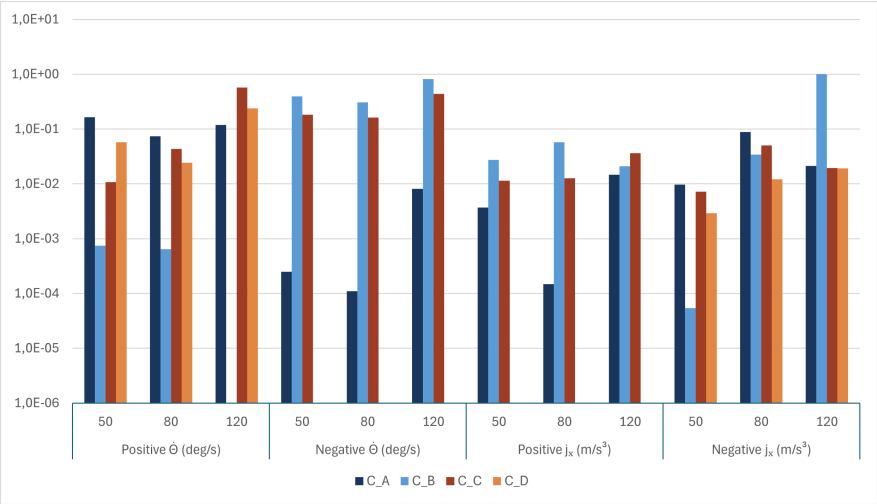
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.642	0.642	0.00
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.376	0.349	6.89
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.819	18.1
Negative $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.088	-0.061	29.9
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.705	-0.612	13.4
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.660	34.1
Positive $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.112	0.113	-0.526
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.992	0.834
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.986	0.989	-0.253
Negative $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.739	-0.736	0.268
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.402	-0.402	-0.369
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.009	0.800

6.10 Throttle After Braking

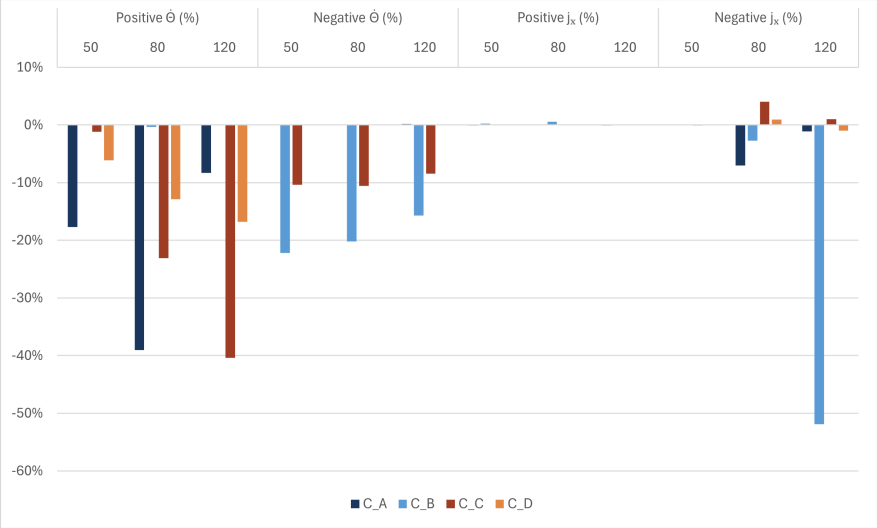
The following tests have been conducted for Throttle After Braking.

6.10.1 First Test

The results can be found in Figure 17. Here, the damper settings that have the highest effect are B and C, while A and D mostly only contribute to decreasing the positive pitch velocity, $\dot{\theta}$. Besides the small percentage increase they create in the negative longitudinal jerk, j_x, they only have negative effects on parameters, and so all the damper settings were increased to $l = 0.8$ and $h = 1.3$.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 17: Results from the first test in Throttle After Braking.

6.10.1.1 Second Test

Within Appendix A in Figure 42 indicates similar results as in the first test, with some smaller effects. A continuation of increased scaling in a similar manner is thus seen as a reasonable next step. All parameters will therefore be increased to $l = 1.1$ and $h = 1.6$.

6.10.2 Third Test

In the final test of Throttle After Braking, run 9 gave the best performance. The setup used is seen in Table 23. The C-values and percentage differences can be found within Appendix A in Figure 43, while the most crucial results are in Table 24.

Table 23: Testing and final setup for Throttle After Braking

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1

Table 24: Normalized final result parameter values and their percentage change in Throttle After Braking

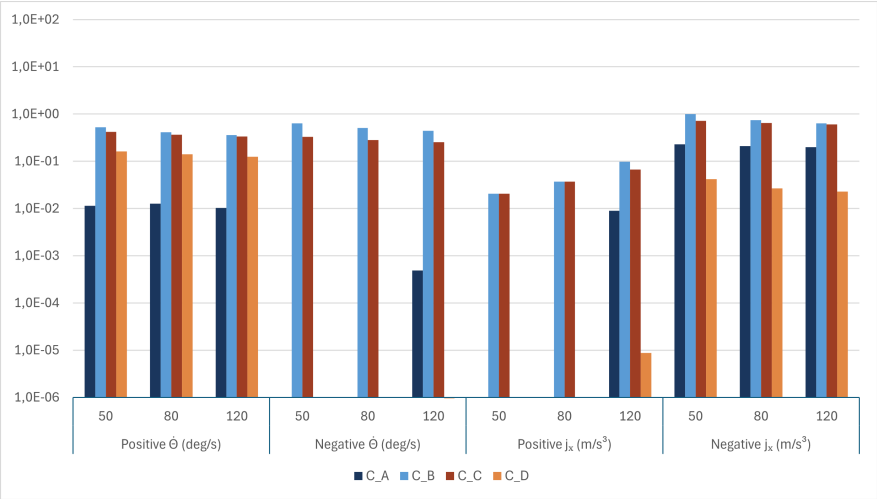
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	0.653	0.545	16.4
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	0.132	0.097	26.7
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.839	16.1
Negative $\dot{\theta}\,(\mathrm{deg/s})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.343	-0.329	4.22
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.295	-0.282	4.39
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.959	4.14
Positive $j_x\,(\mathrm{m/s^3})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	0.398	0.398	0.064
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	0.309	0.310	-0.161
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.000	0.087
Negative $j_x\,(\mathrm{m/s^3})$	$50\ \frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.004	-0.322
	$80\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.160	-0.160	0.415
	$120\ \frac{\mathrm{km}}{\mathrm{h}}$	-0.248	-0.244	1.39

6.10.3 Not Easing Off

The following tests have been conducted for Not Easing Off.

6.10.4 First Test

Damper modification effects from the initial condition can be viewed in Figure 18. Here, positive and negative pitch velocities, as well as positive and negative j_x have been studied. What is considered most important here is negative j_x, that is due to this being the jerk perceived when stopping without lifting off of the brake pedal, jolting the vehicle backwards. After the first test B, C and D were changed to $l = 0.8$ and $h = 1.3$, while A was changed to $l = 0.2$ and $h = 0.8$.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 18: Results from the first test in Not Easing Off.

6.10.5 Second Test

Within Appendix A in Figure 44 a clear improvement can be seen, where for example, the contribution of B has decreased significantly. Some positive contribution has also presented itself in D, resulting in the next test being run with A set to $l = 0.2$ and $h = 0.8$, while B and C are set to $l = 1.1$ and $h = 1.6$ and D set to $l = 0.8$ and $h = 1.3$.

6.10.6 Third Test

The results for the final test in Not Easing Off can be seen within Appendix A in Figure 45. From the results of the last test, run 7 has been decided as the best possible setup, the final setup is presented in Table 25 and the improvements from reference to modified can be seen in Table 26.

Table 25: Testing and final setup for Not Easing Off

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.2, High = 0.8	Low = 0.2, High = 0.8	0.2
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 0.8, High = 1.3	0.8

Table 26: Normalized final result parameter values and their percentage change in Not Easing Off

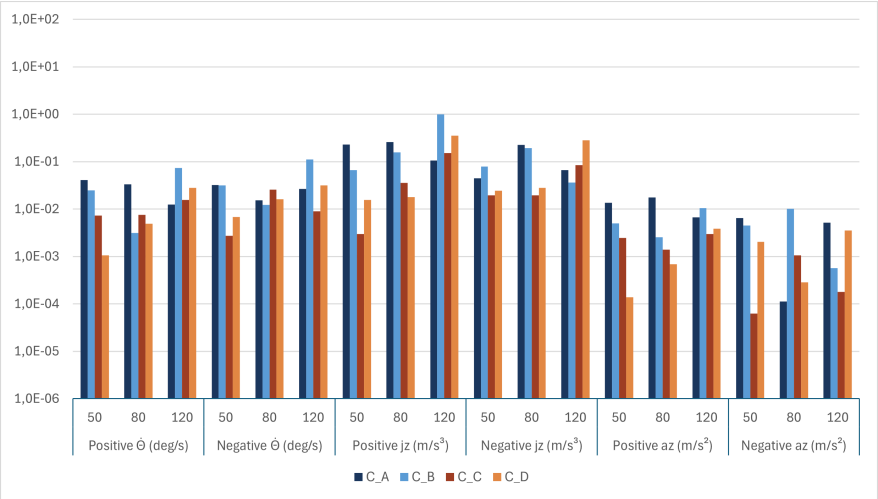
Parameter	Speed	Reference setup	Final setup	Improvement in percent (%)
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.254	74.6
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.826	0.237	71.4
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.708	0.221	68.7
Negative $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.834	16.7
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.859	-0.708	17.4
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.771	-0.633	17.9
Positive $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.000	0.00
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.924	0.924	0.00
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.860	0.860	0.00
Negative $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.542	45.7
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.892	-0.431	51.9
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.827	-0.359	56.6

6.11 Speed Mismatch With Road Surface

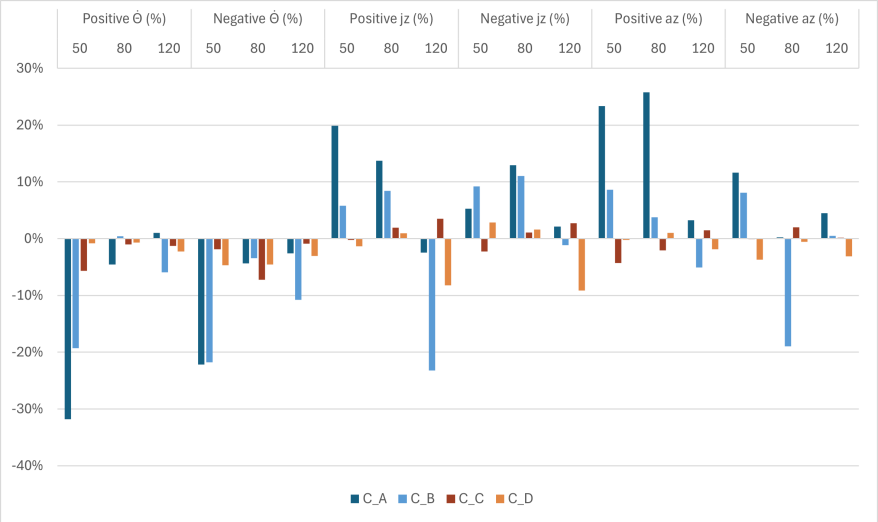
The following tests have been conducted for Speed Mismatch With Road Surface.

6.11.1 First Test

For Speed Mismatch With Road Surface the pitch velocity, vertical jerk and acceleration in both directions were analysed. The C-values and their percentage affect vs the reference are presented in Figure 19. The vertical jerk, j_z, and the vertical acceleration, a_z, were prioritized and thus, A was decreased to $l = 0.2$ and $h = 0.7$, C was left unchanged and D was increased to $l = 0.8$ and $h = 1.3$. For C a compromise had to be made between the negative and positive effects it has for the different parameters and speeds. The decision was made to focus on the results with the largest values, such as positive j_z, and so it was increased to $l = 0.8$ and $h = 1.3$.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 19: Results from the first test in Speed Mismatch With Road Surface.

6.11.2 Second Test

For the second test of Speed Mismatch With Road Surface, the damper settings were not changed much as both increasing or decreasing the settings would have severe trade-offs as can be seen within Appendix A in Figure 46. Due to this, the decision was made to instead change the span on the low and high settings while only making minor changes to the setup. For the third test A was modified to $l = 0.2$ and $h = 0.5$, B and D to $l = 1$ and $h = 1.4$, while C was kept the same as before.

6.11.3 Third Test

The best performance was seen in run 8, the setup used in this trial can be found in Table 27. The detailed results for the parameter values can be found in Table

28, and the C-values and percentages of the effects during the final test can be found within the Appendix A in Figure 47.

Table 27: Testing and final setup for Speed Mismatch With Road Surface

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.2, High = 0.7	Low = 0.2, High = 0.5	0.2
B	Low = 0.5, High = 1	Low = 0.8 High = 1.3	Low = 1, High = 1.4	1.4
C	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1, High = 1.4	1.4

Table 28: Normalized final result parameter values and their percentage change in Speed Mismatch With Road Surface

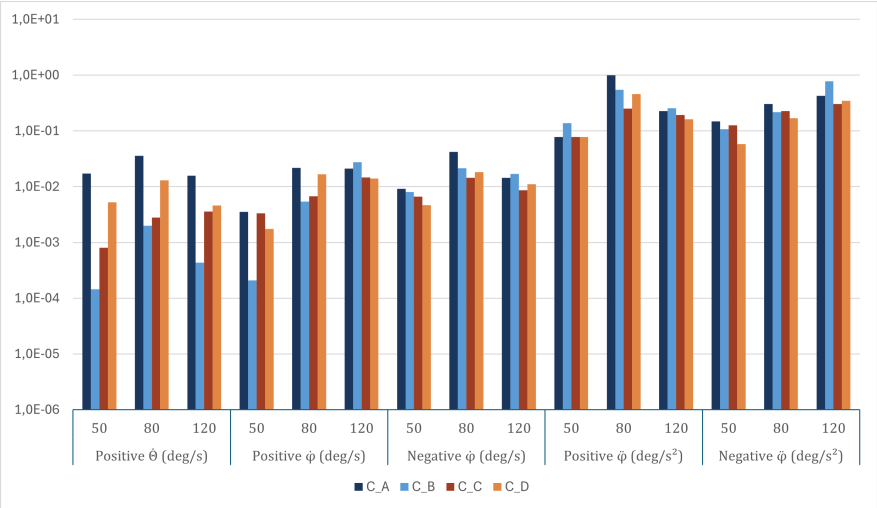
Parameter	Speed	Reference setup	Final setup	Improvement in percent (%)
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$10\,\frac{\mathrm{km}}{\mathrm{h}}$	0.105	0.135	-29.0
	$25\,\frac{\mathrm{km}}{\mathrm{h}}$	0.595	0.640	-7.31
	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.935	6.36
Negative $\dot{\theta}\,(\mathrm{deg/s})$	$10\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.140	-0.152	-8.45
	$25\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.341	-0.323	5.36
	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.976	2.40
Positive $j_z\,(\mathrm{m/s^3})$	$10\,\frac{\mathrm{km}}{\mathrm{h}}$	0.268	0.194	27.6
	$25\,\frac{\mathrm{km}}{\mathrm{h}}$	0.437	0.388	11.1
	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.900	9.77
Negative $j_z\,(\mathrm{m/s^3})$	$10\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.273	-0.257	6.11
	$25\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.563	-0.503	10.7
	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.993	0.269
Positive $a_z\,(\mathrm{m/s^2})$	$10\,\frac{\mathrm{km}}{\mathrm{h}}$	0.282	0.208	26.4
	$25\,\frac{\mathrm{km}}{\mathrm{h}}$	0.330	0.242	26.7
	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.886	11.1
Negative $a_z\,(\mathrm{m/s^2})$	$10\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.482	-0.434	9.91
	$25\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.463	-0.452	2.12
	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.938	6.11

6.12 Accelerating Into Turn

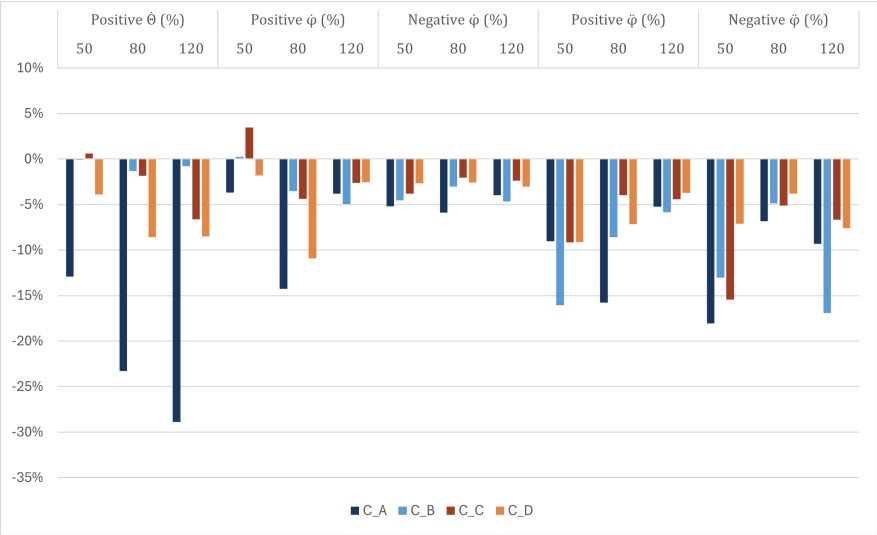
The following tests have been conducted for Accelerating Into Turns.

6.12.1 First Test

The first test conducted on Accelerating Into Turns gave a clear indication that increased scaling is positive for all parameters, shown in Figure 20. Here, pitch velocity as well as roll velocity ($\dot{\varphi}$) and roll acceleration ($\ddot{\varphi}$) have been considered. An increase to all parameters was seen as reasonable, changing all parameters to $l = 0.8$ and $h = 1.3$ respectively.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 20: Results from the first test in Accelerating Into Turn.

6.12.2 Second Test

As can be seen within Appendix A in Figure 48, the only parameter that gave a positive correlation was positive roll velocity, $\dot{\varphi}$, which was dominated by C damper setting. This leads to no increased modification for that setting, while increasing all the others, as they almost only reduced the parameter values. For the third test it was therefore decided to increase A, B, and D to $l = 1.1$ and $h = 1.6$ while keeping C at $l = 0.8$ and $h = 1.3$.

6.12.3 Third Test

The third test showed that the setup with all the dampers on their high settings, being run 16, performed the best. The final setup and results are presented in Tables 29 and 30, and within Appendix A in Figure 49.

Table 29: Testing and final setup for Accelerating Into Turns

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 0.8, High = 1.3	1.1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6

Table 30: Normalized final result parameter values and their percentage change in Accelerating Into Turn

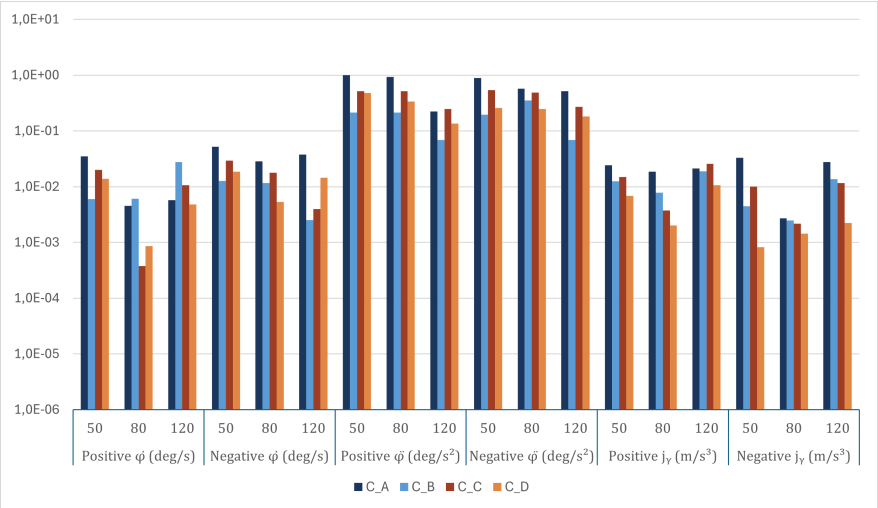
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.878	0.732	16.6
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.774	22.7
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.354	0.279	21.4
Positive $\dot{\varphi}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.175	0.159	9.17
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.277	0.282	-1.77
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.893	10.9
Negative $\dot{\varphi}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.250	-0.213	14.8
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.897	10.3
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.517	-0.464	10.3
Positive $\ddot{\varphi}\,(\mathrm{deg/s^2})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.134	0.104	22.7
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.800	20.0
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.686	0.606	11.7
Negative $\ddot{\varphi}\,(\mathrm{deg/s^2})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.178	-0.154	13.6
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.965	-0.880	8.84
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.736	26.1

6.13 Varying Angle in Constant Turn Radius

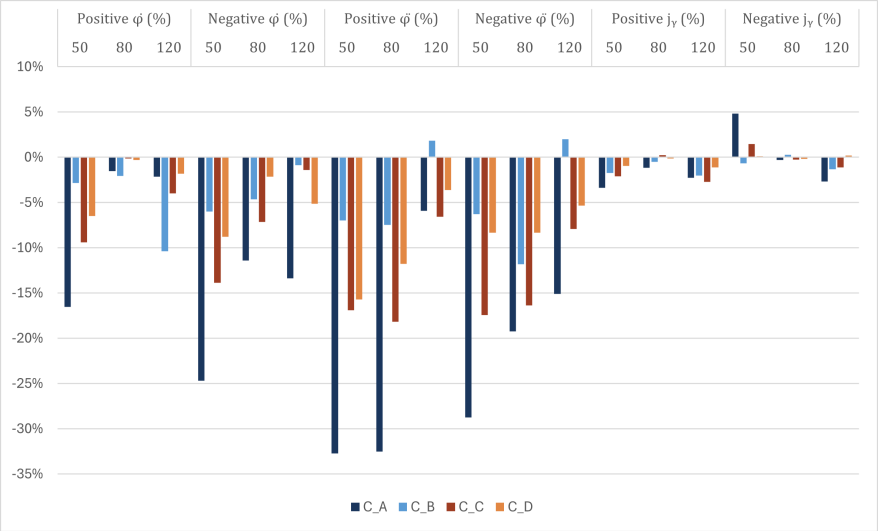
The following tests have been conducted for Varying Angle in Constant Turn Radius.

6.13.1 First Test

After analysis of the first results of Varying Angle in Constant Turn Radius, as seen in Figure 21, a clear correlation between an increase in the damper settings and improved comfort can be seen. This is almost universal, except for a few outliers in negative jerk; therefore, all the settings are increased for the second test, specifying new values to $l = 0.8$ and $h = 1.3$ respectively.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 21: Results from the first test in Varying Angle in Constant Turn Radius.

6.13.2 Second Test

After analysing the results within Appendix A in Figure 50, it is clear that C and D solely reduce the relevant parameters and so will be set to increase to $l = 1.1$ and $h = 1.6$. Settings A and B are more complicated as these have large negative effects on most parameters but also affect positive roll acceleration, $\ddot{\varphi}$, and lateral jerk, j_y, in a positive way. Here, it was decided to compromise on those parameters and rather focus on the larger reduction that would result with increased A and B settings, so they were also set to $l = 1.1$ and $h = 1.6$.

6.13.3 Third Test

After the final test for Varying Angle in Constant Turn Radius had been performed, the effects were as shown within Appendix A in Figure 51. With these effects and the results for all parameters studied for VA, the decision was taken

to use the setup from run 9 shown in Table 31. The result of all examined parameters for run 9 can be seen in Table 32.

Table 31: Testing and final setup for Varying Angle in Constant Turn Radius

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
C	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1

Table 32: Normalized final result parameter values and their percentage change in Varying Angle in Constant Turn Radius

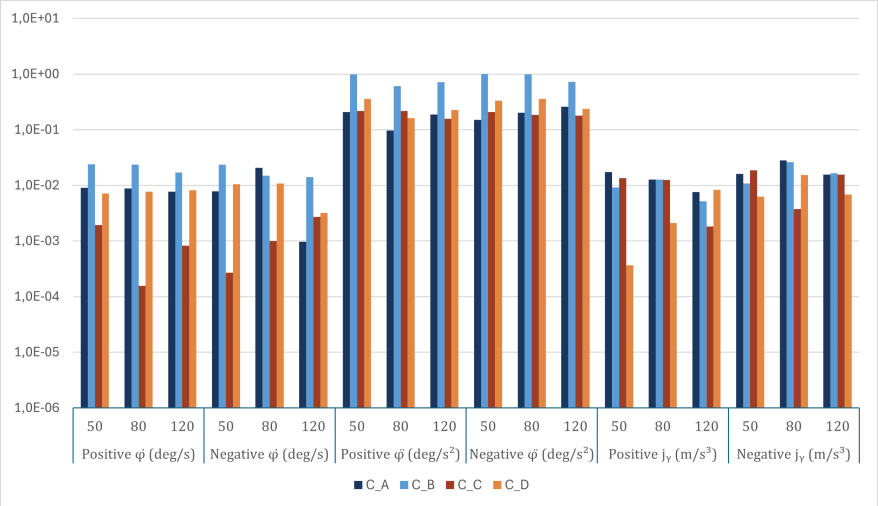
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\varphi}\,(\mathrm{deg/s})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	0.718	0.702	2.05
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.921	7.94
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	0.901	0.762	15.2
Negative $\dot{\varphi}\,(\mathrm{deg/s})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.750	-0.608	18.8
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.883	-0.767	13.2
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.879	12.1
Positive $\ddot{\varphi}\,(\mathrm{deg/s^2})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	0.814	0.613	24.7
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	0.758	0.657	13.4
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	1.000	1.066	-6.56
Negative $\ddot{\varphi}\,(\mathrm{deg/s^2})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.907	-0.725	20.1
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.866	-0.801	7.26
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.849	15.2
Positive $j_y\,(\mathrm{m/s^3})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	0.450	0.449	0.154
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.993	0.785
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	0.590	0.596	-1.14
Negative $j_y\,(\mathrm{m/s^3})$	50 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.660	-0.679	-2.91
	80 $\frac{\mathrm{km}}{\mathrm{h}}$	-0.841	-0.847	-0.773
	120 $\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.008	-0.839

6.14 Jerky Steering

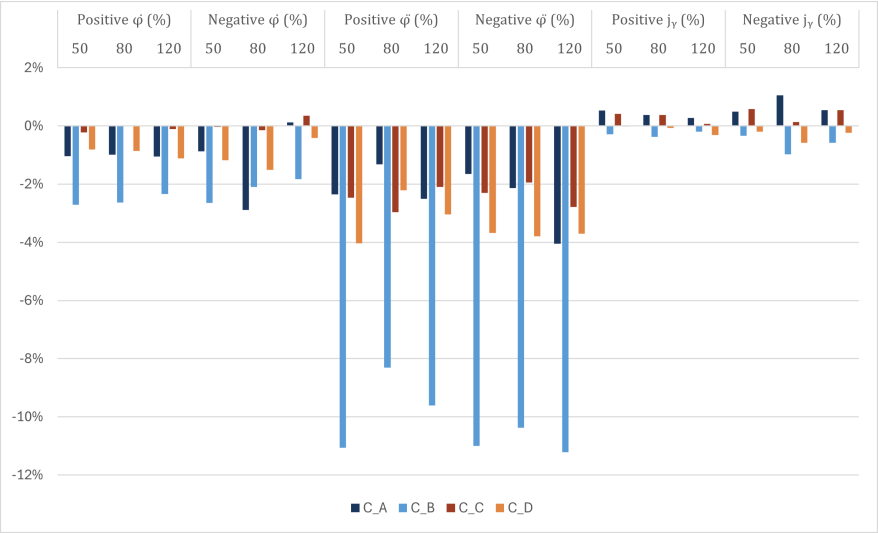
6.14.1 First Test

When considering the results for the first test of Jerky Steering as seen in Figure 22, a clear relation between all factors considered and increased B emerges. A, C, and D have smaller effects, generally positive effects from increased values when considering roll velocity and roll acceleration, and some positive effect for lower values when considering lateral jerk. With this in mind and the fact

that increased D is also useful in lateral jerk, B and D will be set to $l = 0.8$ and $h = 1.3$ while A and C are kept in their default configuration.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 22: Results from the first test in Jerky Steering.

6.14.2 Second Test

Results for the second test of Jerky Steering can be evaluated within Appendix A in Figure 52, here a similar picture emerges as in the first test. Although with increased B and D already applied, the negative effects of A on roll acceleration increase. With increased A, j_y would increase, this makes the change unjustifiable. Instead, B and D are further increased in order to decrease roll velocity and acceleration. Damper settings B and D are set to $l = 1.1$ and $h = 1.6$ whilst A and C are kept in the former configuration.

6.14.3 Final Test

When evaluating the results for the last Jerky Steering test, Figure 53 and the parameter values of each of the final runs were assessed. Best results were deemed to have been achieved in run 6, where the setup was as shown in Table 33. Each considered parameter's results from run 6 is presented in Table 34.

Table 33: Testing and final setup for Jerky Steering

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	0.5
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	0.5
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6

Table 34: Normalized final result parameter values and their percentage change in Jerky Steering

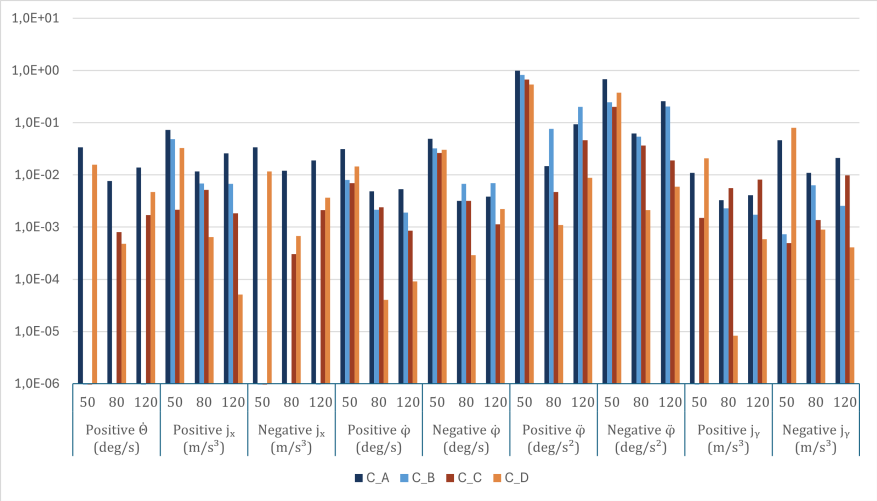
Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\varphi}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.985	0.970	1.50
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.985	1.63
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.817	0.807	1.14
Negative $\dot{\varphi}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.985	1.70
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.801	-0.801	0.259
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.862	-0.862	0.505
Positive $\ddot{\varphi}\,(\mathrm{deg/s^2})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.928	7.17
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.826	0.815	1.18
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.836	0.831	0.949
Negative $\ddot{\varphi}\,(\mathrm{deg/s^2})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.957	-0.885	7.69
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.923	7.47
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.675	-0.656	2.56
Positive $j_y\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.966	0.952	1.40
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.986	1.37
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.805	0.799	0.905
Negative $j_y\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.980	1.98
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.832	-0.814	2.31
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.888	-0.871	1.93

6.15 Steering and Braking

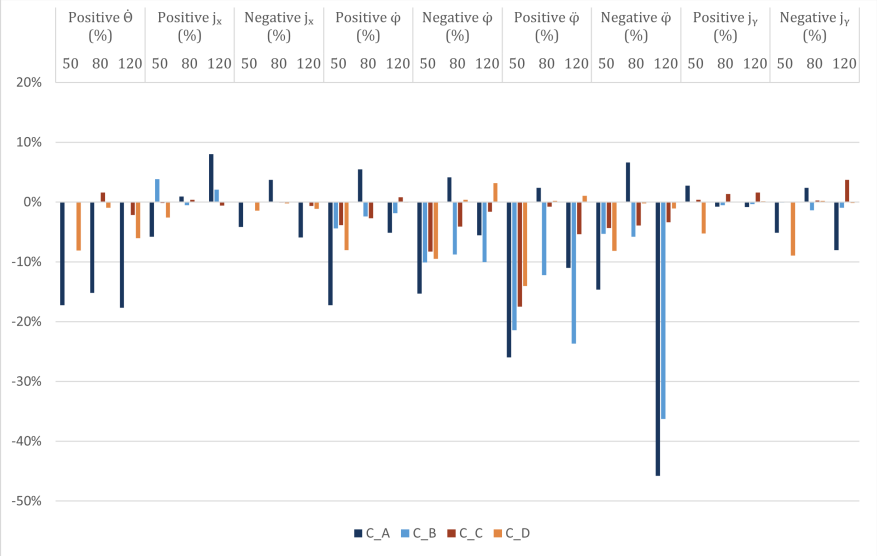
The following tests have been conducted for Steering and Braking.

6.15.1 First Test

When closely inspecting Figure 23, it can be seen that generally for A, the runs performed at the speed 50 $\frac{km}{h}$ have the highest value and these all have an inverse relation. On the other hand, the A effect values from the 80 $\frac{km}{h}$ runs have the lowest values, when comparing the three different speeds, and also they are the ones that lead to a worse ride with increased value. With the higher effect outweighing the lower effect, it was decided to increase A. Also, an increased D has a negative effect on almost all of the chosen parameters; one exception is in negative $\dot{\varphi}$, but its magnitude is low and thus disregarded, and D is chosen to be increased for the second test. Similarly, for B, it almost only has negative correlation, with the exception of positive j_x, so it will be increased. In summary, for the second test the values of A, B and D were set to $l = 0.8$ and $h = 1.3$, while C was kept the same.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 23: Results from the first test in Steering and Braking.

6.15.2 Second Test

The same phenomena occurred as in the first test involving A, and so will be reasoned in a similar fashion to be increased even more. The rest of the damper settings have mainly negative relations to the parameter values, and so they will be increased except for C, as that would increase the negative lateral jerk, j_y, with around 4% relative to the reference value. The settings for the third test are the following: A, B, and D are set to $h = 1.6$ and $l = 1.1$ while C is kept at $h = 1$ and $l = 0.5$.

6.15.3 Third Test

The best performing simulation from the test of Steering and Braking was deemed to be run 8. The damper settings used in this run are presented in Table 35 and results can be observed in Table 36 and within Appendix A in Figure 55.

Table 35: Testing and final setup for Steering and Braking

	First test	Second Test	Third Test	Final Setup
A	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.1
B	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6
C	Low = 0.5, High = 1	Low = 0.5, High = 1	Low = 0.5, High = 1	1
D	Low = 0.5, High = 1	Low = 0.8, High = 1.3	Low = 1.1, High = 1.6	1.6

Table 36: Normalized final result parameter values and their percentage change in Steering and Braking

Parameter	Speed	Reference setup	Final setup	Percentage change
Positive $\dot{\theta}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.940	6.02
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.258	0.239	7.18
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.399	0.389	2.65
Positive $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.992	1.003	-1.11
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.997	0.121
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.255	0.257	-0.666
Negative $j_x\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-1.000	0.00
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.400	-0.403	-0.719
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.396	-0.396	0.00
Positive $\dot{\varphi}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.927	7.27
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.492	0.473	3.73
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.575	0.548	4.58
Negative $\dot{\varphi}\,(\mathrm{deg/s})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.841	15.9
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.241	-0.228	5.04
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.219	-0.202	7.07
Positive $\ddot{\varphi}\,(\mathrm{deg/s^2})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.756	24.7
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.161	0.139	14.2
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	0.223	0.223	0.169
Negative $\ddot{\varphi}\,(\mathrm{deg/s^2})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.881	12.0
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.201	-0.201	0.480
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.122	-0.116	5.2
Positive $j_y\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	0.793	0.759	4.52
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	0.834	0.828	0.892
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	1.000	0.993	0.316
Negative $j_y\,(\mathrm{m/s^3})$	$50\,\frac{\mathrm{km}}{\mathrm{h}}$	-1.000	-0.923	7.51
	$80\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.519	-0.519	0.171
	$120\,\frac{\mathrm{km}}{\mathrm{h}}$	-0.295	-0.293	0.490

7 Results From Evaluation of Dynamic Algorithm

In this section the answers from the dynamic experiment surveys are presented. As described in Chapter 5.2 the survey contained 10 questions about how the car felt to drive and each question could be scored from 0 to 10 points. Here, the first 5 lap outing will be referred to as Test 1 and similarly, the second 5 lap outing, will be mentioned as Test 2. During the postprocessing, the answers were split into different categories such as Test 1, Test 2, the difference between the tests where the algorithm was turned off and on, and the difference between the first and the second test. The deviation in answers were also a point of interest and so error bars were added to each graph, where they represent one standard deviation.

The average scores for each question in Test 1 can be seen in Figure 24.

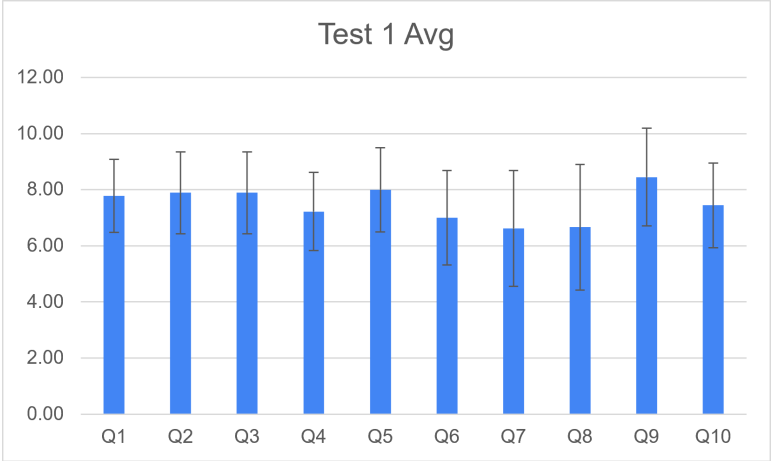


Figure 24: Average results from Test 1

The average scores for each question in Test 2 can be seen in Figure 25.



Figure 25: Average results from Test 2

The difference between the average scores for each question in the test where active damper adjustments were being made and the test without it are presented in Figure 26. Here, the average score for all the questions was 0,31, meaning that the test where the algorithm was off scored higher than when it was turned on.

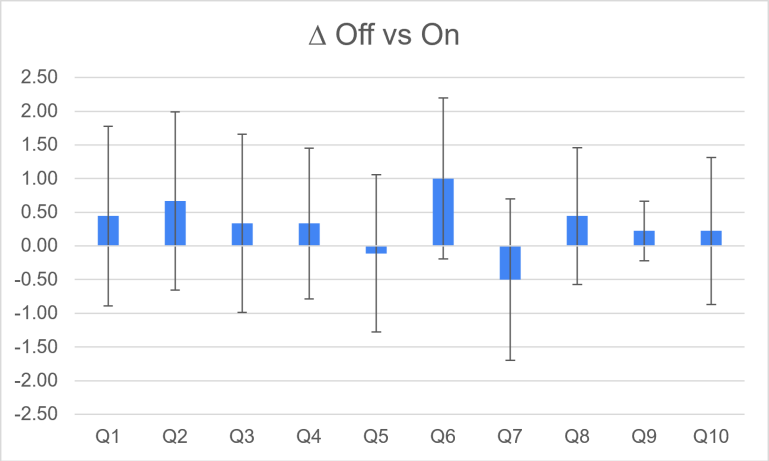


Figure 26: Average difference between the 5 lap outings where the dynamic algorithm off vs on.

The difference between the average scores for each question in the Test 1 and Test 2 are presented in Figure 27. Here, the average score for all the questions was -0,15, which indicated that the Test 2 was scored higher than Test 1.

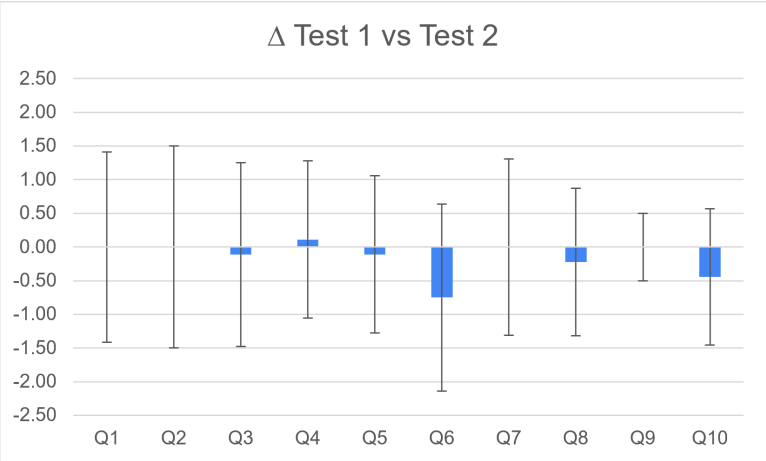


Figure 27: Average difference between Test 1 and Test 2.

The answers from experienced drivers, those who had driven over 10 hours on a track or in a simulator, were picked out in order to see if experience affected the results.

Presented in Figure 29 are the differences between the tests when the algorithm was off and on, driven solely by the experienced drivers. Here, the average score for all the questions was -0,27, meaning that the test where the algorithm was on scored higher than when it was turned off.

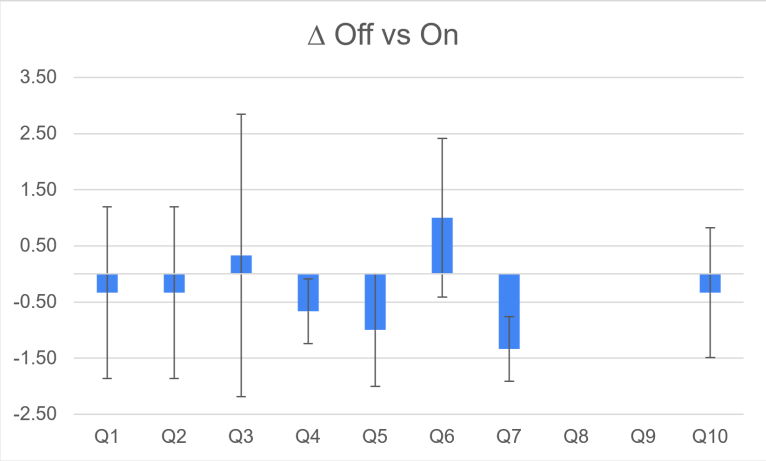


Figure 28: Average difference between the 5 lap outings where the dynamic algorithm off vs on from the experienced drivers.

Driven by the experienced drivers, the difference between the average scores for each question in the Test 1 and Test 2 can be seen in Figure 29. Here, the average score for all the questions was 0.07, which indicated that Test 1 was generally preferred over Test 2.

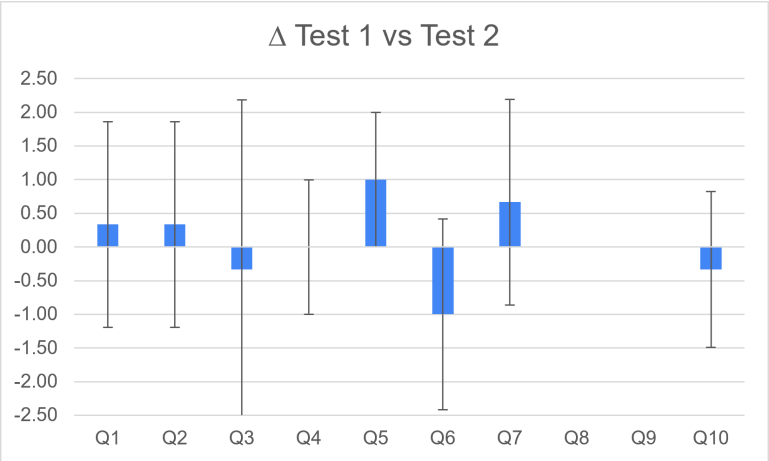


Figure 29: Average difference between Test 1 and Test 2 from the experienced drivers.

8 Discussion

In this section, the results as well as other aspects of the project will be discussed. Here, arguments about what went well, what did not go well, and why so were the case are analysed.

8.1 Comfort Results Discussion

In the following sections, the results from the comfort evaluation will be discussed.

8.1.1 General Results

Although it does not apply to all cases, the most prominent common denominator is the reduction of pitch velocity. Especially in all longitudinal cases, it was evident early on during testing that this is where big improvements were possible. Most other parameters that were considered had no significant results when changes were made. It therefore became clear that the early belief to minimize jerk and accelerations might not be the most efficient parameters to analyse when trying to maximize comfort. Focus was instead shifted to improving those factors where damper changes had the largest effect, which in most cases were pitch velocity. However, one state where jerk was considered and resulted in large improvements, about 50 %, albeit with smaller values, was Not Easing Off. Probably due to the nature of the state, where negative jerk occurs from weight transfer and loading of the front springs during braking, which then springs back to equilibrium at the moment the vehicle stops. This is a very positive result, as this jerk is what was referenced as particularly uncomfortable in Chapter 2.2.3.3. Besides the jerk improvements in the case of Not Easing Off, the pitch velocity was also decreased immensely, around 70 % in dive and 17 % in lift. It should be noted that these velocities had small values.

Large improvements in pitch velocity were made in Aggressive Throttle and Brake, Uneven Throttle and Braking of around 15-20 %, Throttle After Braking and Braking After Throttle had around 15-30%, and great improvements in Brake Tapping of around 50 %. Due to the states being purely longitudinal, these reductions can be attributed to the increased scalings that was introduced for their damper settings. The higher scaling creates an equal force at a lower damper velocity, and as the displacement at the dampers is directly correlated to the pitch.

In general, accelerations have been affected very little, as on an ideally flat and smooth road, such as the one in our simulations, the dampers' settings will not have much impact on the dynamics of the car. In reality, roads have certain irregularities, bumps, potholes, etc. This means that the car will have reduced grip if the dampers are not set up correctly. This loss in grip would affect the accelerations and thus the changes in A, B, C, D would result in larger changes. The sole case where the acceleration was significantly impacted was in Speed Mismatched With Road Surface, a peak reduction of roughly 26 %, all the others had below a 2% change between the reference and the final setup. In this state, the vertical acceleration changed due to the bumps and irregularities on the road present in the testing of this state, resulting in excitations in the vertical direction, which is where the dampers play a crucial role.

When also considering lateral cases, jerk even in the y-direction still had quite small effects from damper changes. The focus instead shifted to roll velocity and roll acceleration, where large improvements were detected as possible. Looking at the four lateral cases, AIT, VA, JS, and SB, all have seen an improvement in roll velocity and roll acceleration, mostly in the range from 5% to 20%, which can be considered a success. Especially as only a very small number of other parameters had a negative outcome, with the most worsened being negative j_y for $50\frac{km}{h}$ in JS, where the jerk was increased by 2.91%. An acceptable worsening when considering the number of parameters that have been improved.

An unwanted side-effect that could come up based on the damper changes that have been made is the effect of road irregularities. When increasing the damper settings, road unevenness, such as potholes and expansion joints, results in higher accelerations in the cabin and thus worse comfort. Even though the states are only active during certain driving events, there is no certainty that they will not be triggered when hitting a pothole. This is a stochastic behaviour and is difficult to simulate, which is why it has been left out of this study. However, a real-life test where road comfort is evaluated for this algorithm should take this into account in order to fully assess comfort improvement.

8.1.2 Identification Accuracy

Another topic that is of interest is the identifications. As no damper changes are made unless the identification is active, this activation point will be of utmost importance. If it triggers too late, the changes might have small to no effect, and if it triggers too early, it might either worsen the driving experience or interfere with other identifications. To achieve as good a result as possible has a multitude of outing been analysed, wherein parameters have been tweaked multiple times with the help of data inspection. However, this approach might not be optimal as there is no real-life feedback. An expansion in which live output of active states can be observed while in the vehicle would be helpful in order to validate the accuracy of the activated states.

8.1.3 C- Value's Percentage Possible Misrepresentation

Through the evaluation of the results from comfort tests, it was important to investigate the C-values and the percentage difference as a whole for each parameter and compare them to measured data from the reference drive. If not careful, one could see the high percentage effect that one of the damper settings generates and believe that this is vital information. Why this could be misleading is that if the measured values of a certain parameter are minuscule, then even tiny effects can produce high percentage changes, all the while the change itself is irrelevant.

8.1.4 Delay of Identification

A limitation of our solution is that, due to the noise present in the signals used for the algorithm, they need to be passed through a low-pass filter in order for the identification to be more accurate. This filter introduces some delay, roughly 30 ms. Potential solutions could be to redesign the filter itself, but this was not investigated, as the algorithm was tested using this filter, and the delay was not deemed to be a crucial improvement, this delay should be kept in mind.

8.1.5 Initial Condition

An assumption of using the scaling $l = 0.5$ and $h = 1$ as the initial conditions was applied due to the inherent idea of soft damping being more comfortable. This is, in many cases, the truth, especially when considering road irregularity or bumpiness; however, many of the states identified in this project are transient errors made by the driver. Errors such as Aggressive Braking and Jerky Steering introduce weight transfers in the vehicle, making it pitch and roll due to the induced accelerations. As shown in the results, these movements are then counteracted by increasing the scaling. As the testing in most cases (not SM) has been conducted without elevation and on a flat surface, there is increased scaling and an improvement in comfort.

8.1.6 RMS Jerk

RMS jerk was initially considered as a comfort parameter, especially due to its inclusion in multiple sources such as Huang and Wang, as well as Winkel et al., but was quickly disregarded when test results were analysed [14, 10]. This was due to the extremely small differences between high and low scalings, the largest changes detected were improvements of 0.05% where other parameters mentioned in the results had improvements 3 to 4 magnitudes larger. Instead, peak jerk values were considered as these showed much greater differences between settings, and are more relevant to the driver errors presented in the report. RMS jerk might be a valuable parameter for testing, for example, constantly occurring vibrations due to resonance in the vehicle or rough roads; however, for these transient driver errors, such as Aggressive Braking, RMS jerk is not the right parameter to use in order to evaluate comfort.

8.2 Dynamic Results Discussion

In the following sections, the results from the dynamic evaluation will be discussed.

8.2.1 General Results

When focusing on the results from the dynamic experiment performed at the Brembo simulator, it is clear that no conclusive result can be established. There were small to no differences between the two evaluated outings, both when evaluating results from all drivers as well as the three more experienced ones. As presented in Chapter 7, when studying all the drivers' answers, the averaged results showed that in general they scored the outing without the dynamic algorithm higher than when it was active, and also preferred Test 2 over Test 1. While for experienced drivers, the results showed the opposite. These differences in scores are extremely small, thus making it impossible to draw any conclusion about performance.

Another goal of the thesis was the development of the identification algorithm. Although the damper changes had a small to no perceivable effect for the driver, the identification was successful. This was not necessarily the case, as the identifiers are different in the dynamic case than the comfort case. However, while the experiment was running, live telemetry showed the active state, establishing that the states triggered at the correct points. Damper changes were also applied correctly, as a damper force was plotted live during each experiment.

8.2.2 Simulator

The main reason for the inconclusive results is thought to be the level of feedback from the simulator. The simulator used in this study, in contrast to real-life racing, only give feedback through the steering wheel and not accelerations. Therefore, if the steering model is inadequate or the steering wheel is poorly calibrated, achieving an accurate evaluation becomes extremely difficult. It is exactly this that is thought to be the major reason for the inconclusive result. During our time at Brembo, we had our supervisor and Öhlins test driver Emil Sällberg with us, who early on described the feedback from the wheel as poor and lacking in fidelity, and also pointed out that when the algorithm was on versus it being off, there were no to minuscule differences. Even when the scaling was increased by 300% the changes were minimal. As a test driver who specializes in dampers, Sällberg stated that this level of change should be extremely noticeable. It is therefore believed that the simulator is a major source of error in these results, explaining the inconclusive results.

8.2.3 Participation

Another topic that should be highlighted is participation. As a result of time constraints, only having one full day to run the experiment, participation had to be limited. With each experiment taking about 45 minutes, including introduction, driving, and survey taking, nine participants were the maximum

possible. However, this is, of course, too few to make definitive conclusions about the driving experience. The low number of participants is a likely explanation for the high standard deviation seen on each question in the dynamic evaluation.

We also saw from the experiment results that drivers with more experience perceived smaller differences. This might allude to the fact that those who are experienced and know what nuances to look for felt that the changes were unnoticeable, while those who had less experience either knew that something was going to change and therefore made guesses for which they preferred, or more likely got more comfortable with the simulator for the second outing and thus ranked it higher.

8.2.4 Simulation Implementation

The implementation of the developed CarMaker algorithm into Brembo's, VI Grade-based simulation environment was initially considered a potential source of issues. However, a solution to bridge the gap between the two softwares was found with the help of Mats Widmark. The whole Simulink subsystem, wherein the identifications were made, could be ported into VI Grade, and no problems emerged from this solution.

8.3 Driveability

One issue that could present itself if pursuing the development of this semi-active damping would be handling. As the tests conducted have been targeted at maximizing comfort, could an adverse effect result in the deterioration of the vehicle's handling characteristics. This is something that has to be kept in mind when further developing this model and should be evaluated in an eventual real-life study. However, due to the high costs of non-simulation testing as well as the time scale of this project, the handling of the vehicle will have to be neglected as a parameter.

8.4 Response Time for Dampers and the Algorithm

Due to the way the algorithm functions, certain behaviour identifications trigger quicker than others. Nonetheless, all the states that are designed to work actively activate immediately once the criteria are fulfilled. Within the criteria, some triggers depend on waiting a few time steps. For example, in order to make sure a signal has settled, this in itself delays the activation or deactivation of the state, but it is crucial for the algorithm's accuracy. Also, all the input signals are passed through a low-pass filter, which helps relieve the noise in the signals, but it also adds a delay of about 30 ms that affects all the identifications. In perspective, according to Gorantiwar, semi-active dampers typically operate at a response time of 40 ms, although when consulting with our supervisor at Öhlins, they claimed this might vary depending on the magnitude of the damper change [20]. Ideally, the cases would trigger as soon as the error occurs, but being in the same range as the dampers themselves, we believe this

will not pose any issues, as humans in general are much slower than this, and it will feel almost instantaneous.

8.5 Driver Overcompensation to the Damper Changes

One issue that could occur when making adaptive changes based on behaviour is the introduction of driver overcompensation. For example, it could mean that if the driver is inherently jerky in their steering and they like to make aggressive manoeuvres and fast lane changes, then a change to counteract this could lead to a stronger action by the driver, as they now feel like they have to do more to achieve the same vehicle behaviour as before. This might be further enhanced by the fact that the drivers are not informed of each identification and damper change that is made. However, this probably will not present itself as an issue, mainly due to the fact that it is supposed to be utilized as an inclusion in a ”comfort mode”. This means that when activated, the driver should know that the vehicle is optimized for comfort and not dynamic driving. In addition, this system must, work in parallel with safety systems such as ABS and ESP, which should take over in the case that the driver goes from making uncomfortable manoeuvres to dangerous ones.

8.6 Could Inner/Outer Wheel Changes Help Comfort

In a future expansion of this project, an improvement could be made by introducing individual damper changes at each wheel in comfort. Having inner and outer wheels that behave independently could help keep the vehicle level in cornering, adding safety during emergency manoeuvres, or even softening the damper on one wheel if that wheel hits a pothole. The vehicle could therefore stay level on the support of three corners while the corner is very soft and follows the excitation without disturbing the cabin. The decision not to consider each wheel independently in comfort was twofold. Firstly, the scope of the project did not consider outer and inner dynamics as necessary and was therefore not considered until later in the project. Secondly, and most importantly, the time constraints made such an addition unattainable. Although the time to add individual damper changes to the algorithm would not be very extensive, adding four extra parameters to the testing would be a massive task. This would take the factorial design table from 4×16 to 4×256, introducing a massive post-processing task. This was therefore seen as a future development in case this project proved to be successful.

9 Conclusions

When setting out on this thesis work, the objective was to determine whether it was possible to identify driver errors through driver inputs. Then, using these identifications to improve either the ride comfort or dynamic handling performance by providing accurate real-time damper adjustment.

In the comfort scenario, the use of a driver behaviour-based algorithm was successful in reducing unwanted parameters by a significant amount. For most cases, the greatest improvement was seen in pitch and roll velocity as a result of generally increased damper settings. A select few showed betterments in longitudinal jerk, which was due to increased B and C, while reducing A and D. There were also improvements in vertical acceleration, which was caused by reducing A, and increasing B and D. Each parameter's reduction is considered to increase comfort noticeably, which was the goal. All these results came from simulated drives, as we did not have the opportunity to test the model in a real car, so the results may vary once integrated into a physical car.

In the dynamic scenario, the results were inconclusive, mainly because the simulator did not provide the level of force-feedback fidelity that was initially expected. However, the identification and damper modifications were seen as a success. The identifications triggered in the correct scenarios when the driver made mistakes, and in turn, the correct damper changes were applied.

In conclusion, the objectives of this thesis were fulfilled. An algorithm has been developed that can identify driver errors through the driver's inputs. It was also successful in the objective to counteract these errors through damper changes, although the results concerning the dynamic evaluation remain uncertain.

10 Future Work

10.1 Real Life Study

One way to more thoroughly evaluate if the comfort has been improved in a noticeable way would be to conduct a real-life study, wherein the algorithm is implemented on a genuine road vehicle equipped with Öhlins dampers. This would create a more lifelike testing scenario in which parameters other than the earlier-mentioned quantitative ones are tested. Subjective feelings based on human experience could then be evaluated in a similar way to how the dynamic testing was conducted, in order to see if and how big a difference the algorithm makes. In real-world tests quantitative data could also be tested on the passenger and not the vehicle, i.e. an accelerometer could be placed on the passenger in order to investigate the accelerations occurring while in a seated position with some damping from the cushioning.

10.2 Levelling of States and Prioritization

A planned future development that, unfortunately, was hindered due to time constraints was the categorization of each state into multiple levels. The idea was to create different levels based on how big of a mistake the driver has made and introduce different severity of damper changes. Small mistakes lead to small corrections, and big mistakes lead to large corrections. They could be helpful in comfort by creating a more seamless change between different settings, as well as not making too large corrections when not needed. However, this is seen as more of an opportunity for improvement in dynamic driving, where finding the best setup for each driver comes down to compromises and finding sweet spots instead of overdoing changes. A small change might be just what the driver wants to help with some rotation on entry, while making a large change would make the vehicle rotate too much and thus be difficult to control. The same driver might then want a large correction on exit, where rotation is too much already and needs to settle the vehicle. If the identification was levelled, this could all be achieved, and the algorithm might be more helpful for the driver.

Another way to increase both comfort and dynamic performance could be to introduce a prioritization of the states based on how big the mistake is. As mentioned earlier, multiple mistakes can be identified at once while only one is activated, activation is currently based on a predetermined ranking of the fault level of each state. If this is instead based on the level of mistake compared

to each state, the most problematic one could always be the active state, thus improving both comfort and dynamic driving experience.

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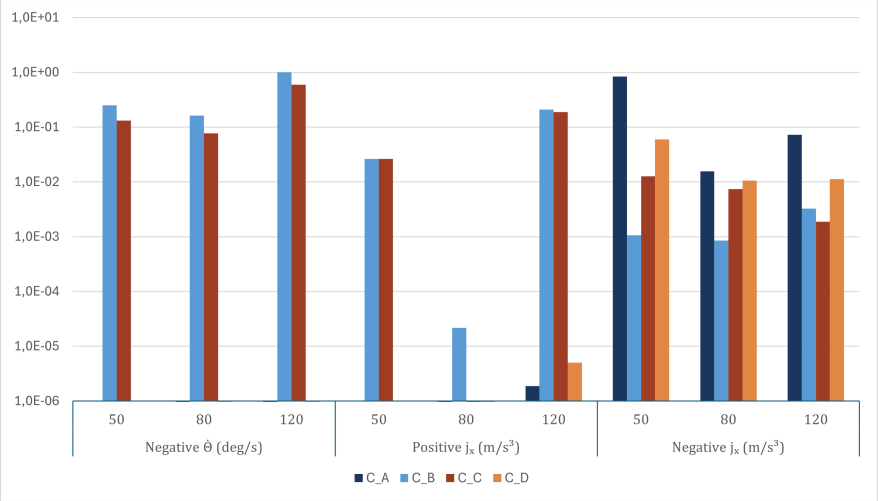
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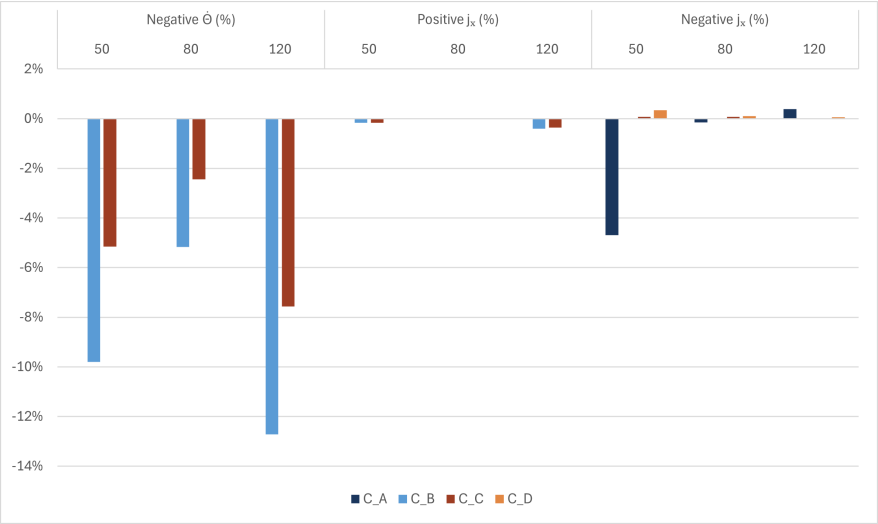
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A Appendix

Additional Figures Aggressive Throttle 2nd and 3rd Test

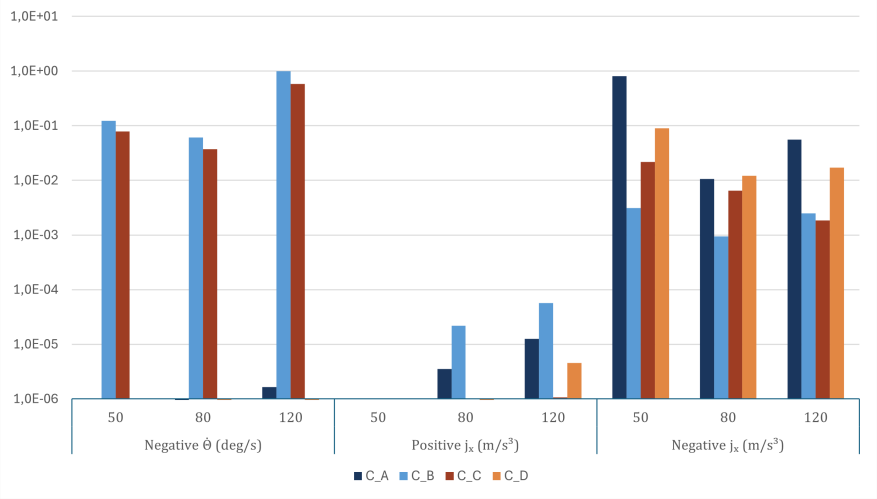


(a) Results in effect (C)

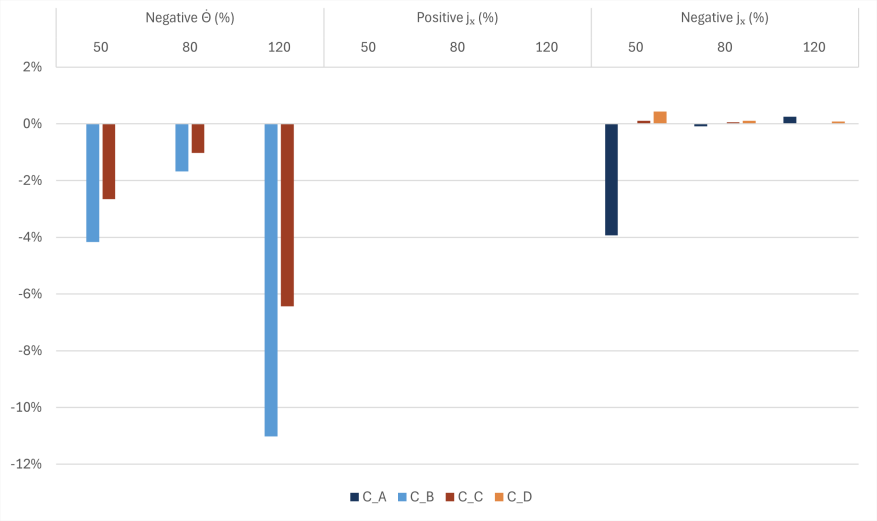


(b) Results in percentage of reference value (%)

Figure 30: Test results from the second test in Aggressive Throttle.



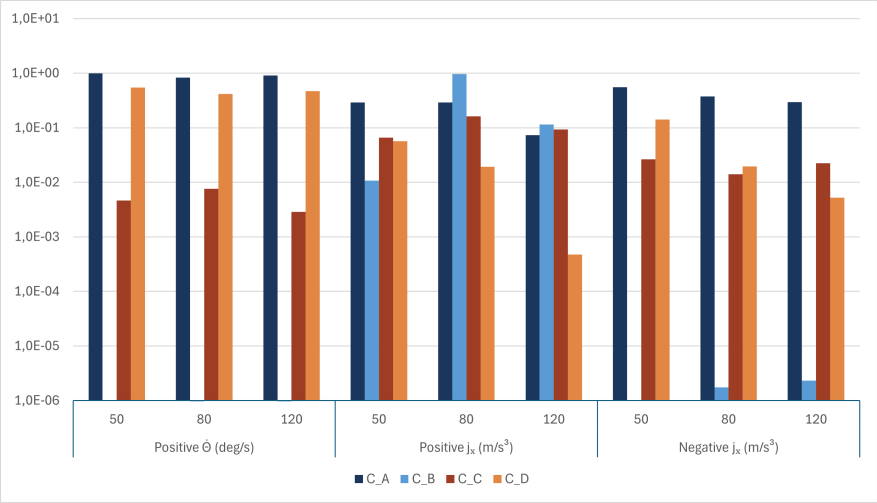
(a) Results in effect (C)



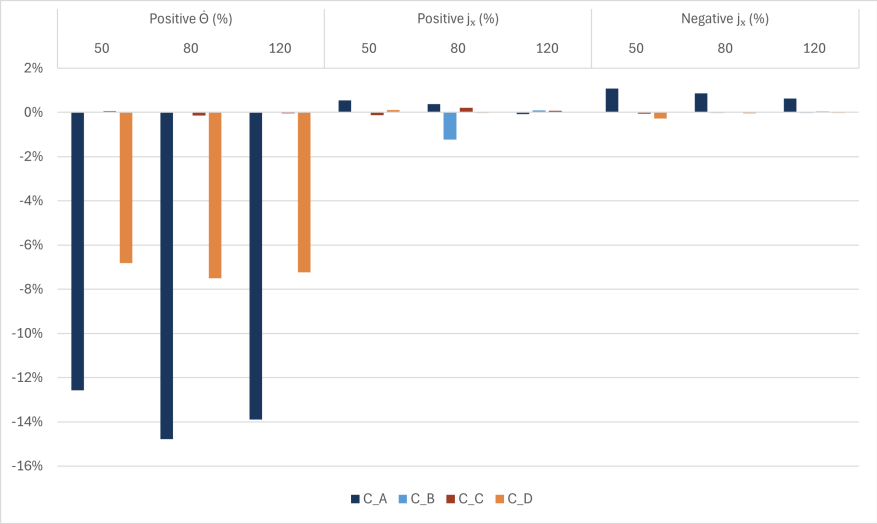
(b) Results in percentage of reference value (%)

Figure 31: Test results from the third test in Aggressive Throttle.

Aggressive Braking 2nd and 3rd Test

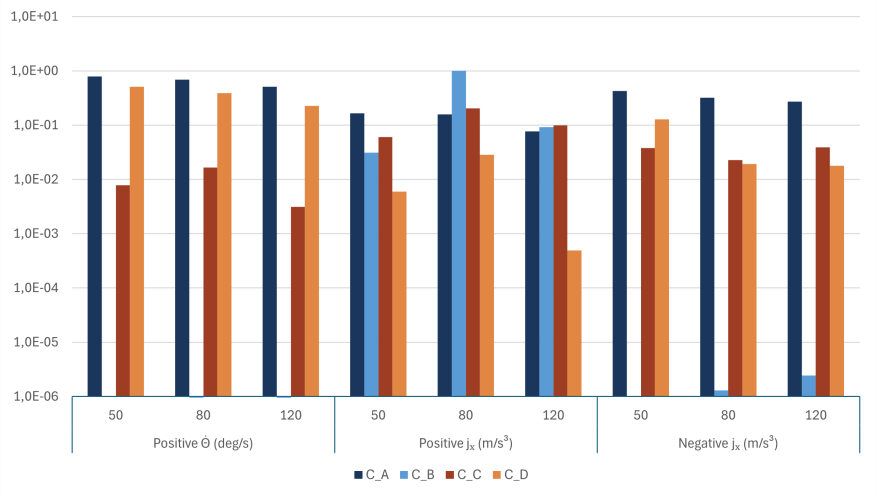


(a) Results in effect (C)

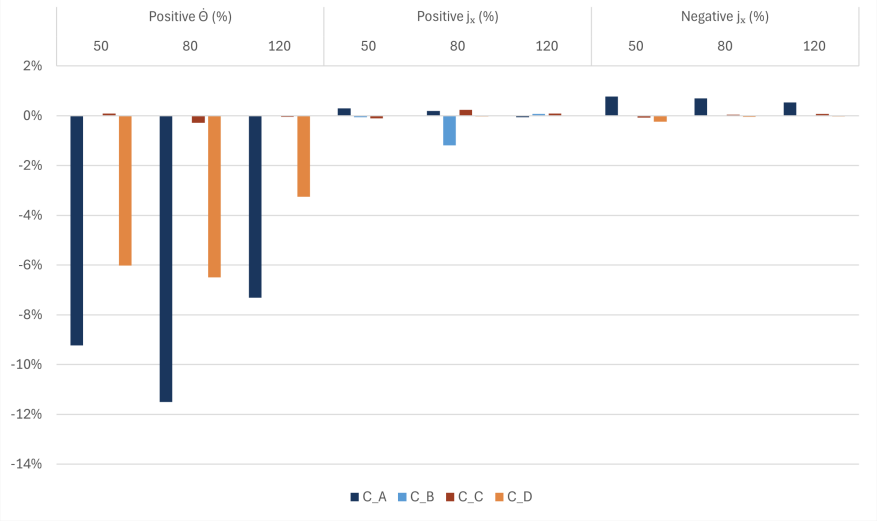


(b) Results in percentage of reference value (%)

Figure 32: Test results from the second test in Aggressive Braking.



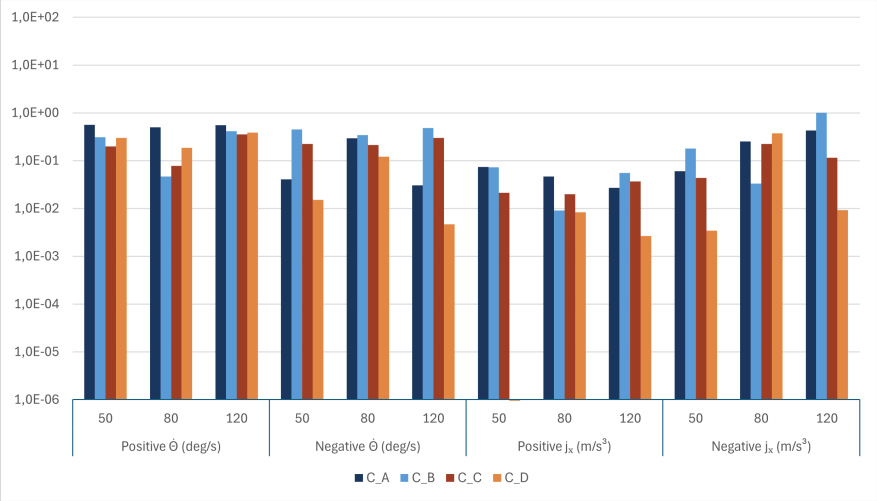
(a) Results in effect (C)



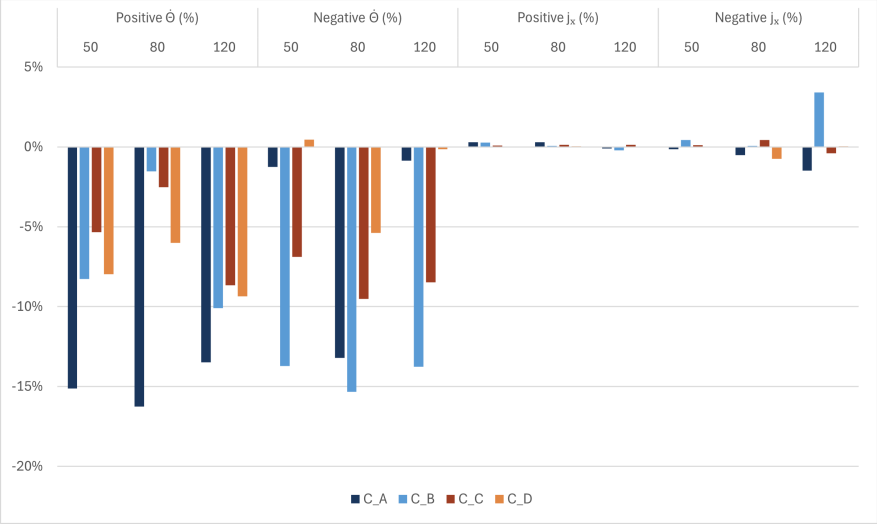
(b) Results in percentage of reference value (%)

Figure 33: Test results from the third test in Aggressive Braking.

Uneven Throttle 2nd and 3rd Test

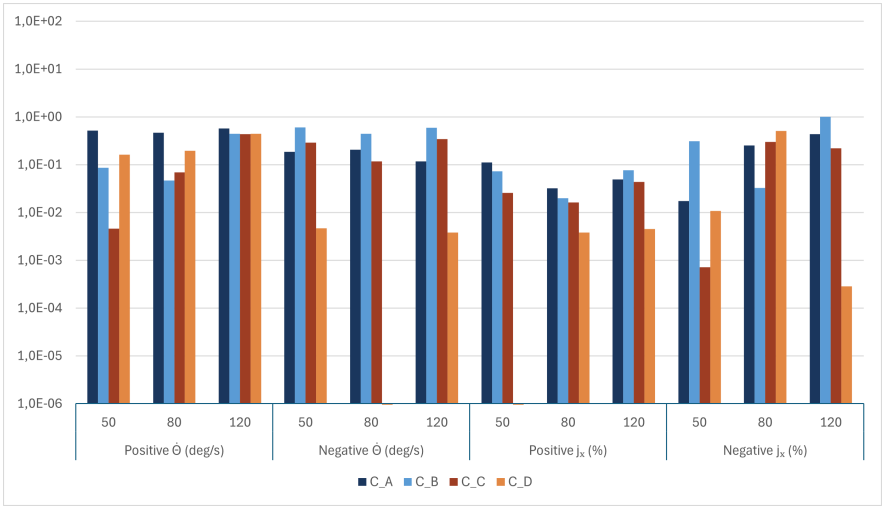


(a) Results in effect (C)

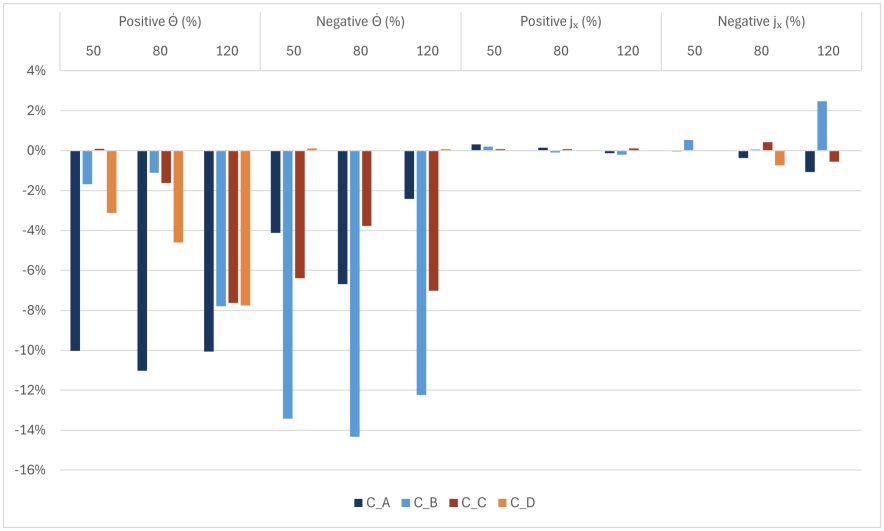


(b) Results in percentage of reference value (%)

Figure 34: Test results from the second test in Uneven Throttle.



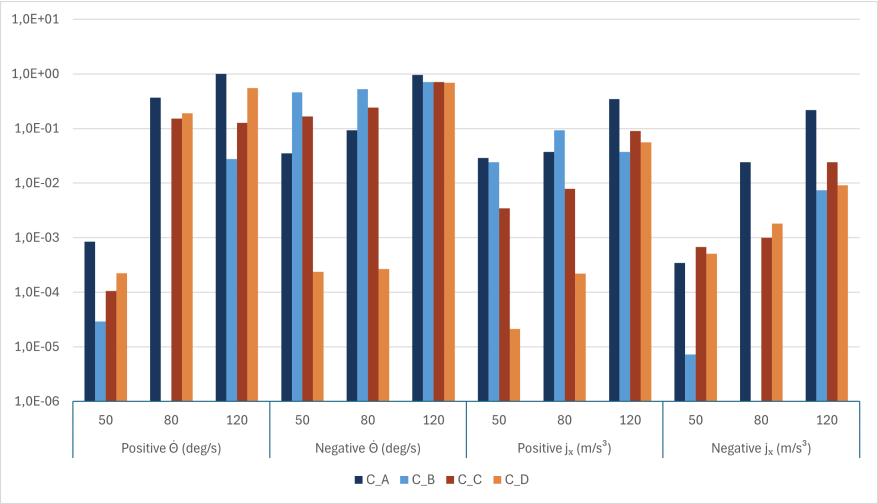
(a) Results in effect (C)



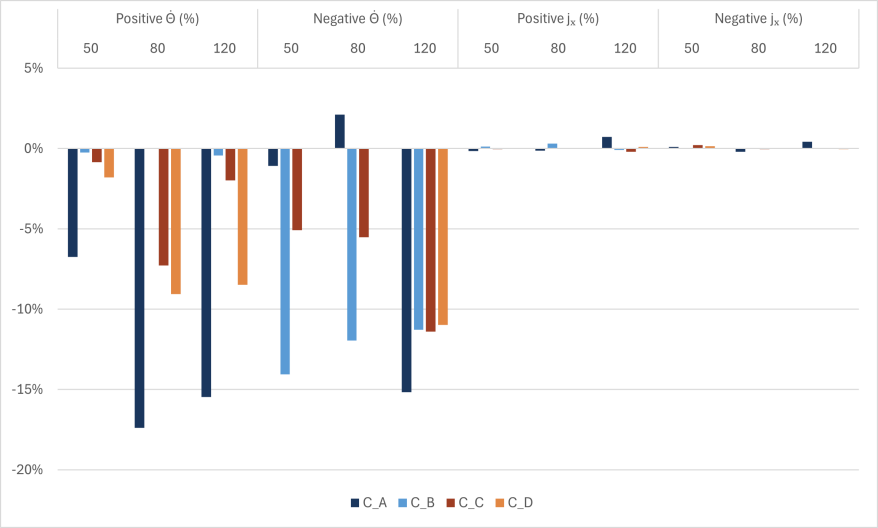
(b) Results in percentage of reference value (%)

Figure 35: Test results from the third test in Uneven Throttle.

Uneven Braking 2nd and 3rd Test

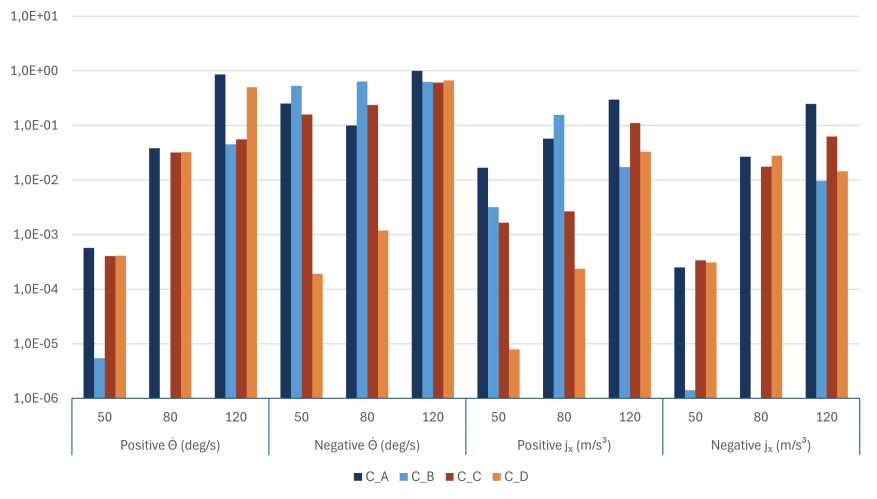


(a) Results in effect (C)

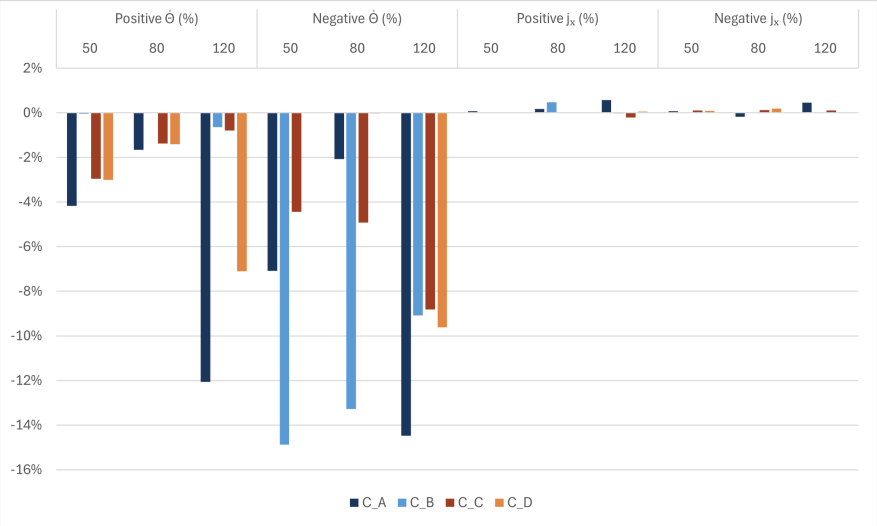


(b) Results in percentage of reference value (%)

Figure 36: Test results from the second test in Uneven Braking.



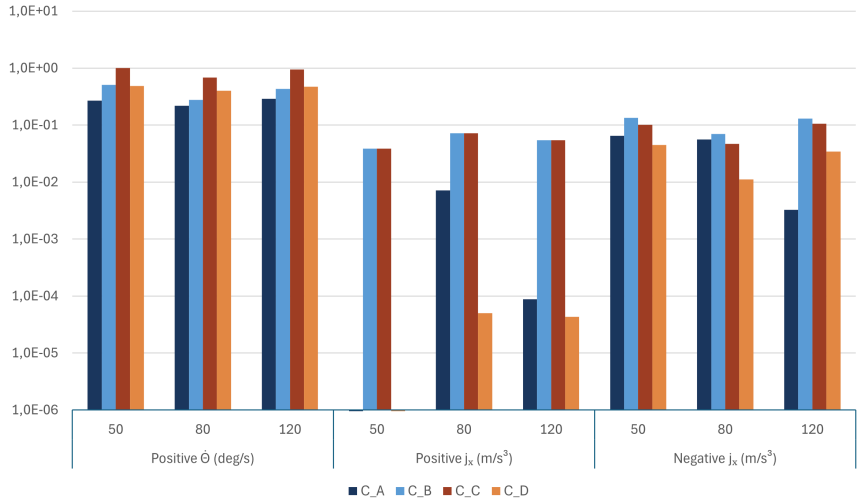
(a) Results in effect (C)



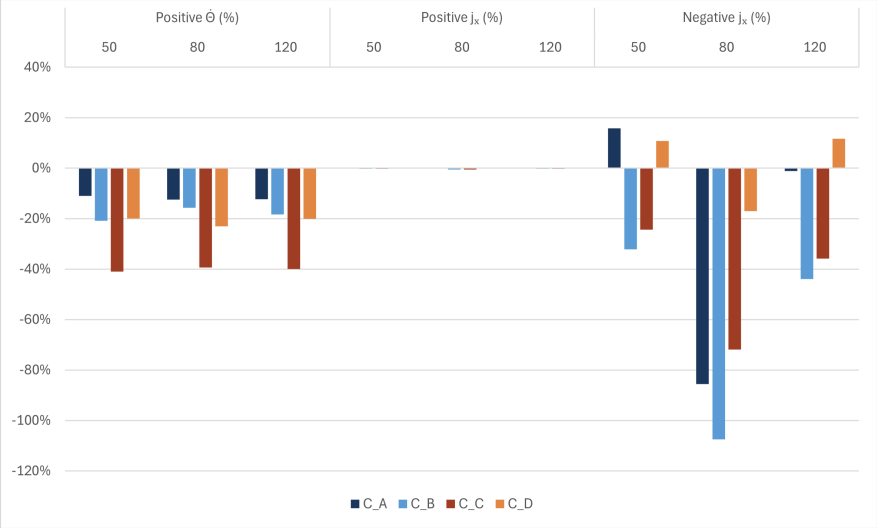
(b) Results in percentage of reference value (%)

Figure 37: Test results from the third test in Uneven Braking.

Brake Tapping 2nd and 3rd Test

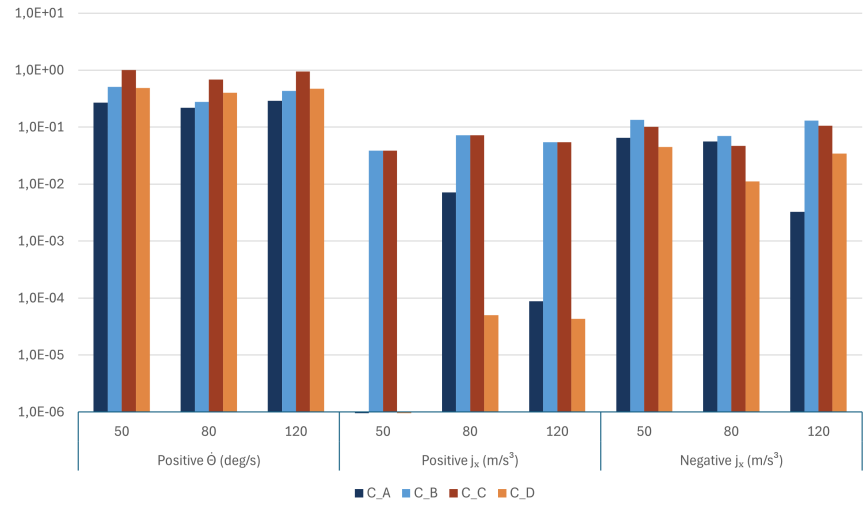


(a) Results in effect (C)

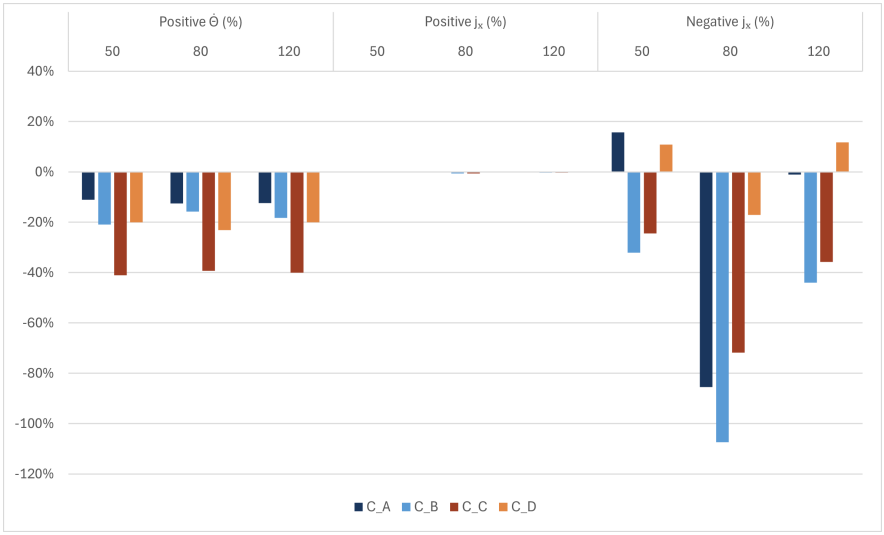


(b) Results in percentage of reference value (%)

Figure 38: Test results from the second test in Brake Tapping.



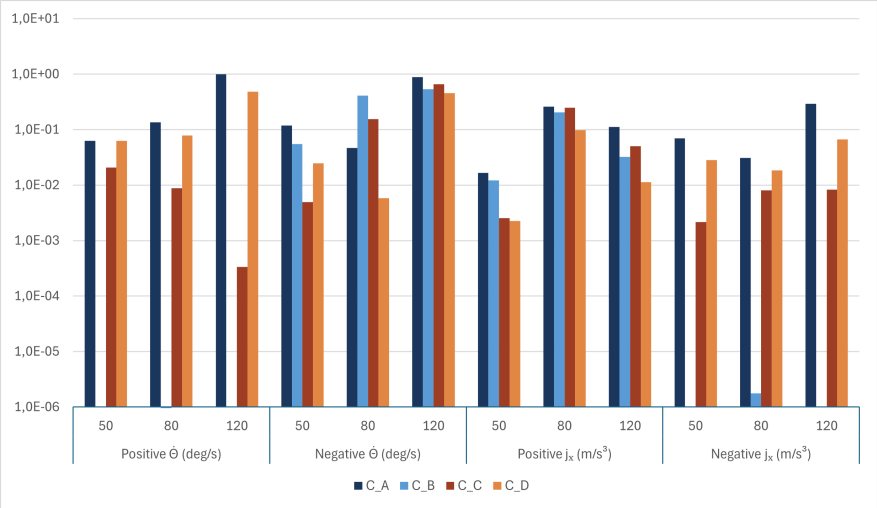
(a) Results in effect (C)



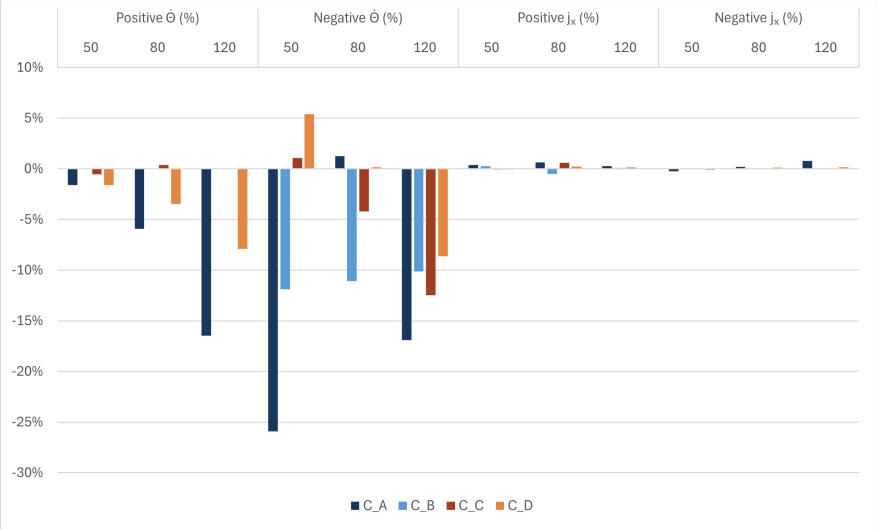
(b) Results in percentage of reference value (%)

Figure 39: Test results from the third test in Brake Tapping.

Braking After Throttle 2nd and 3rd Test

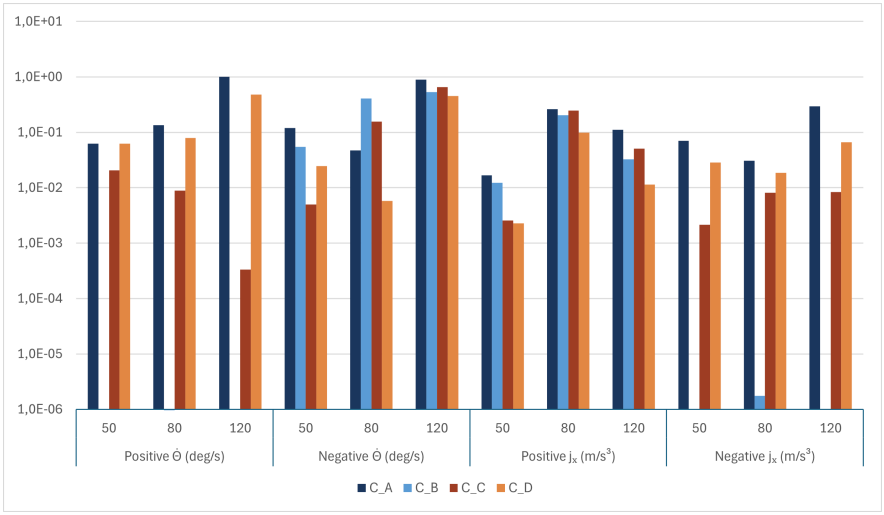


(a) Results in effect (C)

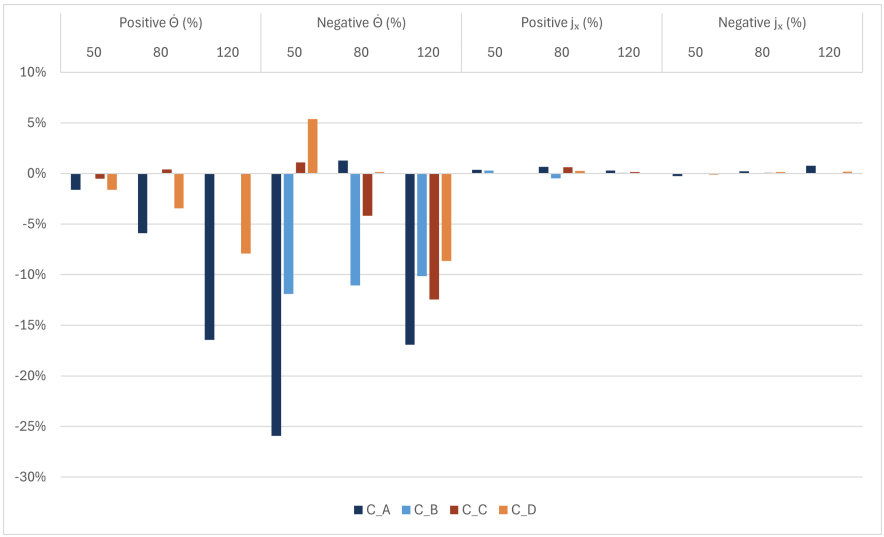


(b) Results in percentage of reference value (%)

Figure 40: Test results from the second test in Braking After Throttle.



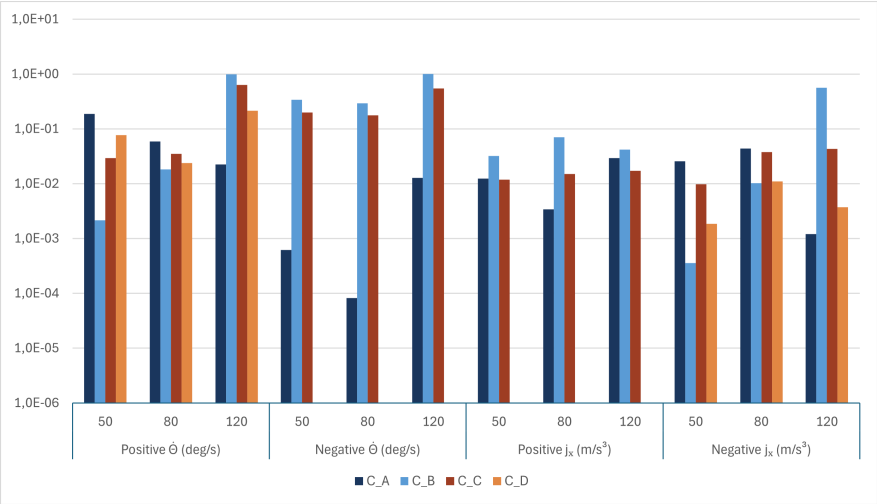
(a) Results in effect (C)



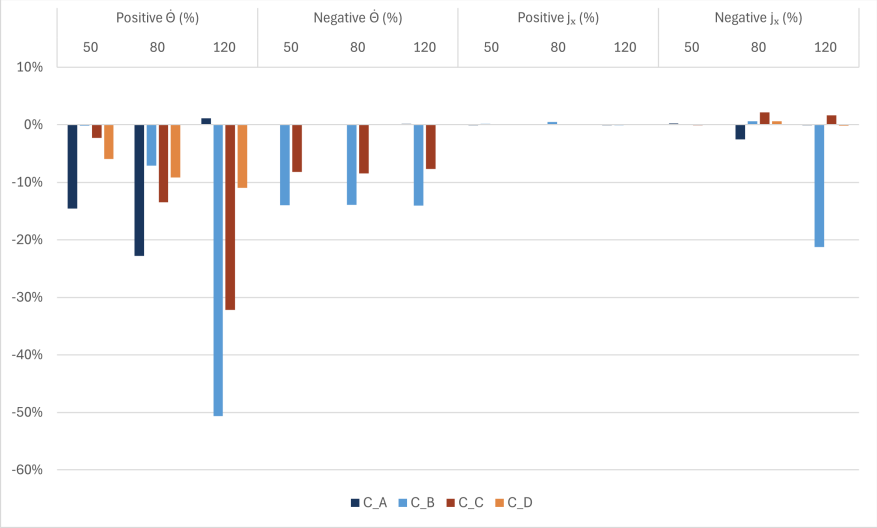
(b) Results in percentage of reference value (%)

Figure 41: Test results from the third test in Braking After Throttle.

Throttle after Braking 2nd and 3rd Test

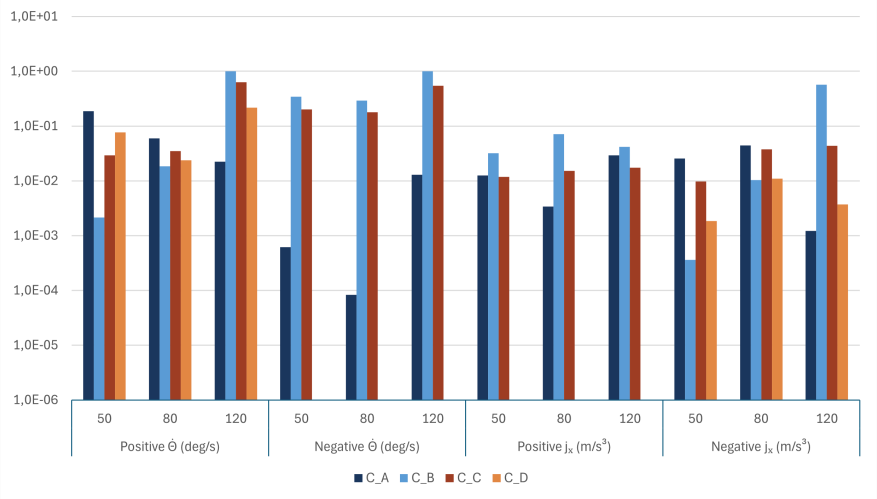


(a) Results in effect (C)

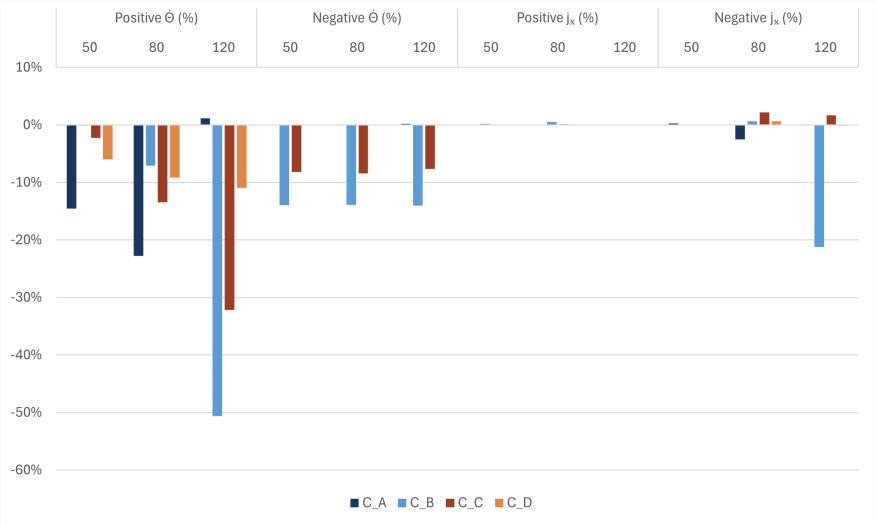


(b) Results in percentage of reference value (%)

Figure 42: Test results from the second test in Throttle After Braking.



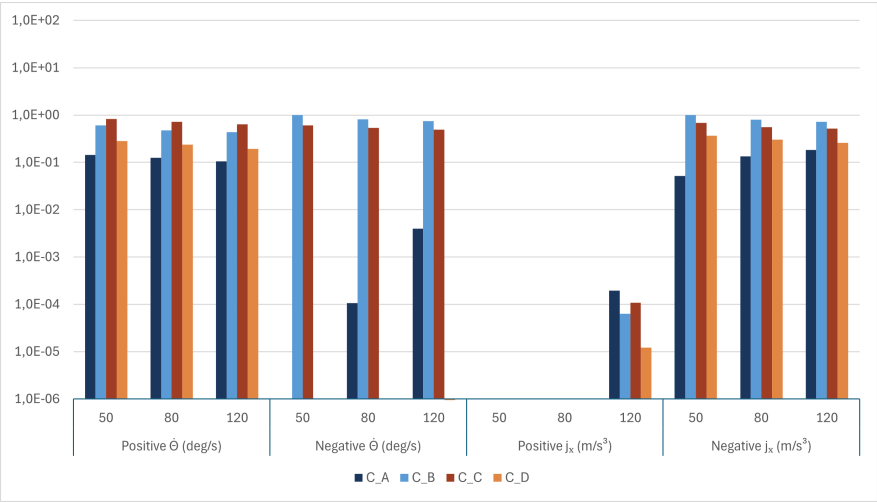
(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 43: Test results from the third test in Throttle After Braking.

Not Easing Off 2nd and 3rd Test

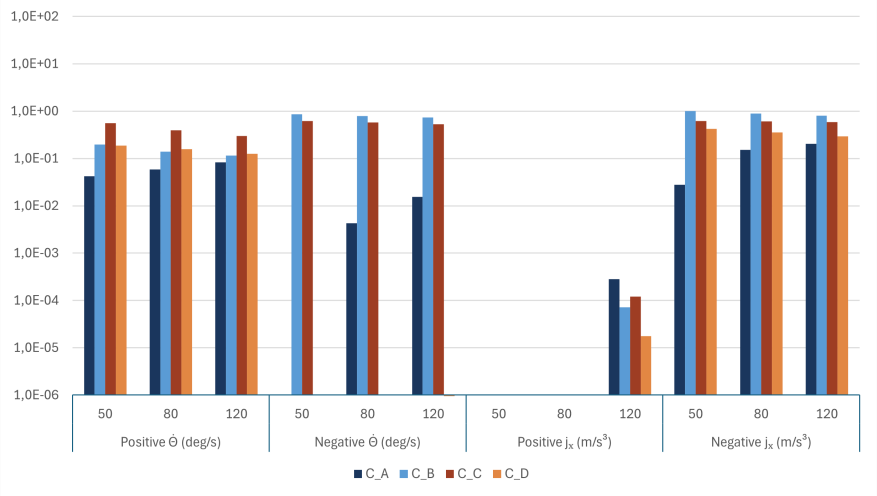


(a) Results in effect (C)

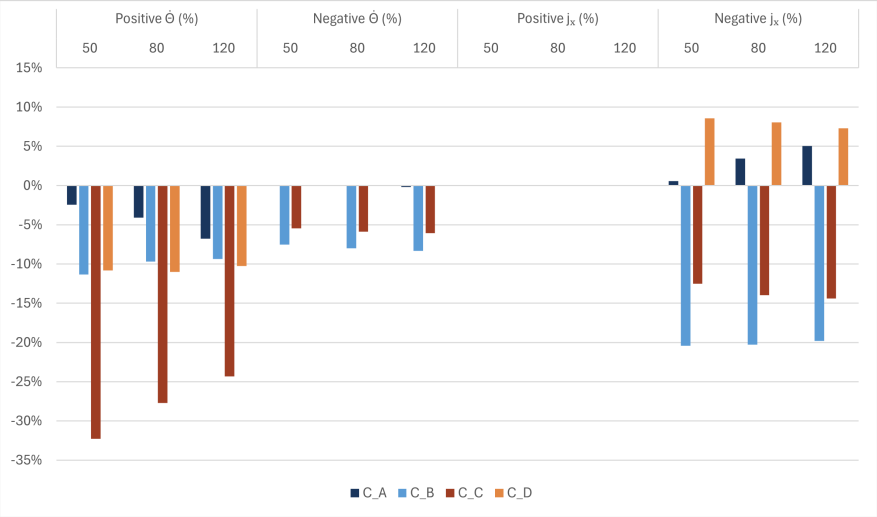


(b) Results in percentage of reference value (%)

Figure 44: Test results from the second test in Not Easing Off.



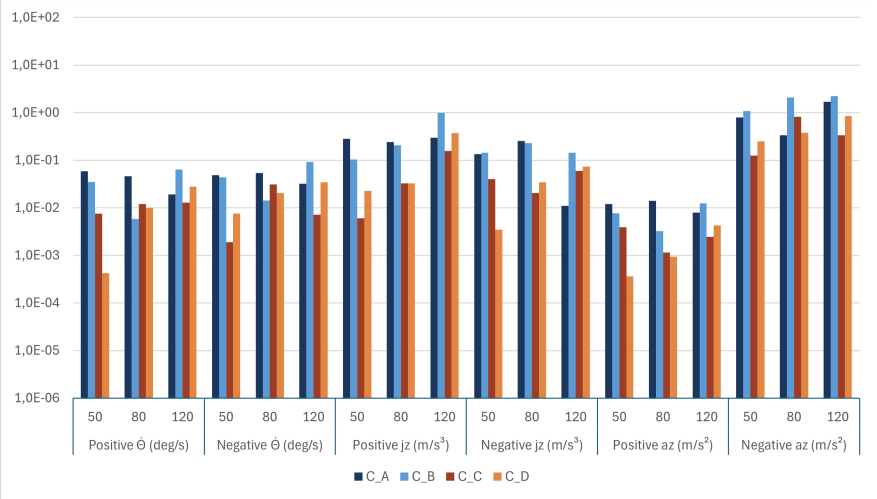
(a) Results in effect (C)



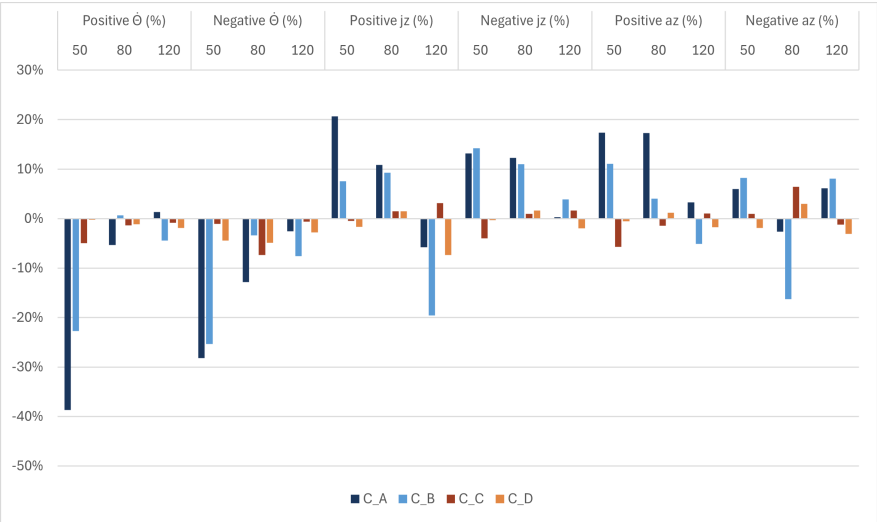
(b) Results in percentage of reference value (%)

Figure 45: Test results from the third test in Not Easing Off.

Speed Mismatch 2nd and 3rd Test

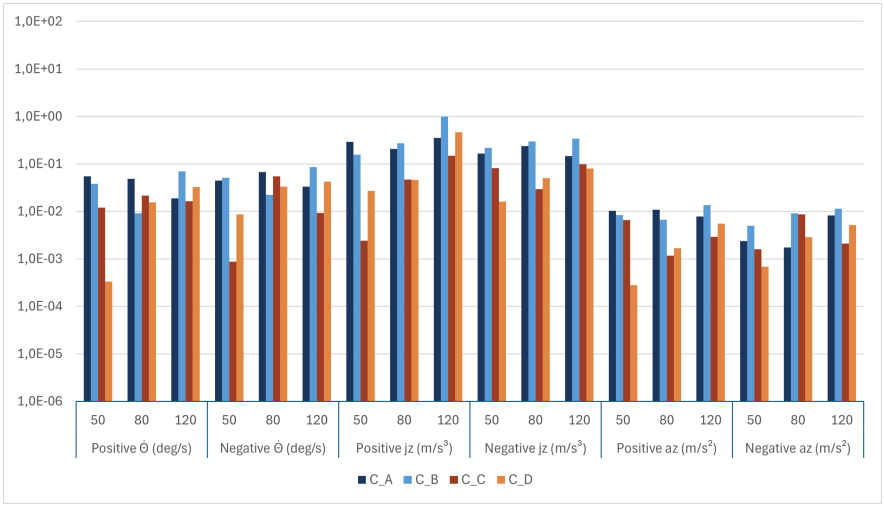


(a) Results in effect (C)

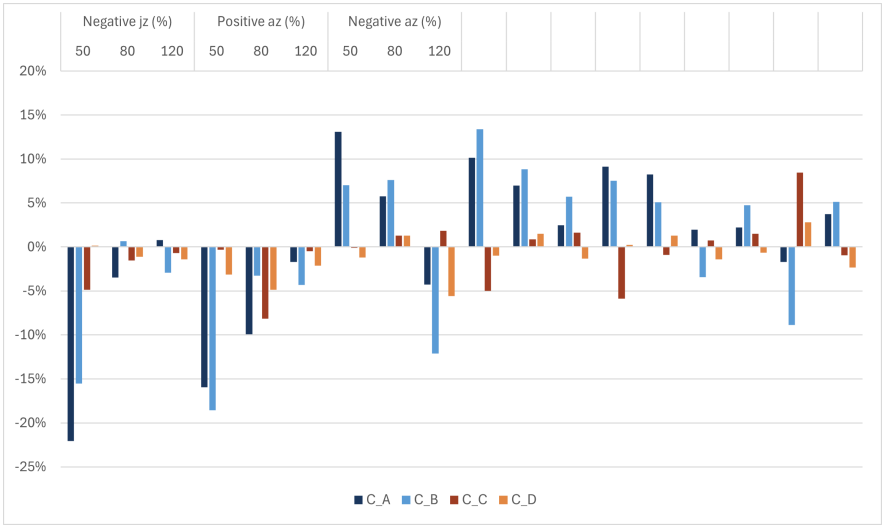


(b) Results in percentage of reference value (%)

Figure 46: Test results from the second test in Speed Mismatch With Road Surface.



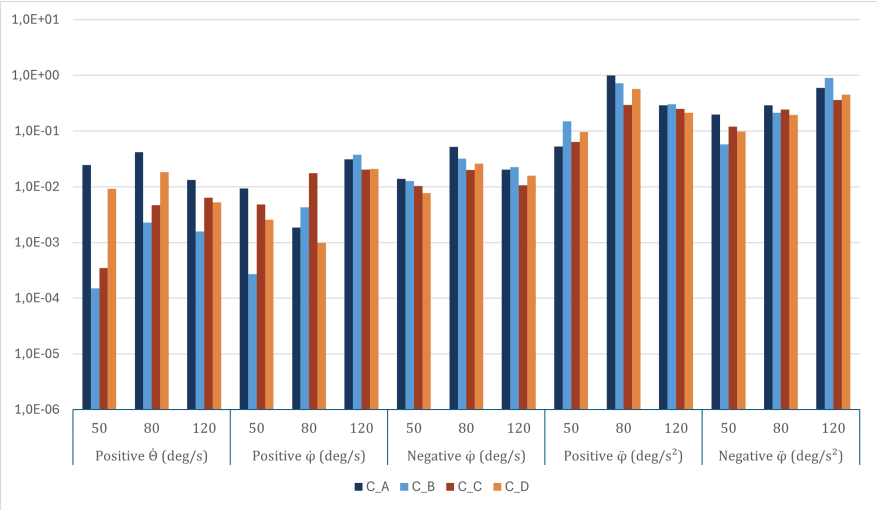
(a) Results in effect (C)



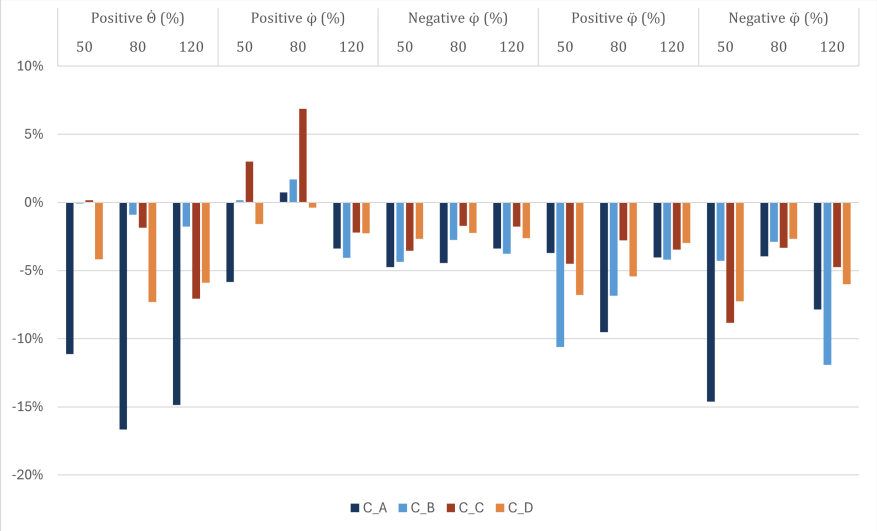
(b) Results in percentage of reference value (%)

Figure 47: Test results from the third test in Speed Mismatch With Road Surface.

Accelerating Into Turn 2nd and 3rd Test

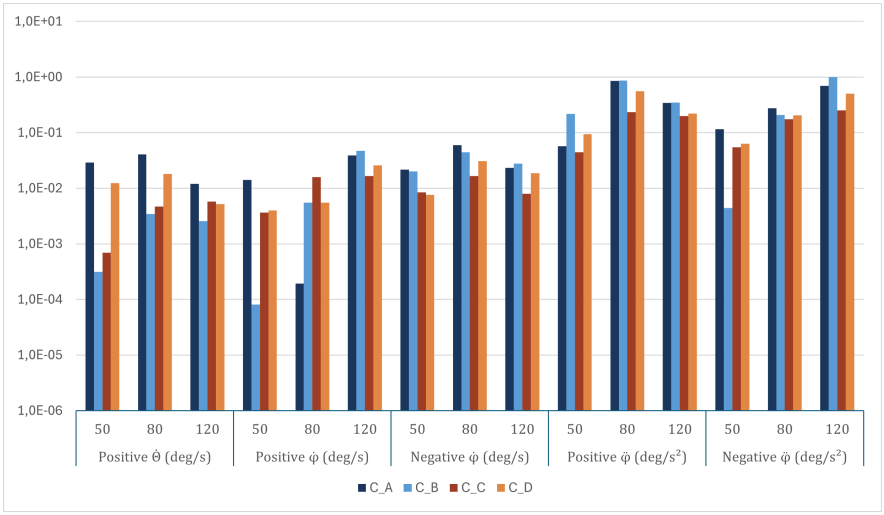


(a) Results in effect (C)

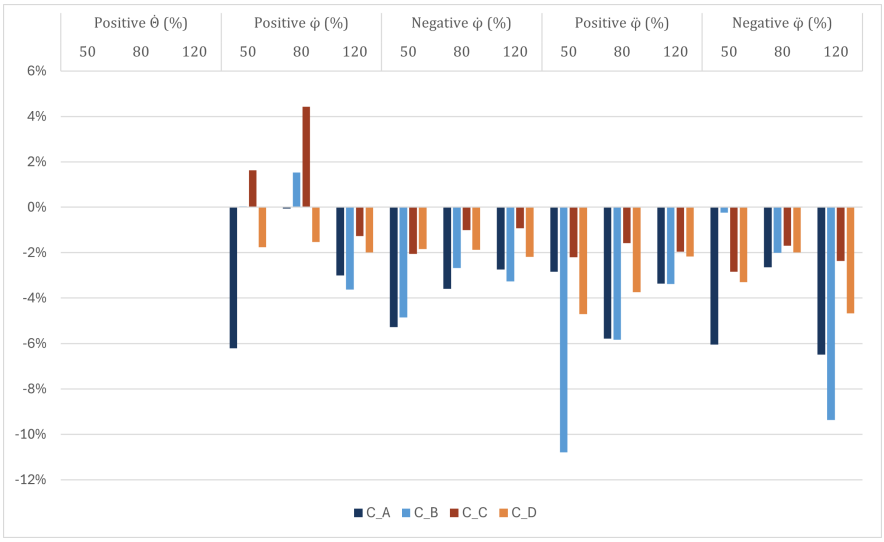


(b) Results in percentage of reference value (%)

Figure 48: Test results from the second test in Accelerating Into Turn.



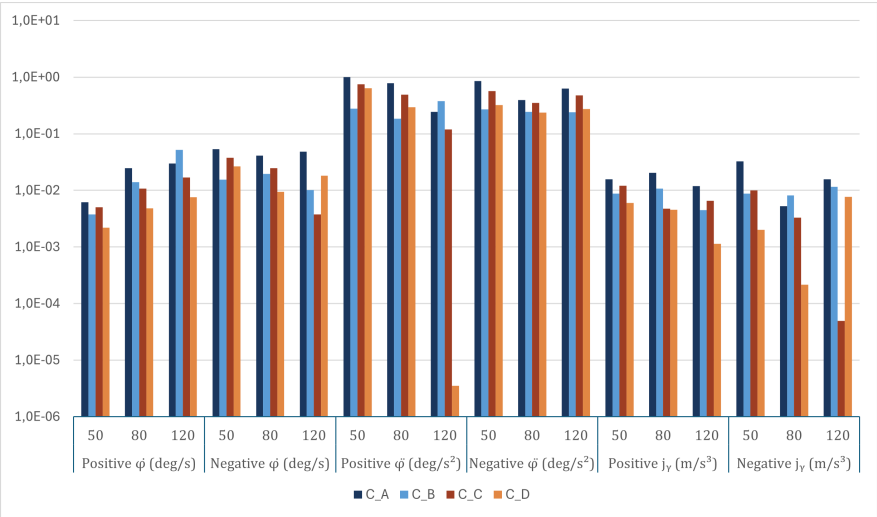
(a) Results in effect (C)



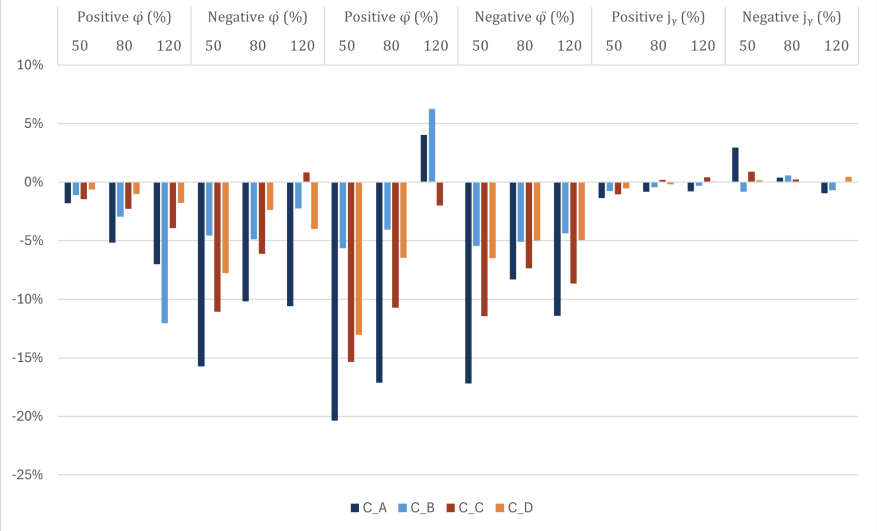
(b) Results in percentage of reference value (%)

Figure 49: Test results from the third test in Accelerating Into Turn.

Varying Angle In Constant Turn Radius 2nd and 3rd Test

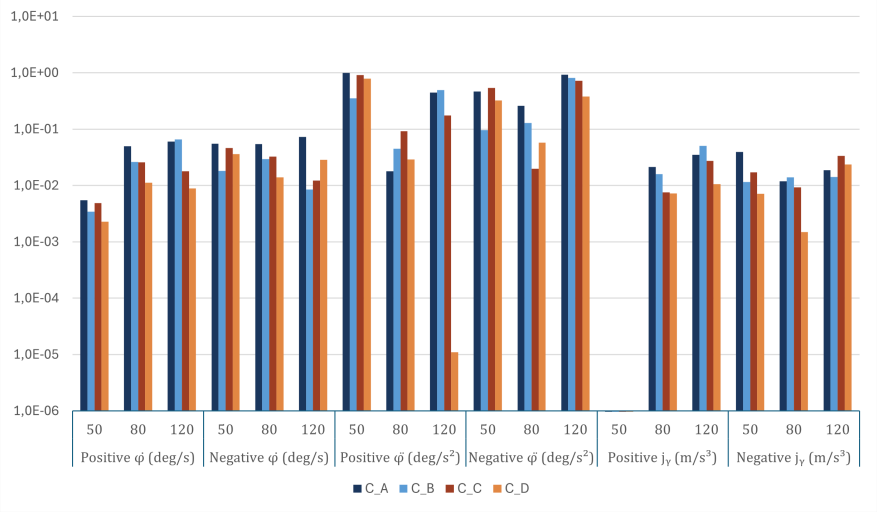


(a) Results in effect (C)

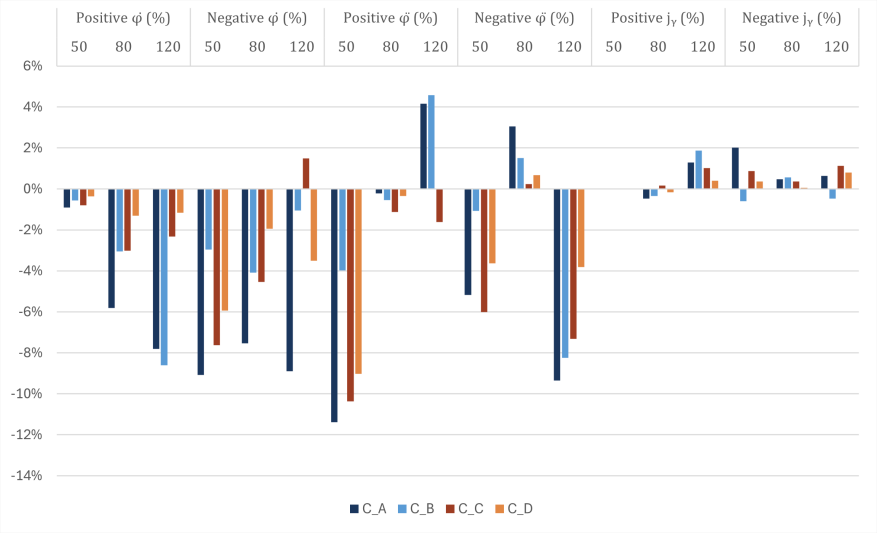


(b) Results in percentage of reference value (%)

Figure 50: Test results from the second test in Varying Angle In Constant Turn Radius.



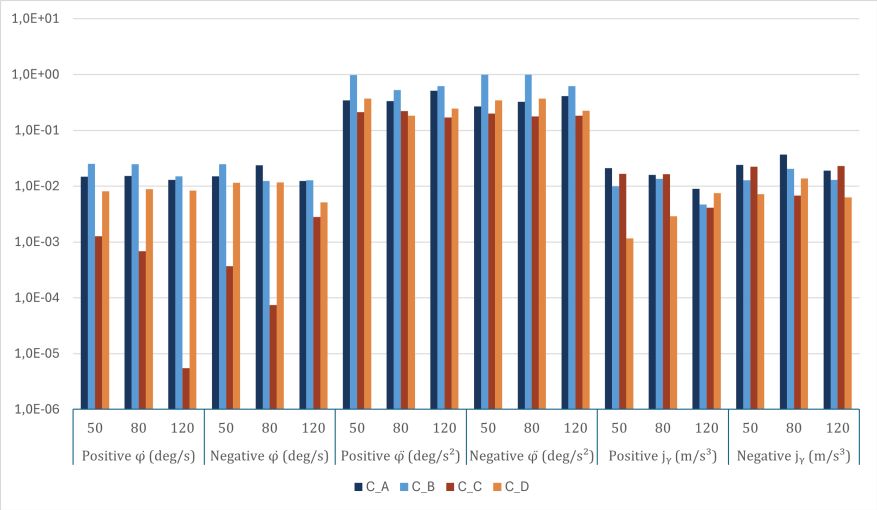
(a) Results in effect (C)



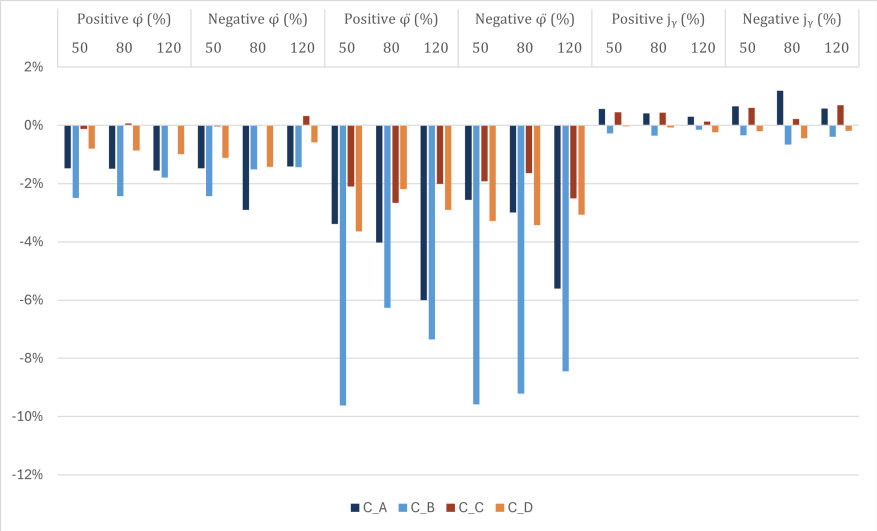
(b) Results in percentage of reference value (%)

Figure 51: Test results from the third test in Varying Angle In Constant Turn Radius.

Jerky Steering 2nd and 3rd Test

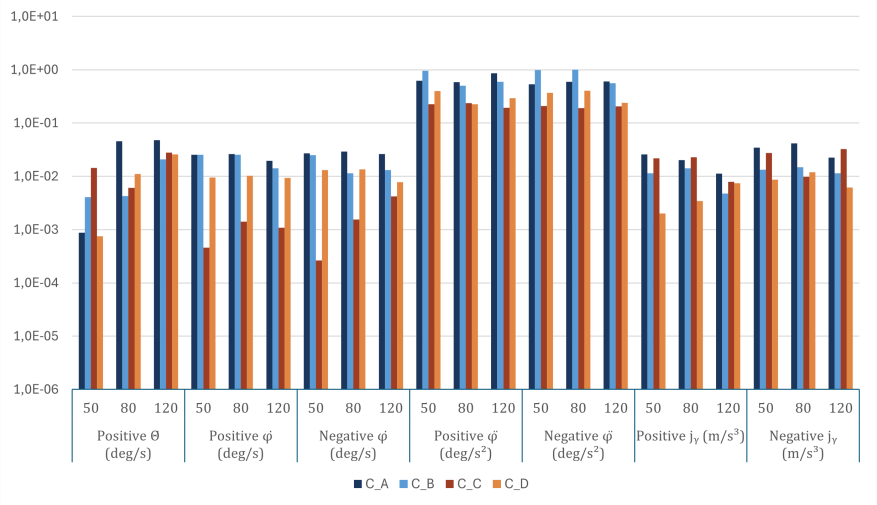


(a) Results in effect (C)

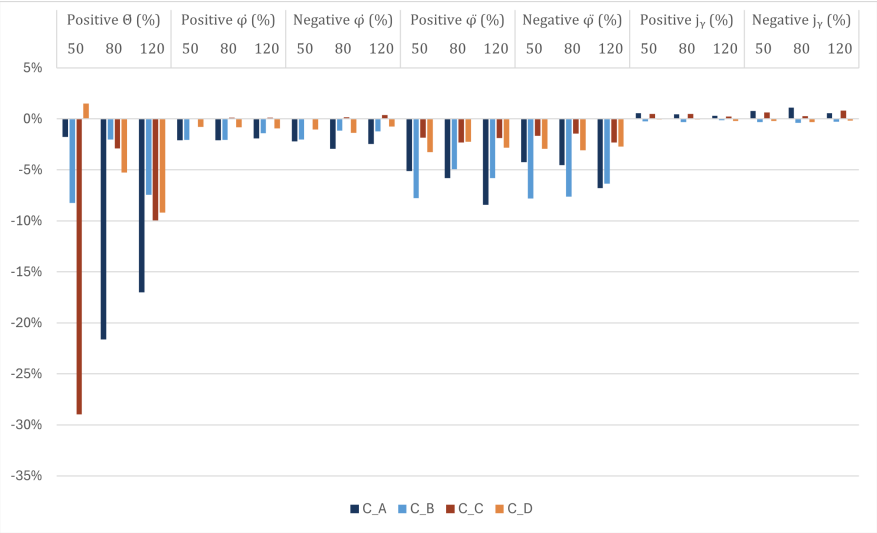


(b) Results in percentage of reference value (%)

Figure 52: Test results from the second test in Jerky Steering.



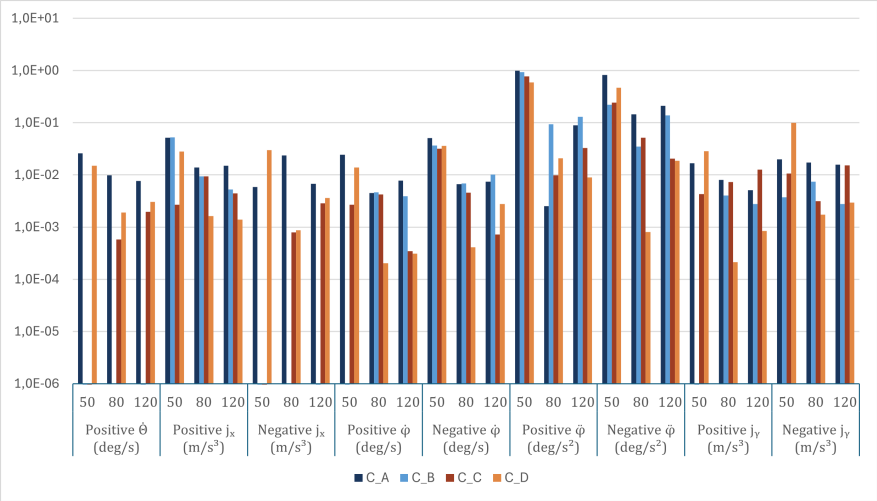
(a) Results in effect (C)



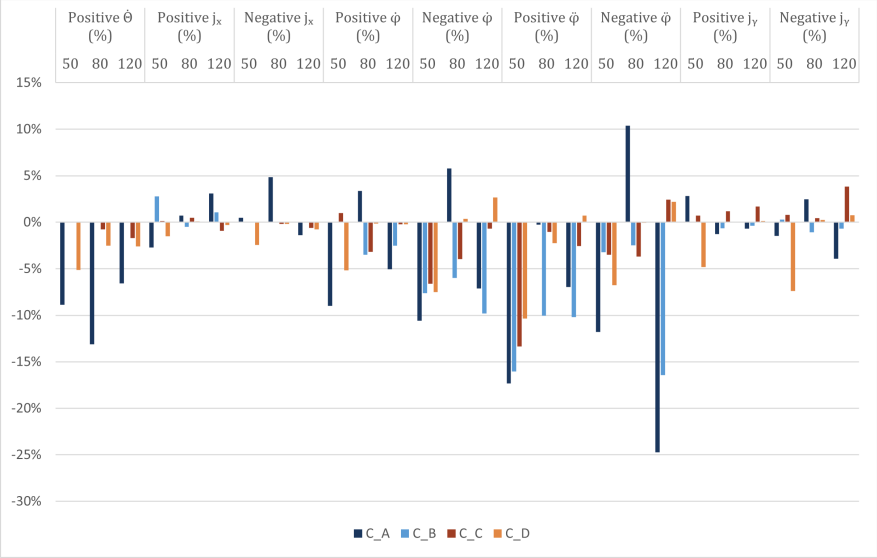
(b) Results in percentage of reference value (%)

Figure 53: Test results from the third test in Jerky Steering.

Steering and Braking 2nd and 3rd Test

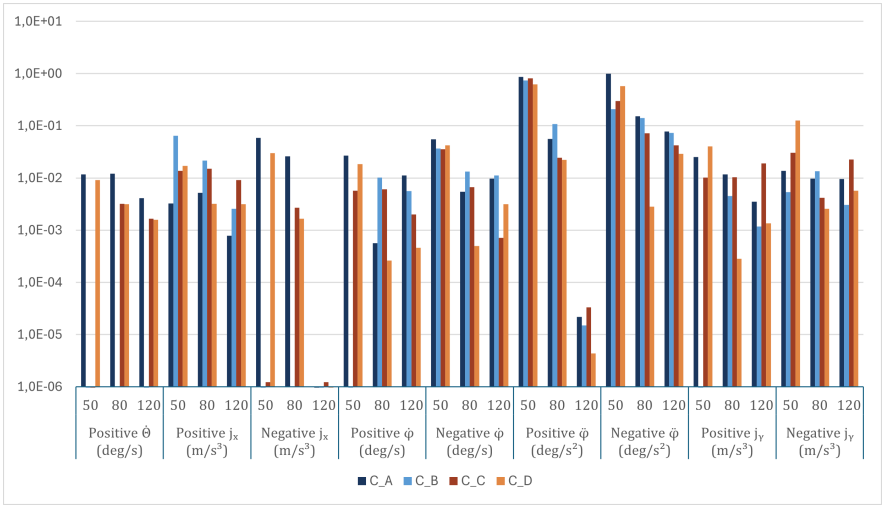


(a) Results in effect (C)

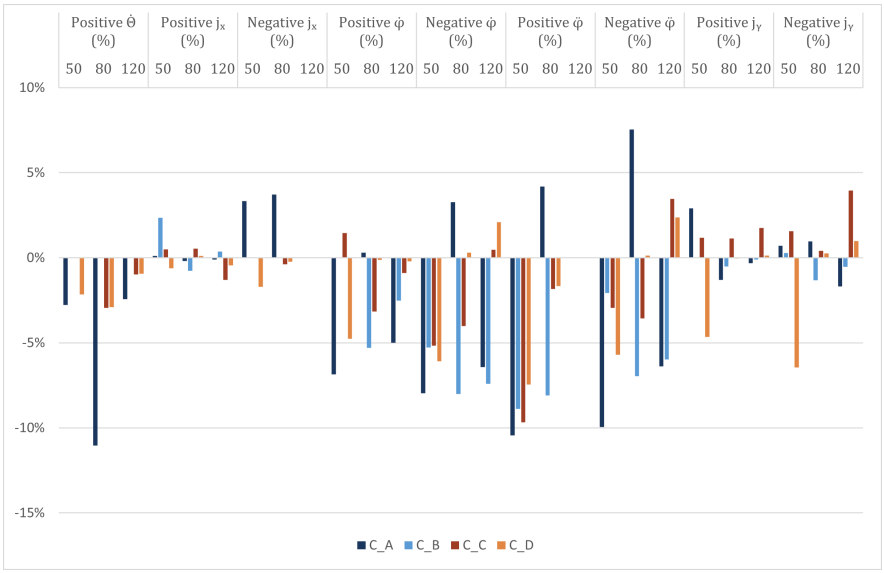


(b) Results in percentage of reference value (%)

Figure 54: Test results from the second test in Steering and Braking.



(a) Results in effect (C)



(b) Results in percentage of reference value (%)

Figure 55: Test results from the third test in Steering and Braking.

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